



GENCC501

COMPUTER SYSTEM AND ARCHITECTURE



Cloud Computing Services

TRAINEE'S MANUAL

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MODULE CODE AND TITLE: GENCC501 CLOUD COMPUTING SERVICES

Learning Outcome 1: Create Cloud Computing Account

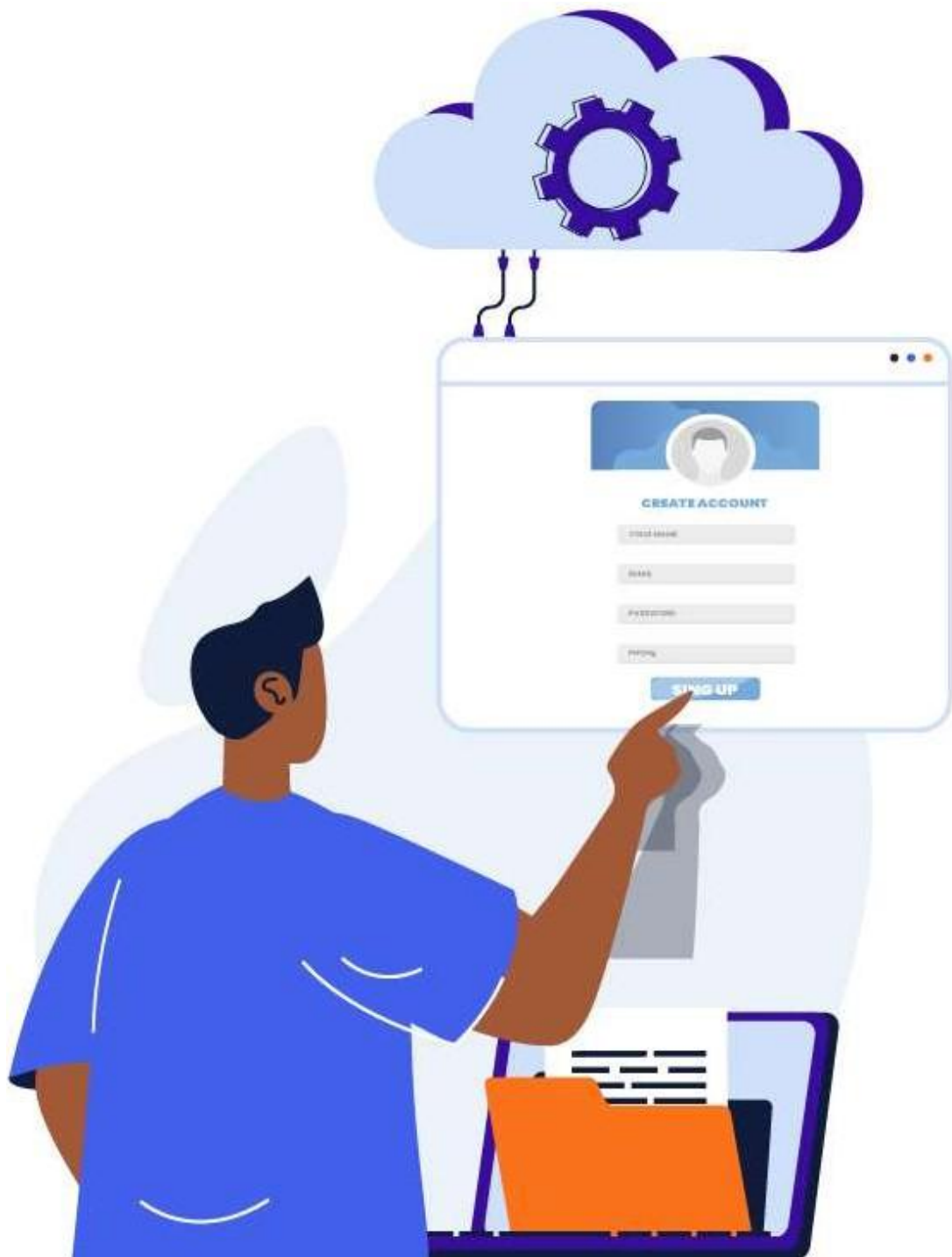
Learning Outcome 2: Manage Cloud Computing

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Learning Outcome 1: Create Cloud Computing Account



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Key Competencies for Learning Outcome: Create an Account with Cloud Service Providers

Knowledge	Skills	Attitudes
• Description of cloud computing service models. • Description of Service Level Agreement • Description of cloud computing service providers. • Description of customer's account in cloud computing. • Description of Identity and Access Management (IAM)	• Selecting cloud computing service providers. • Creating customer's account on cloud computing. • Creating Identity and Access Management (IAM). • Performing Backup and Disaster Recovery	• Being passionate • Being attentive • Being creative. • Being innovative. • Having critical thinking skills



1. Definition of key terms in Cloud Computing

1.1. Cloud computing

Cloud computing is defined as the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet (the cloud).

Instead of owning and maintaining physical infrastructure, users can access these resources on-demand from cloud providers, allowing for scalability, flexibility, and cost efficiency.

1.2. Cloud service model

Cloud service model refers to the different ways or methods in which cloud

computing services can be provided to customers or organizations.

2. Cloud computing service models

Common cloud service models include Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

2.1. Software as a Service

Provides fully managed software applications over the internet. Users simply access and use the software without worrying about installation, maintenance, or infrastructure management.

Example of Software as a Service (SaaS) is Google Workspace (formerly G Suite), which includes applications like Gmail, Google Drive, Google Docs, and Google Sheets.

How It Works:

Gmail (Email Service): Instead of setting up and maintaining email servers, users can simply log in to Gmail through a web browser or app to send, receive, and manage emails.

Google Docs (Word Processing): Users can create, edit, and share documents online without needing to install any software on their computer.

Google Drive (Cloud Storage): Users can store, access, and share files from anywhere, using an internet connection, without worrying about managing servers or storage devices.

In this example, the software (email, word processing, storage) is fully managed by Google, meaning the user doesn't need to install, update, or maintain anything.

2.2. Platform as a Service (PaaS)

Provides a platform that allows users to build, run, and manage applications without dealing with the underlying infrastructure. The cloud provider manages the infrastructure, while users focus on application development and management.

Example of Platform as a Service (PaaS) is Heroku. a cloud platform used by developers to build, run, and manage applications without worrying about the underlying infrastructure.

How It Works:

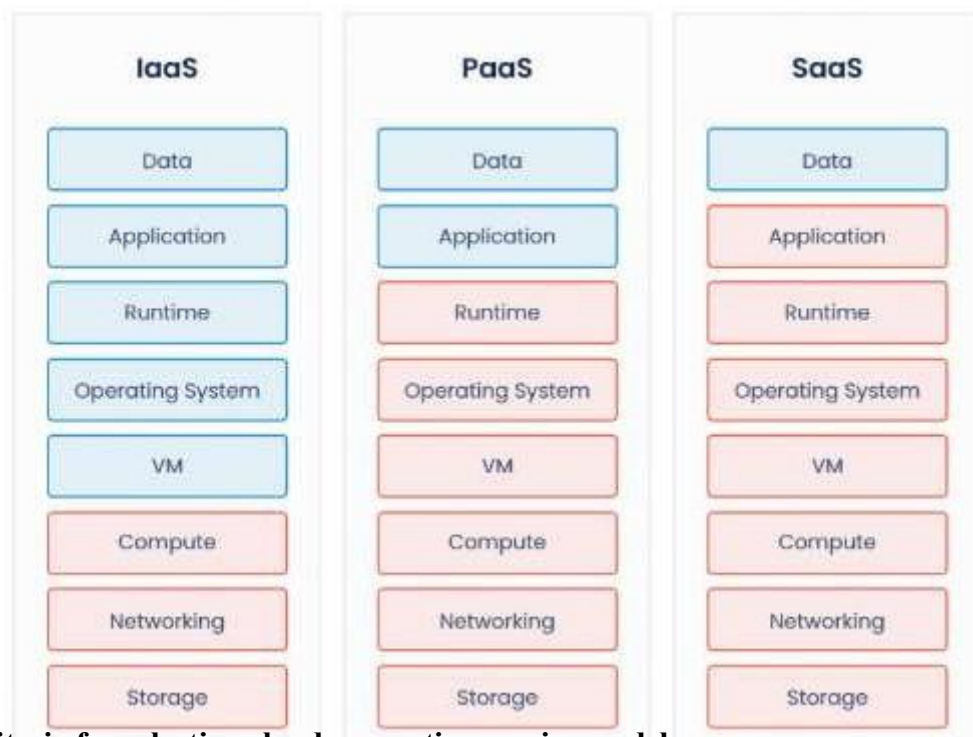
Developers can write code in different programming languages (e.g., Python, Ruby, Java), then upload it to Heroku. Heroku automatically manages the servers, operating systems, and runtime environments needed to run the application.

Developers don't need to set up or maintain servers, databases, or networks. Heroku takes care of all the infrastructure in the background.

2.3. Infrastructure as a Service (IaaS)

Provides virtualized hardware resources like servers, storage, and networking. Users have control over operating systems, applications, and storage but the cloud provider manages the underlying physical infrastructure.

Example: Amazon Web Services (AWS EC2), Microsoft Azure.



3. Criteria for selecting cloud computing service models

Selecting the right cloud computing service model depends on various criteria that align with the specific needs of your organization or project. The following key criteria can help guide the decision:

3.1. User needs

IaaS: If you need full control over the infrastructure (e.g., managing virtual machines, operating systems), IaaS is ideal.

PaaS: If you want to focus on developing applications without managing the underlying infrastructure, PaaS is a better choice.

SaaS: If you only need to use software without worrying about infrastructure or development, SaaS is the simplest option.

3.2. Price

IaaS: Typically involves more upfront costs, but you pay for what you use and scale as needed. Suitable for long-term infrastructure investments.

PaaS: Can reduce costs by eliminating the need to manage servers, storage, or networks, but still provides flexibility for developers.

SaaS: Generally, the most cost-effective for end-users, as you pay for subscription-based software without worrying about infrastructure or maintenance.

3.3. Operation

IaaS: Requires a technical team with expertise in infrastructure, networking, and server management.

PaaS: Requires developers to work on application development, with less need for infrastructure management skills.

SaaS: Requires minimal technical expertise, making it suitable for non-technical users who need ready-to-use applications.



Points to Remember

- Cloud computing is defined as the delivery of computing services, such as servers, storage, databases, networking, software, and analytics, over the internet (the cloud).
- Cloud service model refers to the different ways or methods in which cloud computing services can be provided to customers or organizations.
- Common cloud computing service models include Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).
- Cloud computing service model are selected based on the user needs, pricing, and operation.



Key readings 1.2.1: Description of Service Level Agreement

1. Definition of service level agreement

In cloud computing, a **Service Level Agreement (SLA)** is a legal contract between a cloud service provider and its customer that outlines the expected level of service performance. It defines the specific metrics and standards that the provider must meet, such as:

Uptime: The percentage of time the service is available and operational.

Performance: The speed, response time, and throughput of the service.

Data security: The measures taken to protect customer data from unauthorized

access or breaches.

Technical support: The availability and responsiveness of the provider's support team.

Disaster recovery: The procedures in place for restoring service in case of a failure or outage.

2. Components of service level agreement

2.1. Service Definition

- **Services Included:** Specifies the exact services covered by the SLA, such as infrastructure as a service (IaaS), platform as a service (PaaS), or software as a service (SaaS).
- **Service Boundaries:** Clearly defines the scope(opportunity) of the services provided and any limitations or exclusions.

2.2. Performance Metrics

- i. **Key Performance Indicators (KPIs):** Identifies measurable metrics that will be used to assess service performance, such as:
 - **Uptime:** Percentage of time the service is available.
 - **Response Time:** Time it takes for the service to respond to requests.
 - **Throughput:** The amount of data or transactions a service can handle.
 - **Data Loss Rate:** The frequency of data loss or corruption.
- ii. **Targets:** Sets specific targets for each KPI that the provider must meet.

2.3. Service Credits

- **Penalties:** Specifies the penalties or service credits that the provider will owe to the customer if the service fails to meet the agreed-upon performance metrics.
- **Calculation Methods:** Outlines how service credits will be calculated, such as based on the severity and duration of service disruptions.

2.4. Reporting and Monitoring

- **Reporting Frequency:** Determines how often the provider will report on service performance.
- **Monitoring Tools:** Specifies the tools or methods that will be used to monitor service performance.
- **Data Retention:** Defines how long performance data will be retained.

2.5. Dispute Resolution

- **Process:** Outlines the process for resolving disputes or disagreements related to the SLA.
- **Mediation or Arbitration:** May specify whether mediation or arbitration will be used to resolve disputes.

2.6. Term and Termination

- **Duration:** Specifies the length of the SLA.
- **Termination Conditions:** Defines the conditions under which either party can terminate the SLA.

2.7. Confidentiality

- **Data Privacy:** Addresses the protection of customer data and information.
- **Non-Disclosure:** Specifies the obligations of both parties to maintain confidentiality.

2.8. Force Majeure

- **Exclusions:** Defines events that may excuse the provider from meeting its obligations, such as natural disasters or acts of war.

3. Types of service level agreement in cloud computing

3.1. Basic SLA

- Provides a general overview of service performance expectations.
- Often includes standard metrics like uptime and response time.
- Suitable for customers with less stringent requirements.

3.2. Tiered SLA

- Offers multiple levels of service with varying performance guarantees and costs.
- Customers can choose the tier that best suits their needs and budget.
- Provides flexibility for organizations with diverse requirements.

3.3. Customized SLA

- Tailored to meet the specific needs of a particular customer or use case.
- Can include unique metrics, performance targets, and service credits.
- Ideal for organizations with complex or mission-critical applications.

3.4. Financial SLA

- Focuses on the financial implications of service disruptions.
- May include penalties or service credits based on the financial impact of downtime.
- Suitable for businesses where service outages can have significant financial consequences.

3.5. Operational SLA

- Emphasizes the operational aspects of service delivery, such as data security, compliance, and disaster recovery.
- Often includes specific requirements related to data protection, regulatory compliance, and business continuity planning.

- Important for organizations with stringent regulatory or compliance requirements.

3.6. Hybrid SLA

- Combines elements of different types of SLAs to address the unique needs of a customer.
- Can include a mix of basic, tiered, customized, financial, and operational components.
- Provides flexibility to accommodate various requirements within a single agreement.

4. Benefits of Service Level Agreements (SLAs) in Cloud Computing

4.1. For Cloud Service Providers:

- **Enhanced Customer Satisfaction:** SLAs help providers meet customer expectations by providing clear performance guarantees. This can lead to increased customer satisfaction and loyalty.
- **Improved Service Quality:** By defining specific performance metrics and targets, SLAs incentivize providers to continuously improve their service quality.
- **Risk Mitigation:** SLAs can help providers mitigate risks associated with service disruptions by outlining clear responsibilities and consequences.
- **Competitive Advantage:** Offering strong SLAs can give providers a competitive edge in the market, attracting more customers and differentiating their services.

4.2. For Cloud Customers:

- **Guaranteed Service Quality:** SLAs ensure that customers receive a certain level of service performance, reducing the risk of unexpected outages or performance issues.
- **Reduced Risk:** SLAs can help customers mitigate risks associated with service disruptions by providing clear remedies or compensation.
- **Improved Business Continuity:** SLAs can support business continuity planning by defining service availability and recovery procedures.
- **Cost Control:** SLAs can help customers control costs by providing a clear understanding of service expectations and penalties for non-compliance.
- **Legal Protection:** SLAs provide a legal framework for resolving disputes and ensuring that both parties fulfil their obligations.



Key readings 1.2.2. Description of cloud computing service providers

1. Definition of cloud computing service providers

Cloud computing service providers are defined as companies that deliver cloud computing services over the internet. They manage and maintain the underlying infrastructure, making it accessible to users on a pay-as-you-go basis.

2. Types of Cloud Computing Service Providers

2.1. Public Cloud Providers

- Offer services to the general public over the internet.
- Typically, large-scale operations with extensive data centres and infrastructure.
- Examples: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)

2.2. Private Cloud Providers

- Offer services to a single organization or a small group of organizations.

- The infrastructure is dedicated to the exclusive use of the customer(s), ensuring higher levels of security and control.
- Often deployed on-premises or in a data centre managed by the provider.
- **Hybrid Cloud Providers**
- Combine public and private cloud services to create a flexible and scalable solution.
- Allows organizations to leverage the benefits of both public and private clouds, such as cost-effectiveness and security.
- Provides options for workload placement and data management.

2.3. Multi-Cloud Providers

- Manage and integrate multiple cloud platforms from different providers.
- Offer a unified management interface and the ability to optimize resource allocation across different clouds.
- Provide flexibility and redundancy, reducing reliance on a single provider.

2.4. Community Cloud Providers

Offer services to a specific group of users, such as government agencies, academic institutions, or industries.

The infrastructure is shared by members of the community, leading to cost savings and improved collaboration.

3. Advantages and disadvantages of cloud computing service provider

3.1. Advantages:

- **Scalability:** Cloud services can be easily scaled up or down to meet changing demands, ensuring optimal resource utilization.
- **Cost-Effectiveness:** Pay-as-you-go pricing models eliminate the need for upfront capital investments and reduce operational costs.
- **Flexibility:** Cloud services offer flexibility in terms of deployment models, location, and service types, allowing organizations to adapt to changing business needs.
- **Efficiency:** Cloud providers manage and maintain the infrastructure, reducing the need for in-house IT resources and freeing up time for core business activities.
- **Innovation:** Cloud providers often offer innovative features and technologies, such as machine learning, artificial intelligence, and big data analytics.
- **Accessibility:** Cloud services can be accessed from anywhere with an internet connection, enabling remote work and collaboration.

3.2. Disadvantages:

- **Dependency:** Relying on a third-party provider can introduce risks related to

service disruptions or data breaches.

- **Security Concerns:** While cloud providers implement security measures, there is still a risk of data breaches or unauthorized access.
- **Vendor Lock-in:** Organizations may become locked into a specific provider's ecosystem, making it difficult to switch to another provider.
- **Control and Customization:** Cloud providers may have limitations on customization and control over the underlying infrastructure.
- **Internet Connectivity:** Cloud services rely on a stable internet connection, which can impact performance and availability.

4. Criteria for selecting cloud computing service providers

4.1. User needs

- **Cloud Models:** Does the provider offer the types of services you need (IaaS, PaaS, SaaS)? Ensure they provide the specific services required for your use case.
- **Service Customization:** Evaluate whether the provider offers customizable options to tailor solutions to your business needs.
- **Pricing**
- **Transparent Pricing:** Ensure the provider offers clear pricing models (e.g., pay-as-you-go, fixed pricing) and that there are no hidden fees.
- **Cost Efficiency:** Compare the costs of storage, computing, and bandwidth. Evaluate if the provider offers cost-saving features such as reserved instances, bulk discounts, or resource optimization tools.
- **Billing Transparency:** Check if the provider offers detailed billing information and usage analysis to avoid unexpected charges.

4.2. Operation

- **Uptime and SLAs:** Ensure the provider offers a high level of uptime (99.9% or more) backed by a Service Level Agreement (SLA) with clear performance guarantees.
- **Latency and Speed:** Check the latency and performance metrics, particularly if your business demands low-latency applications like streaming or real-time services.

Key readings 1.2.3. Selecting Cloud Computing Service Provider based on the SLA

1. Steps for selecting cloud providers through their SLAs

1.1. Identify Your Requirements

- **Business Needs:** Define your organization's cloud service requirements—such as storage, processing power, and specific services (e.g., SaaS, PaaS, IaaS).
- **Compliance:** Determine the regulatory and compliance standards (e.g., GDPR, HIPAA) that your cloud provider must adhere to.
- **Performance:** Establish performance metrics like response time, scalability, and uptime needed to support your operations.

1.2. Review the SLA Terms

- **Availability/Uptime:** Ensure the SLA specifies uptime guarantees (e.g., 99.9% or 99.99% availability), with clear definitions of what constitutes downtime.
- **Downtime Definition:** Examine how downtime is calculated and any exclusions (e.g., scheduled maintenance).
- **Service Credits:** Evaluate what compensation (typically in the form of service credits) is offered in case the provider fails to meet uptime or performance standards.
- **Response Times:** Check if there are guarantees for response times for technical

support or incident resolution.

- **Data Security:** Look for provisions on data security and privacy, including encryption, backup policies, and disaster recovery plans.

1.3. Assess Compliance & Security Commitments

- **Data Privacy:** Ensure that the provider meets required data privacy and protection standards. Review if the SLA addresses data handling, geographic restrictions, and data breach notification.
- **Certifications:** Verify if the provider holds relevant certifications such as ISO 27001, SOC 2, or industry-specific certifications like PCI-DSS for payment systems.
- **Disaster Recovery and Backups:** Check the SLA for guarantees regarding data redundancy, backup frequency, and disaster recovery timelines.

1.4. Evaluate Performance Metrics

- **Latency:** Review guarantees on latency and performance for critical operations. This is especially important for globally distributed applications.
- **Scalability:** Assess whether the provider guarantees scalability (both vertical and horizontal) within a specific time frame as demand grows.
- **Support and Maintenance:** Check how quickly the provider addresses and resolves service issues and maintenance, including whether they offer proactive monitoring.

1.4. Review Legal and Termination Clauses

- **Termination Conditions:** Understand under what conditions either party can terminate the contract, and what happens to your data upon termination (e.g., migration support, data deletion).
- **Liability:** Analyze the liability limitations in the event of data loss, outages, or security breaches. Ensure the financial penalties are clearly stated and acceptable to your organization.
- **Amendment Process:** Review the process for changing the SLA. Determine if amendments are done unilaterally by the provider or if they require customer consent.

1.5. Compare SLAs Across Providers

- **Benchmark Uptime & Penalties:** Compare the uptime guarantees and penalty structures between providers.
- **Pricing vs. SLA Commitment:** Evaluate the SLA in relation to the pricing. A higher

cost may justify higher performance, availability, or customer service guarantees.

- **Industry Reputation:** Research customer reviews and third-party assessments to validate whether providers meet their SLA commitments in real-world scenarios.

1.6. Negotiate SLA Terms

- **Customization:** If needed, negotiate custom terms that reflect your organization's specific operational and compliance needs.
- **Add-ons:** Some providers offer premium support or add-ons that can improve SLA coverage, such as faster response times or enhanced security services.

1.7. Monitor SLA Performance

- **Ongoing Monitoring:** Once selected, ensure you have systems to continuously monitor the provider's performance against the SLA terms.
- **Reporting:** Ask for automated reports and logs on performance metrics, incident handling, and overall service delivery.



Points to Remember

Description of Service Level Agreement

- **Service level agreement (SLA)** is defined as the bond for performance negotiated between the cloud services provider and the client.
- **Components of service level agreement**
 1. Service definition
 2. Performance metrics
 3. Service credits
 4. Reporting and monitoring
 5. Dispute resolution
 6. Term and termination
 7. Confidentiality
 8. Force majeure
- **Types of service level agreement**
 1. Basic SLA
 2. Tiered SLA
 3. Customized SLA
 4. Financial SLA
 5. Operational SLA
 6. Hybrid SLA

- **Benefits of service level agreement**

5. For Cloud Service Providers:

1. Improved service quality
2. risk mitigation
3. competitive advantage

2. For Cloud Customers

1. Guaranteed service quality
2. Reduced risk:
3. Cost control
4. Legal protection

Description of cloud computing service providers.

Cloud computing service providers refer to the companies that deliver cloud computing services over the Internet.

- **Types of cloud computing service provider**

1. Public cloud providers
2. Private cloud providers
3. Hybrid cloud providers
4. Multi-cloud providers
5. Community cloud providers

Advantages of cloud computing service provider

1. Scalability
2. Cost-effectiveness
3. Flexibility
4. Efficiency
5. Innovation
6. Accessibility

Disadvantages of cloud computing service provider

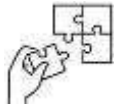
1. Dependency
2. Security concerns
3. Vendor lock-in:
4. Control and customization
5. Internet connectivity

Criteria for selecting cloud computing service model are User needs, cost and operations

Steps for evaluating cloud providers through their SLAs are:

1. Identification your requirements

2. Review the SLA terms
3. Assess compliance & security commitments
4. Evaluation of performance metrics
5. Review legal and termination clauses
6. Compare SLAs across providers
7. Negotiate SLA terms
8. Monitor SLA performance



Application of learning 1.2

You are the IT manager of a mid-sized e-commerce company, Tech Math, which is planning to migrate its infrastructure to the cloud. Your company relies heavily on a 24/7 online storefront with high traffic during seasonal sales. As the IT manager, you are requested to select the best cloud computing service provider based on their SLA after navigating to the different cloud service providers websites.

- Often includes time-bound offers (e.g., free for 12 months) or permanently free services with limited usage.

Use Case: Ideal for individual users, startups, or businesses testing cloud services without a financial commitment.

Example: AWS Free Tier, Google Cloud Free Tier, Microsoft Azure Free Account.

1.2. Pay-As-You-Go (On-Demand) Account

Customers pay only for the resources they consume, with no upfront costs or long-term commitments.

Features:

- Flexibility to use any cloud resources (compute, storage, networking, etc.) on demand.
- Billing is based on actual usage (e.g., per hour for virtual machines, per GB for storage).
- Scalable, allowing users to increase or decrease resources as needed.

Use Case: Suitable for businesses or developers who want full flexibility and control over their cloud usage and costs.

Example: AWS On-Demand Instances, Google Cloud pay-per-use services, Azure pay-as-you-go pricing.

1.3. Reserved or Prepaid Account

Customers commit to using specific resources for a fixed period (e.g., 1-3 years) in exchange for lower pricing.

Features:

- Upfront payment for cloud resources at a discounted rate compared to on-demand pricing.
- Requires a long-term commitment to specific services or instances.
- Predictable costs with significant savings for consistent or heavy usage.

Use Case: Ideal for businesses with steady workloads and predictable resource needs, looking to reduce costs over time.

Example: AWS Reserved Instances, Azure Reserved Virtual Machine Instances, Google Cloud Committed Use Contracts.

1.4. Enterprise Account

Designed for large organizations requiring high levels of support, security, and

customization for cloud services.

Features:

- Advanced features, including enterprise-grade security, compliance tools, and governance.
- Access to specialized support teams, account managers, and service level agreements (SLAs).
- Custom pricing options based on bulk usage or specific business needs.
- Centralized management of multiple teams and departments, with controls for managing resources and users.

Use Case: Large enterprises or organizations needing high levels of customization, security, and dedicated support.

Example: AWS Enterprise Support, Microsoft Azure Enterprise Agreement, Google Cloud Enterprise Account.

1.5. Education or Nonprofit Account

Purpose: Cloud accounts specifically designed for educational institutions, students, or nonprofit organizations, often with discounted or free services.

Features:

- Access to a variety of cloud services, often at reduced prices or with additional benefits.
- Special credits for academic or nonprofit projects.
- Tools and services geared toward research, learning, and community projects.

Use Case: Ideal for schools, universities, research institutions, and nonprofit organizations looking for cloud resources.

Example: Google Cloud for Education, AWS Educate, Microsoft Azure for Students, and Nonprofits.

1.6. Trial Account

Purpose: Short-term access to cloud services for evaluation and testing, usually for new customers.

Features:

- Limited access to cloud services, typically for 30 to 90 days.
- Includes a fixed number of free credits to use on various cloud services.

- After the trial period or upon exhausting the credits, users can either convert to a paid account or stop using the services.

Use Case: Suitable for individuals or businesses exploring cloud options or evaluating cloud providers before making a commitment.

Example: AWS 30-day trial, Azure 30-day trial with \$200 credit, Google Cloud \$300 free credit.

1.7. Developer Account

Accounts used by developers who need cloud resources to build, test, and deploy applications.

Features:

- Access to development and testing environments with cloud resources.
- May include development tools, software libraries, and APIs specific to developers.
- Typically comes with flexible pricing to accommodate experimentation and scaling.

Use Case: For individual developers or development teams building and testing applications in the cloud.

Example: AWS Developer Support, Google Cloud Developer Program, Microsoft Azure Developer Plan.

2. Usage and billing in cloud computing

Usage and billing in cloud computing refers to how cloud service providers measure and charge customers based on their consumption of resources and services.

2.1. Usage

Customers are billed based on their consumption of various cloud services. Common types of resources that affect billing include:

- **Compute:** Virtual machines, containers, or serverless functions (e.g., AWS EC2, Google Compute Engine, Azure Virtual Machines).
- **Storage:** Cloud storage for data (e.g., AWS S3, Azure Blob Storage, Google Cloud Storage).
- **Networking:** Data transfer across regions, inbound/outbound traffic, or using content delivery networks (CDNs).
- **Database Services:** Use of managed database services like Amazon RDS, Google Cloud SQL, or Azure SQL Database.
- **APIs & Microservices:** Use of API gateway, serverless functions (e.g., AWS Lambda), or other managed microservices.

- **Additional Services:** Analytics, machine learning, IoT, security, and developer tools.

2.2. Billing Models

- **Pay-as-You-Go:** Customers are billed based on actual usage. Costs are incurred as resources are consumed (e.g., hours of compute time, GB of storage).
- **Reserved Instances/Commitment Plans:** Discounts for long-term commitments (e.g., reserving instances for 1 or 3 years, prepaid usage).
- **Spot Instances/Preemptible VMs:** Available at discounted rates with the condition that they can be terminated at any time based on availability.
- **Free Tier or Trial:** Some cloud providers offer free usage tiers or trials with limited resources for new customers.

3. security and access control in cloud computing

Security and access control in cloud computing are essential components to protect data, applications, and services from unauthorized access, breaches, and threats.

Below are key concepts related to security and access control in cloud computing:

3.1. Shared Responsibility Model

- **Cloud Provider's Responsibility:** The cloud provider ensures the security of the underlying infrastructure, including physical data centers, hardware, networks, and hypervisors.
- **Customer's Responsibility:** The customer is responsible for securing the data, applications, operating systems, and configuring proper access controls to manage user permissions within the cloud.

3.2. Identity and Access Management (IAM)

IAM is the framework for controlling user access to resources in cloud environments. It is used to define and manage user roles and permissions to ensure that only authorized users can access specific resources.

3.3. Data Protection and Encryption

Data stored in the cloud is encrypted using strong encryption algorithms (e.g., AES-256).

3.4. Network Security

Cloud providers offer virtual firewalls that allow customers to control inbound and outbound traffic to their resources. These firewalls can be configured to restrict traffic based on IP addresses, ports, and protocols.

A VPN provides secure communication between an on-premises network and cloud resources by creating an encrypted tunnel.



Key readings 1.3.2: Creating Cloud Computing Account

1. Choose a Cloud Provider

- Select a cloud provider based on your requirements. Popular choices include AWS, GCP, and Azure.

2. Sign Up for an Account

- Go to the sign-up page of the chosen cloud provider.
- If you don't have an account, you'll typically need to create one using an email address or sign in using an existing account (like a Google or Microsoft account for GCP and Azure, respectively).

3. Enter Personal or Organization Details

Provide necessary information such as Name (or organization name), Email address, Address and contact details, Optionally, you may need to select if you're signing up as an individual or an organization.

4. Select the Free Tier or Paid Plan

- Most cloud providers offer a **free tier** or a **trial plan**. You can select the free tier to explore and use cloud resources within a limit for free (e.g., AWS Free Tier, Google Cloud Free Tier, Azure Free Account).
- Some providers may offer credits (e.g., \$300 free credit for new Google Cloud

users).

5. Provide Payment Information

- Even for free tiers or trials, cloud providers often require a valid payment method (credit card, debit card, or in some cases PayPal) to set up the account.
- This payment information is used for billing in case you exceed the free tier limits or move to a paid plan.
- In most cases, your card will not be charged unless you go beyond the free-tier usage or after the trial period ends.

6. Verify Your Identity

- After entering your payment information, you may need to verify your identity by providing a phone number or email address.
- A verification code (via SMS or email) is usually sent for confirmation.

7. Activate Your Account

- Once you've submitted the necessary details, the cloud provider will process the information and activate your account.
- You may receive a confirmation email informing you that your account is active and ready to use.

9. Set Up Identity and Access Management (IAM)

- For security reasons, it's best to set up Identity and Access Management (IAM) for additional users if this is an organization's account.
- You can create different users and assign roles or policies to manage access to resources securely.

10. Access the Cloud Console

- After activation, log in to your cloud account using your credentials (email and password).
- Access the **Cloud Management Console** where you can create and manage resources such as virtual machines, storage, databases, and other services.

11. Configure Security Settings (Optional but Recommended)

- Set up **Multi-Factor Authentication (MFA)** to add an extra layer of security.
- Review and configure any additional security settings to secure your cloud account.

12. Start Using Cloud Resources

- After everything is set up, you can begin using cloud services like virtual machines, storage, databases, and more, depending on your project requirements.

Example Walkthrough for AWS:

1. Enter an email address and an account name in the below form.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Root user email address

Used for account recovery and some administrative functions.

AWS account name

Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

Sign in to an existing AWS account

2. Add contact conformation



Free Tier offers

All AWS accounts can explore 3 different types of free offers, depending on the product used.



Always free
Never expires



12 months free
Start from initial sign-up date



Trials
Start from service activation date

Sign up for AWS

Contact Information

How do you plan to use AWS?

- ☐ Business - for your work, school, or organization
☐ Personal - for your own projects

Who should we contact about this account?

Full Name

Phone Number

+1 222-333-4444

Country or Region

United States

Address

Apartment, suite, unit, building, floor, etc.

City

State, Province, or Region

Postal Code

☐ I have read and agree to the terms of the [AWS Customer Agreement](#).

Continue (step 2 of 5)

3. Add Payment Method



Secure verification

i We will not charge you for usage below AWS Free Tier limits. We may temporarily hold up to \$1 USD (or an equivalent amount in local currency) as a pending transaction for 3-5 days to verify your identity.



Sign up for AWS

Billing Information

Credit or Debit card number



AWS accepts all major credit and debit cards. To learn more about payment options, review our [FAQ](#).

Expiration date

Cardholder's name

Billing address

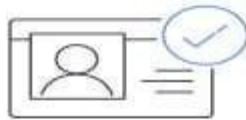
☒ Use my contact address

☐ Use a new address

Verify and Continue (step 3 of 5)

You might be redirected to your bank's website to authorize the verification charge.

4. Confirm your Identity



Sign up for AWS

Confirm your identity

Before you can use your AWS account, you must verify your phone number. When you continue, the AWS automated system will contact you with a verification code.

How should we send you the verification code?

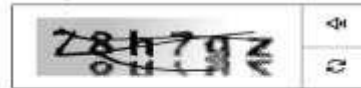
- ☒ Text message (SMS)
☐ Voice call

Country or region code

United States (+1)

Mobile phone number

Security check



Type the characters as shown above

Send SMS (step 4 of 5)

5. Sign in as a root user



Sign in

☒ Root user

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ IAM user

User within an account that performs daily tasks. [Learn more](#)

Root user email address

username@example.com

Next



Security check



Submit



Root user sign in

Email: someone@example.com

Password

[Forgot password?](#)

Sign in

[Sign in to a different account](#)

[Create a new AWS account](#)

Set up users IAM the identity center



Root user sign in

Email:

Password

[Forgot password?](#)

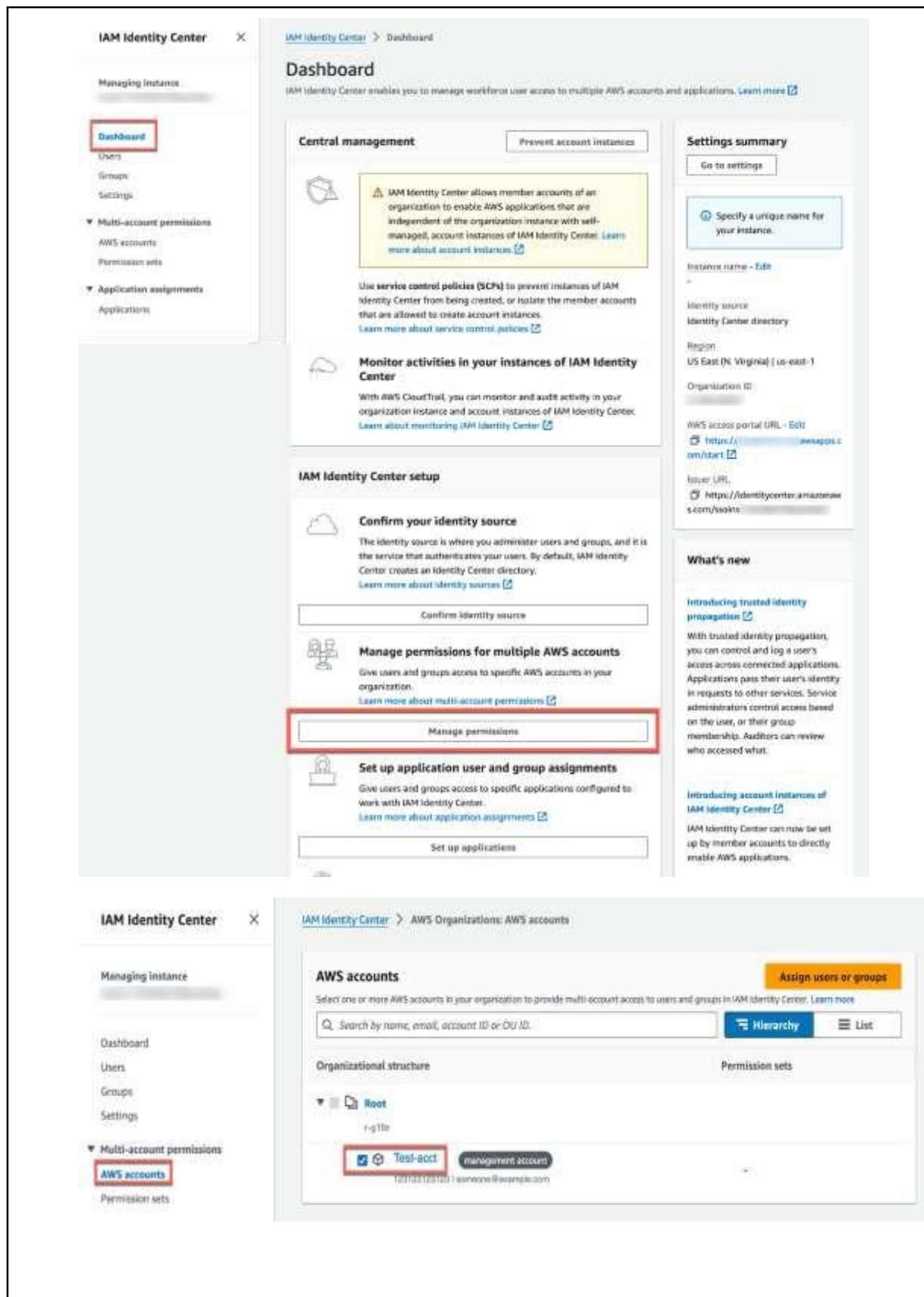
Sign in

[Sign in to a different account](#)

[Create a new AWS account](#)



Manage access console



Test- acct

Overview

Account name
John Smith

Account ID
640702963591

Email
someone@example.com

Users and groups

Permission sets

Assigned users and groups (0)

[Change permission sets](#)[Remove access](#)[Assign users or groups](#)

The following users and groups in IAM Identity Center can select this AWS account from within their AWS access portal. [Learn more](#)

**1**

Username / group name

Permission sets

Type

No users or groups assigned to this account
You have not yet assigned any users or groups to this account.

[Assign users or groups](#)

Step 1

Select users and groups

Step 2

Select permission sets

Step 3

Review and submit

Assign users and groups to "Test-acct"

Select one or more users or groups in IAM Identity Center that you want to give mu

Users

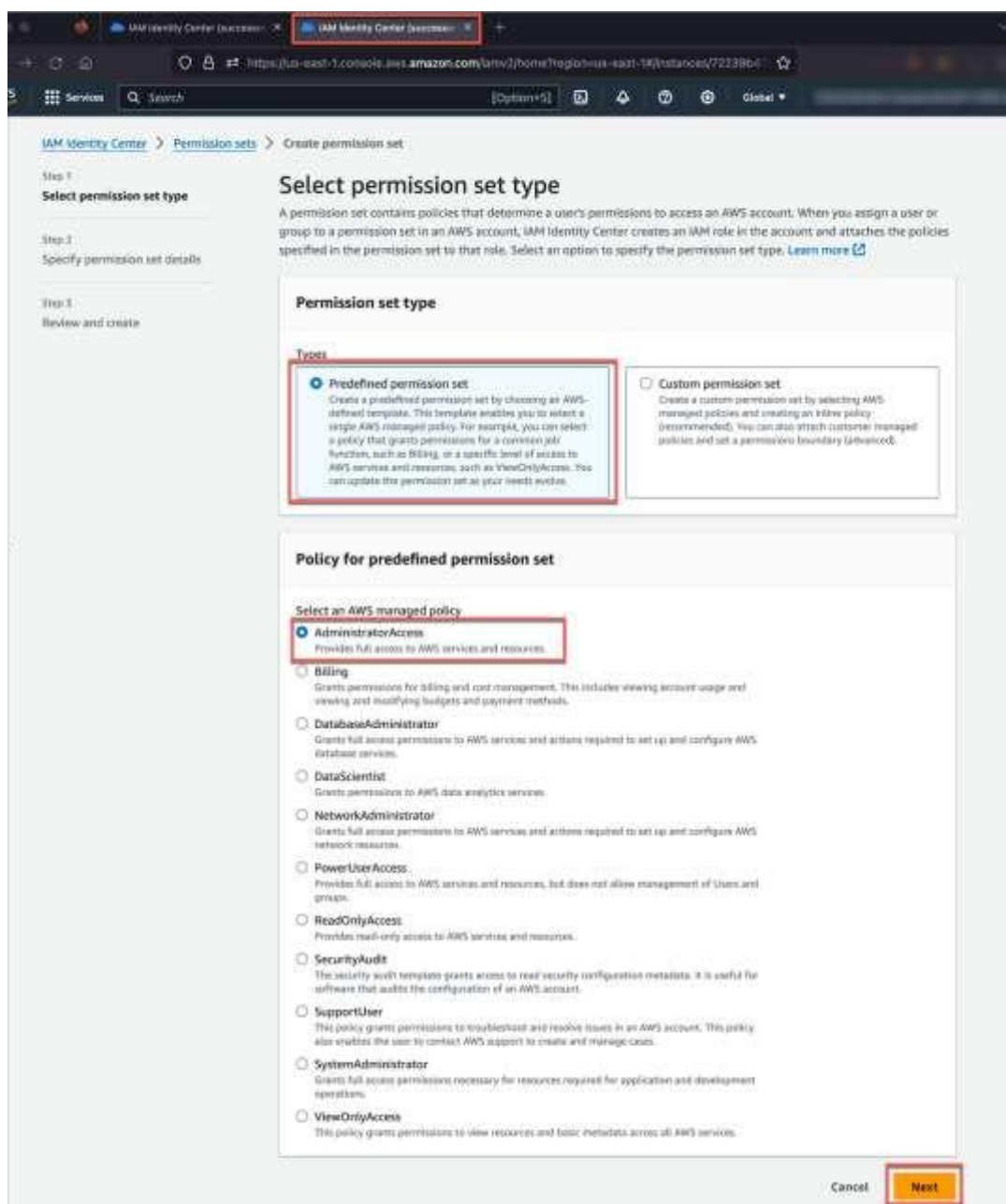
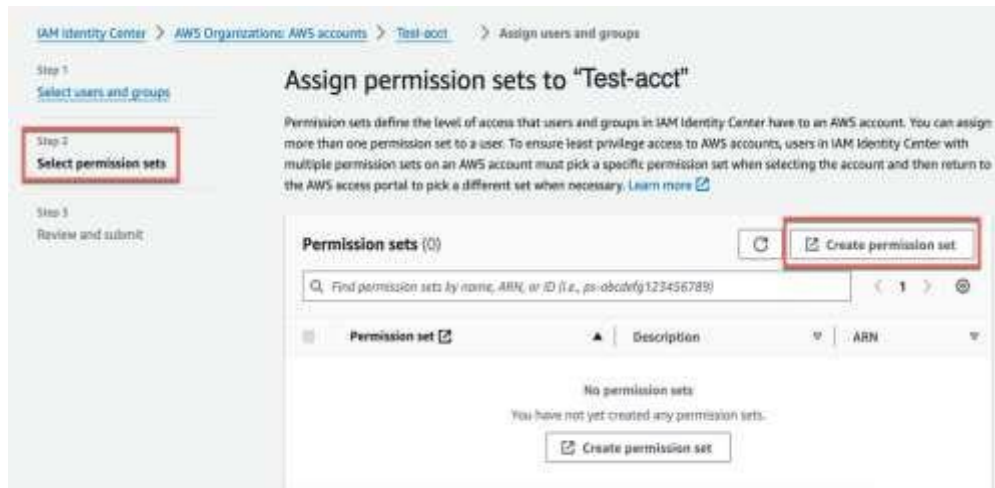
Groups**Groups (1/1)**[Create groups](#)**1**☒ Group name

Description

☒ Admins

-

Selected users and groups (1)[Remove](#)[Cancel](#)[Next](#)



Key readings 1.3.3: Description of identity and access management

1. Define Identity and Access Management (IAM)

Identity and Access Management (IAM) refers to policies, processes, and technologies used to manage and control digital identities and regulate who has access to what resources within an organization.

It plays a crucial role in ensuring the right individuals and systems can access specific resources (such as data, applications, networks) at the appropriate times for the right reasons.

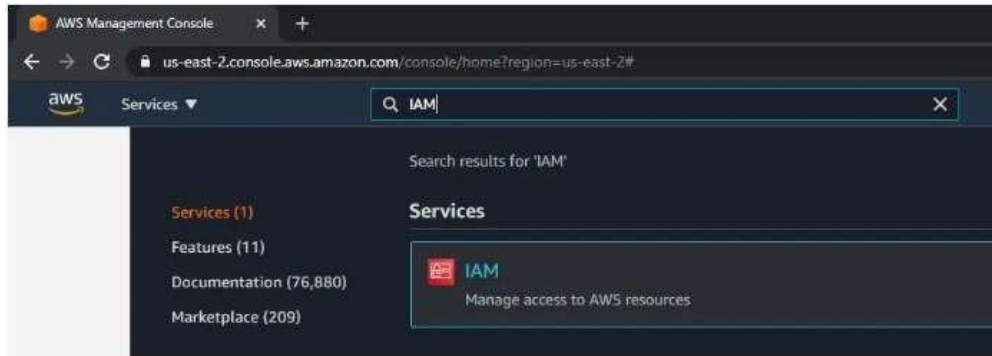
2. Components of IAM

- 2.1. Identification:** Establishes a unique identity for each user, system, or device within a network (e.g., using usernames, IDs).
- 2.2. Authentication:** Verifies that an identity is valid, typically through methods like passwords, biometrics, or multi-factor authentication (MFA).
- 2.3. Authorization:** Determines the level of access a verified identity has to resources. This is often managed through role-based access control (RBAC) or attribute-based access control (ABAC).
- 2.4. Accountability (Auditing/Monitoring):** Tracks and logs activities of identities to ensure compliance with security policies and monitor for any suspicious activities.

3. Benefits of IAM

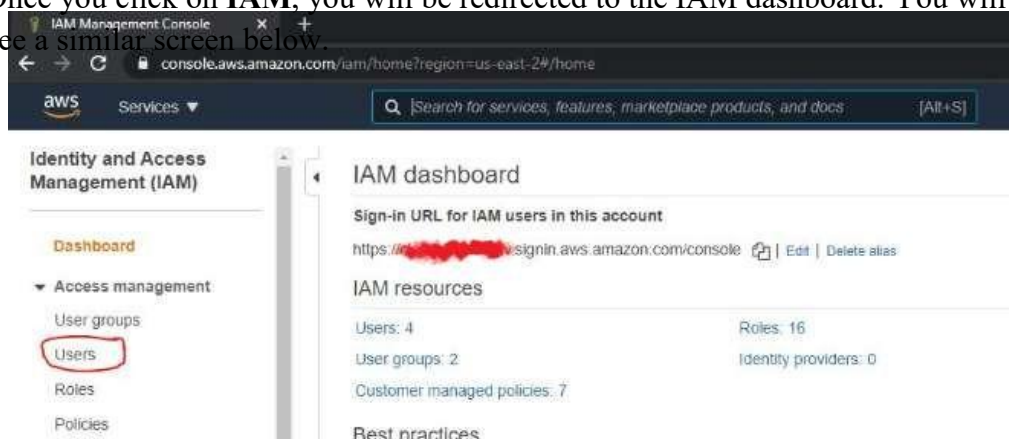
Security: Minimizes the risk of unauthorized access to sensitive information.

Efficiency: Streamlines user access processes through automation (e.g.,



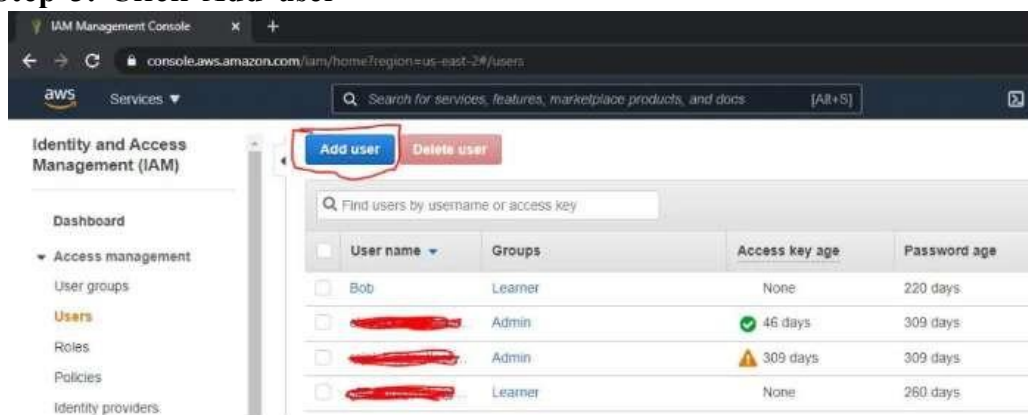
Step 2: Click on Users in the IAM Dashboard

Once you click on IAM, you will be redirected to the IAM dashboard. You will see a similar screen below.



Click on **Users** in the left navigation menu as highlighted in the above screenshot.

Step 3: Click Add user



Step 4: Fill Up the Details

Provide a **User name** (Must be unique within the account)

Select an access type for your user based on the requirement

If you are creating a user for someone else, it makes sense to keep **Require password reset** checkbox checked

Note: You can create more than one user at a time by using add another user button

Add user

1 2 3 4 5

Set user details

You can add multiple users at once with the same access type and permissions: [Learn more](#)

User name* demouser

[Add another user](#)

Select AWS access type

Select how these users will access AWS. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Access type* ☐ Programmatic access
Enables an access key ID and secret access key for the AWS API, CLI, SDK, and other development tools

☒ AWS Management Console access
Enables a password that allows users to sign-in to the AWS Management Console.

Console password* ☒ Autogenerated password
☐ Custom password

Require password reset ☒ User must create a new password at next sign-in

* Required

Cancel **Next: Permissions**

Once you have filled up details, click on **Next: Permission**

Step 5: Provide Required Permission to the User

There are three ways to set permission for the user

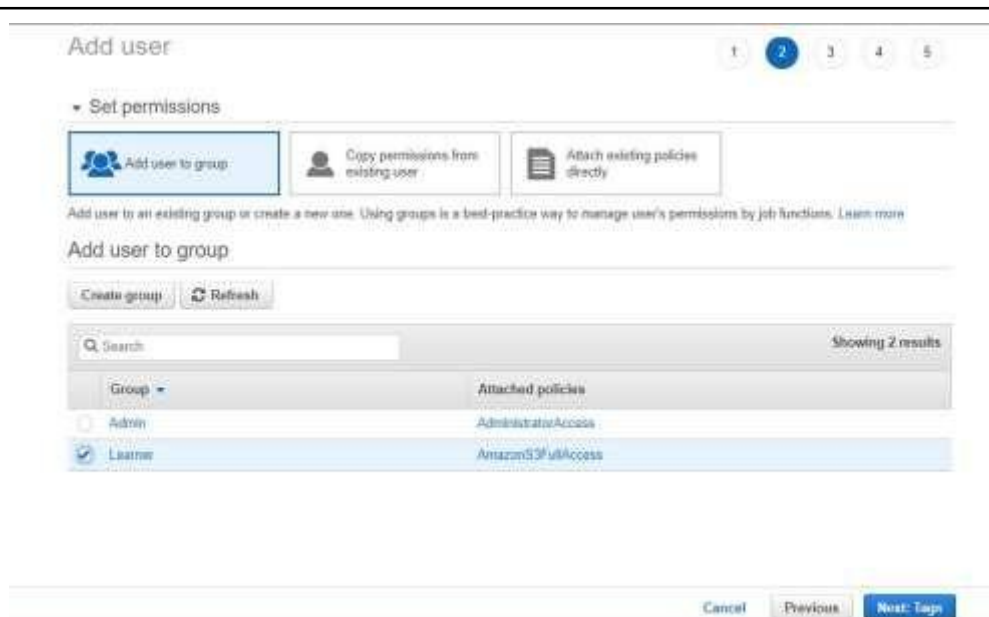
Add user to group

Copy permission from existing user

Attach existing policies directly

You can use any of the options at your convenience. However, it's a good practice to create a group for users who need similar permission, attach policies to the group and then add the user to the group.

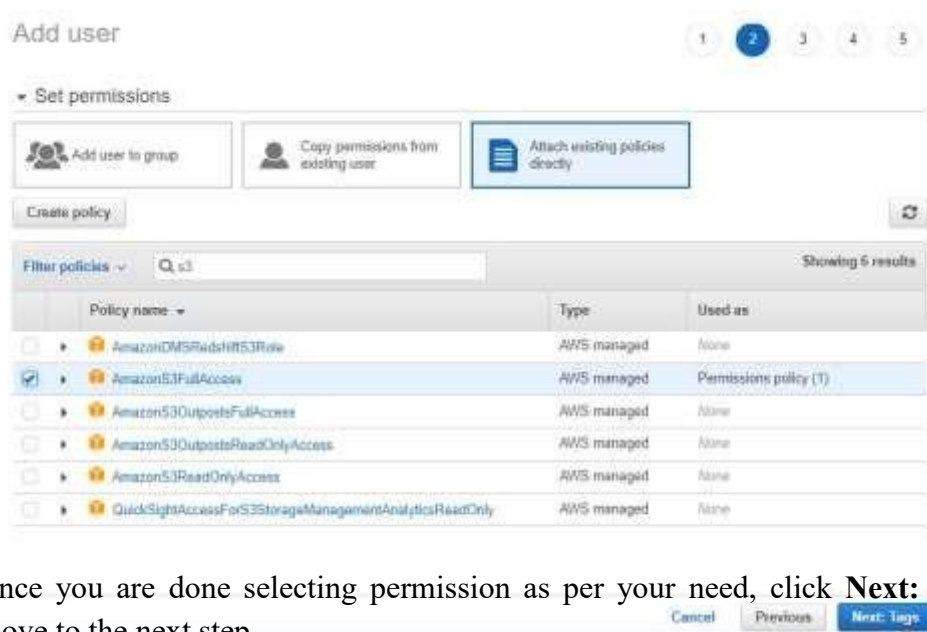
I have already created two groups, so I will add this user to the **Learner** group which has S3 full access



On the other hand, if you wish to attach existing policies directly, click on Attach existing policies Directly tab

Filter appropriate policy by searching and selecting what you need. For example, I have searched for S3 and selected AmazonS3FullAccess as you can see in the below screenshot.

Note: If you think already existing policies don't fit your needs, you can create a custom policy and assign that instead. You can do that using the Create policy button mentioned in the below screenshot.



Once you are done selecting permission as per your need, click **Next: Tags** to move to the next step

Step 6: Add a tag to your user(Optional)

Adding tags to your user is completely optional and you may skip this step altogether. However, tags can provide extra information about the user, for example, department and later can be used to provide permission based on tag.

Add user 1 2 **3** 4 5

Add tags (optional)

IAM tags are key-value pairs you can add to your user. Tags can include user information, such as an email address, or can be descriptive, such as a job title. You can use the tags to organize, track, or control access for this user. [Learn more](#)

Key	Value (optional)	Remove
Team	Dev.	✕
Department	Automation	✕
Add new key		

You can add 48 more tags.

[Cancel](#) [Previous](#) **Next: Review**

After you are done, Click on **Next: Review**

Step 7: Review User Details and Click Create user

Review user details to crosscheck if you have entered all the information correctly. If you think something is not correct or as per your need, you may decide to go back using the **Previous** button and modify it.

Add user 1 2 3 4 5

Review

Review your choices. After you create the user, you can view and download the autogenerated password and access key.

User details:

User name	demouser
AWS access type	AWS Management Console access - with a password
Console password type	Autogenerated
Require password reset	Yes
Permissions boundary	Permissions boundary is not set

Permissions summary

The user shown above will be added to the following groups:

Type	Name
Group	Learner

Tags

The new user will receive the following tags:

[Cancel](#) [Previous](#) [Create user](#)

If everything is good, you can click **Create user**

Add user 1 2 3 4 5

✓ **Success**

You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: <https://signin.aws.amazon.com/console>

[Download .csv](#)

User	Password	Email login instructions
demouser	***** Show	Send email

The user is successfully created and you see a success message as above. Now comes the important part, Download the .csv file which has your user credentials and sign-in link that you can share with your IAM user.

Note: Please note that as the message says, this is the last time you can download the .csv file so make sure to download it before closing the dialog. Although, you can create new credentials anytime just that they are not available for downtime after closing this dialog.

If you are an admin creating an IAM user for your team member, your job is done here. Just share the .csv(Credentials) with your user and you are done.

However, if you are someone, who is doing this for learning purposes, you can use the downloaded creds to log in as an IAM user to better understand end-to-end flow

Step 8: Login as an IAM User (Actual IAM user will do this)

Open the credentials file (.csv file) you downloaded and grab details such as username, password and sign-in link.

Click on the sign-in link present in the .csv file

You will be redirected to the AWS console login page with the AccountId/Alias field already filled up.



Enter your **username**, and password as shown in the above screenshot and click **Sign in**

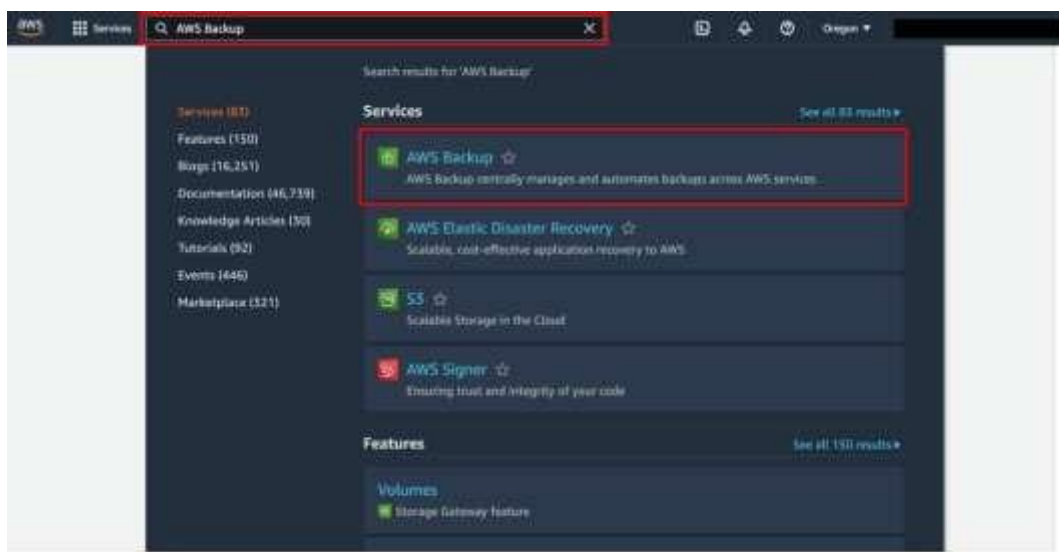
1. Read key reading 1.3.5
2. Perform the above task by following instructions related to the task.
3. Ask more clarifications if any and assistance where needed.
4. Present your work to the trainer and whole class.



Key readings 1.3.5: Performing Backup and Disaster Recovery.

Step 1: Go to the AWS Backup console.

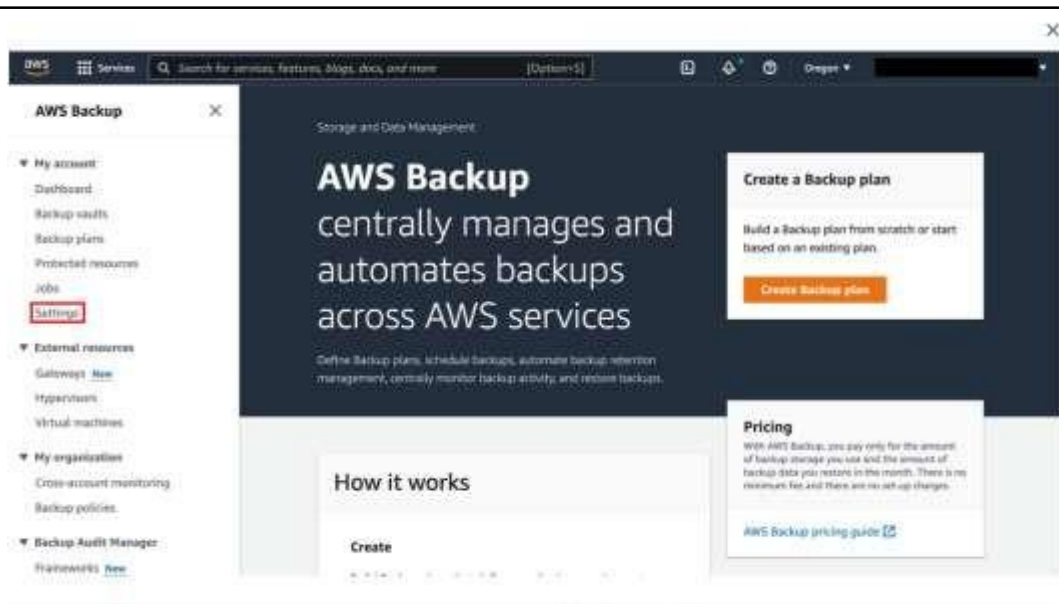
Log in to the **AWS Management Console**, and open the **AWS Backup console**.



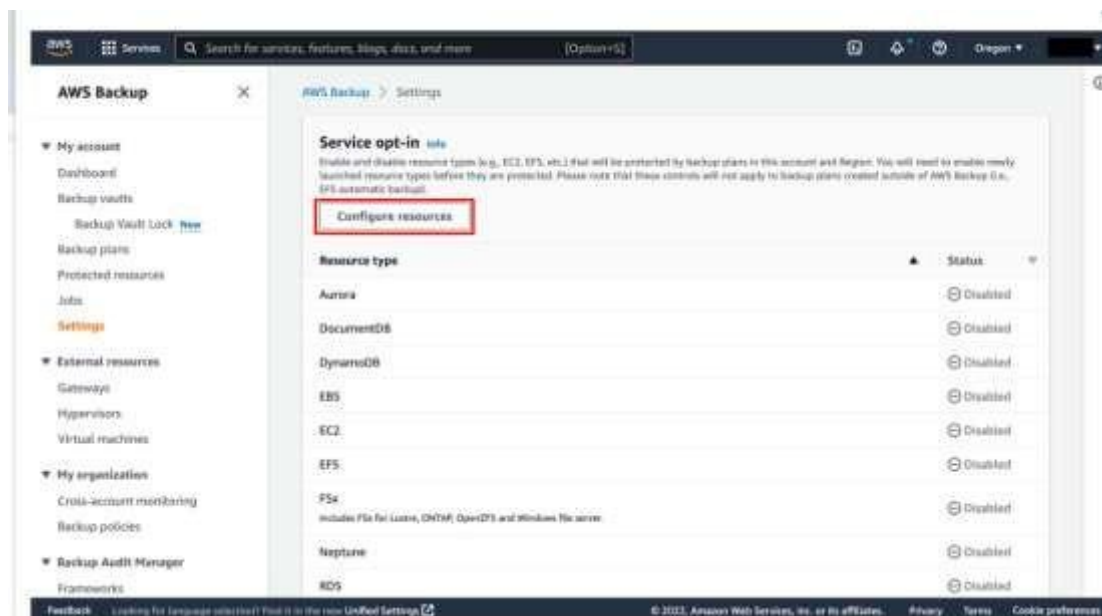
Step 2: Click on Users in the IAM Dashboard

2.1 — Configure the services used with AWS Backup

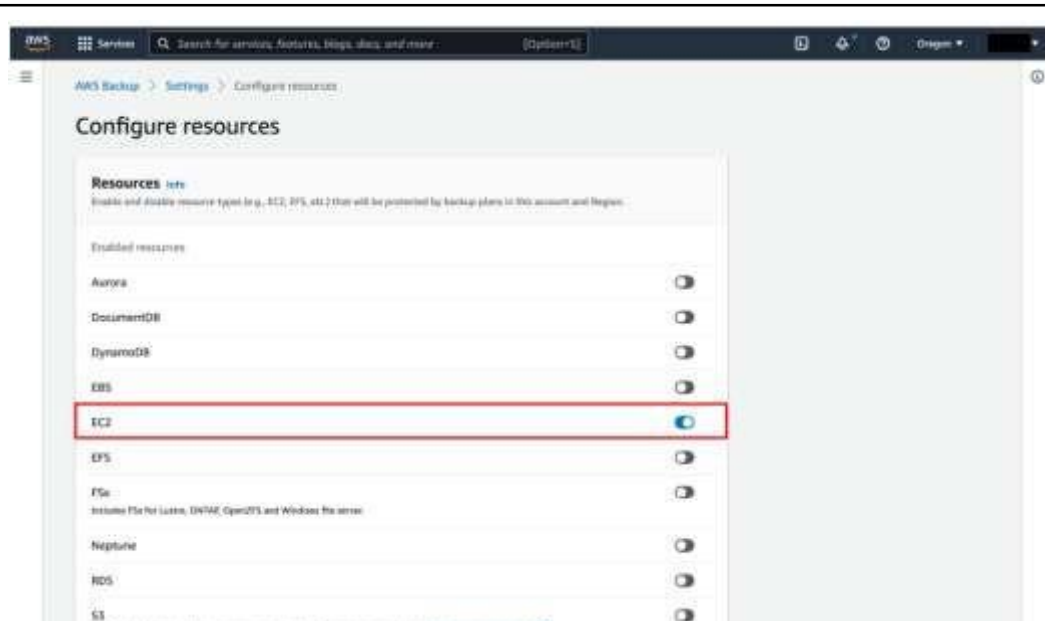
- a. Configure an on-demand AWS Backup job of an Amazon EC2 instance



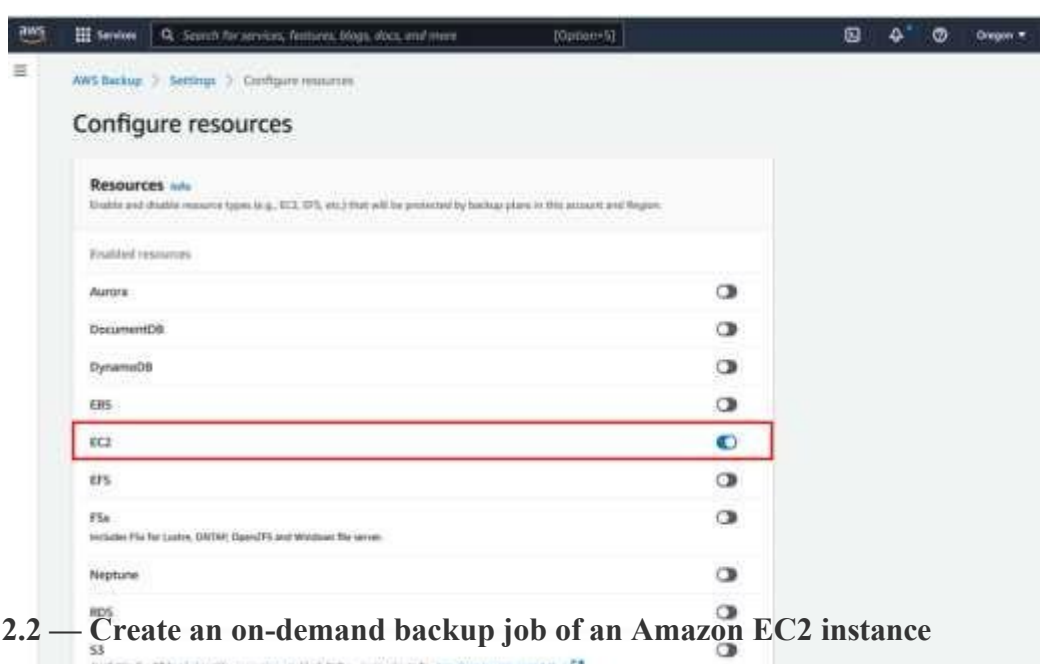
b. On the Service opt-in page, choose Configure resources.



c. On the Configure resources page, use the toggle switches to enable or disable the services used with AWS Backup. In this case, select EC2. Choose Confirm when your services are configured.

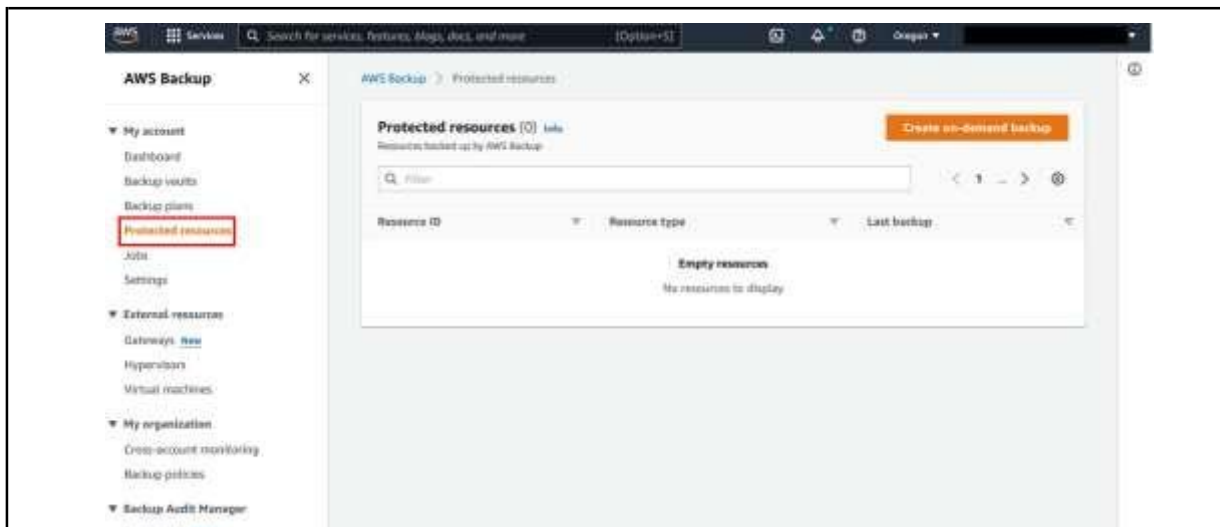


C. On the Configure resources page, use the toggle switches to enable or disable the services used with AWS Backup. In this case, select EC2. Choose Confirm when your services are configured.

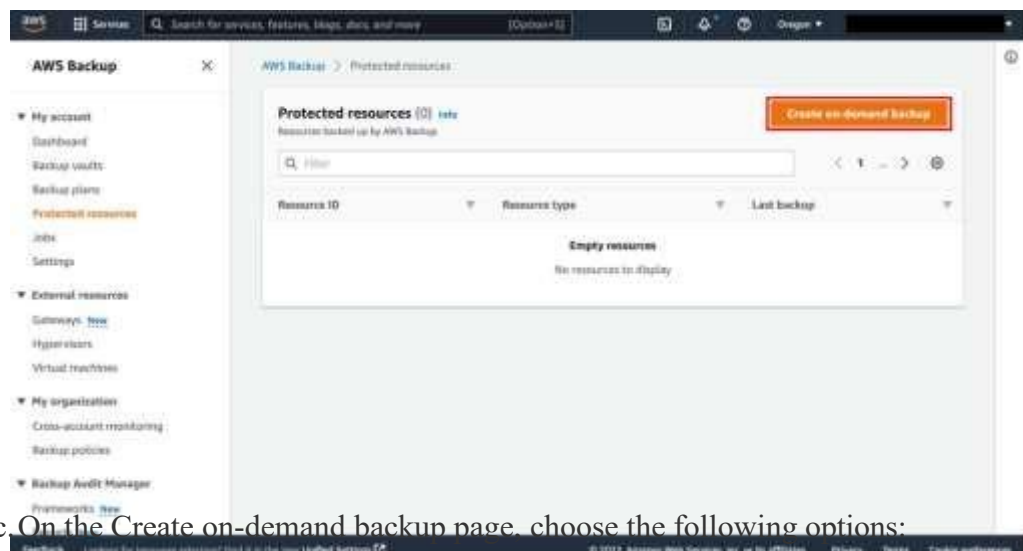


2.2 — Create an on-demand backup job of an Amazon EC2 instance

a. Back in the [AWS Backup console](#), under My account in the left navigation pane, select Protected resources.



b. From the dashboard, choose the Create on-demand backup button.

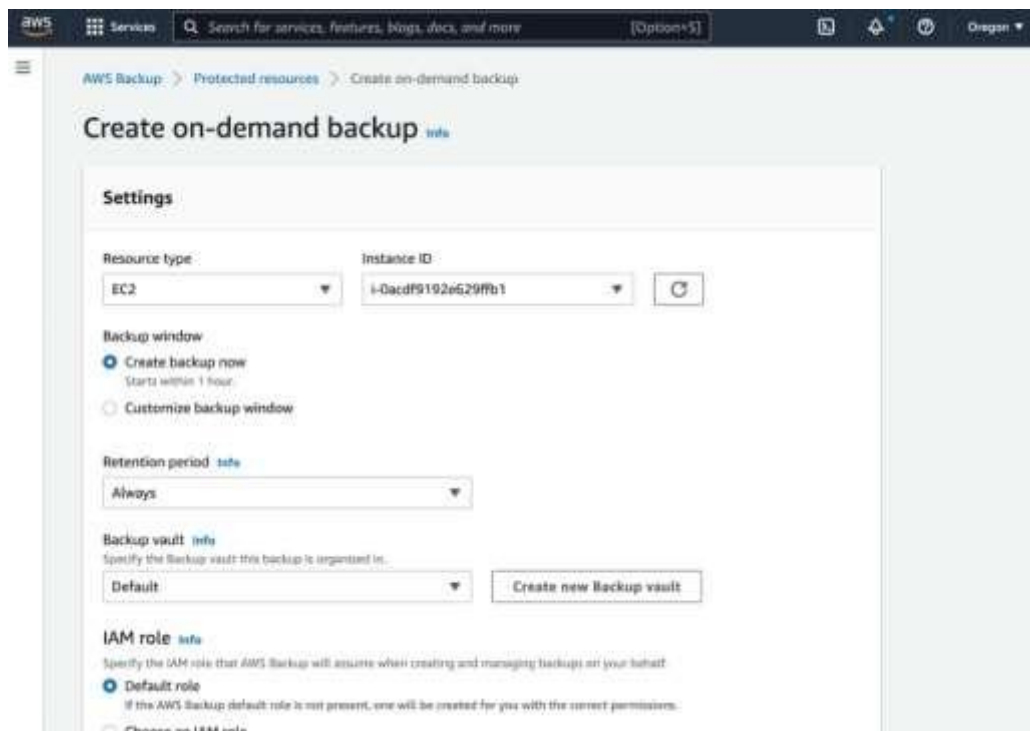


c. On the Create on-demand backup page, choose the following options:

- Select the resource type that you want to back up; for example, choose EC2 for Amazon EC2.
- Choose the Instance ID of the EC2 resource that you want to protect.
- Ensure that Create backup now is selected. This initiates your backup job immediately and enables you to see your saved resource sooner on the Protected resources page.
- Select the desired retention period. AWS Backup automatically deletes your backups at the end of this period to save storage costs for you.
- Choose an existing backup vault. Choosing Create new Backup vault opens a new page to create a vault and then returns you to the Create on-demand

backup page when you are finished.

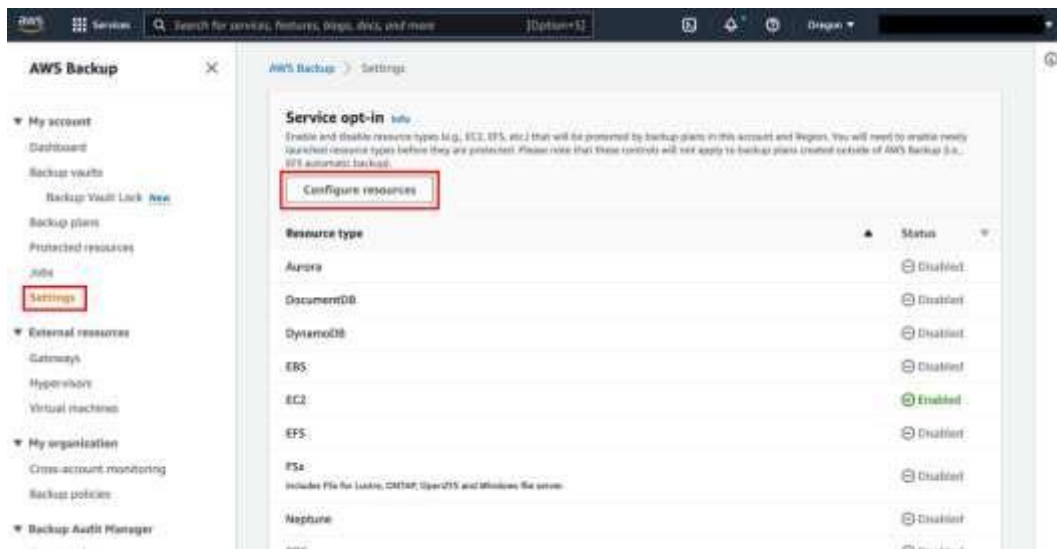
- Under IAM role, choose Default role.
- Note: If the AWS Backup Default role is not present in your account, then an AWS Backup Default role is created with the correct permissions.
- Choose the Create on-demand backup button. This takes you to the Jobs page, where you will see a list of jobs.

The screenshot shows the AWS Backup console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and a user profile dropdown. Below the navigation bar, the breadcrumb trail reads 'AWS Backup > Protected resources > Create on-demand backup'. The main heading is 'Create on-demand backup' with an 'info' link. The 'Settings' section contains several fields: 'Resource type' is set to 'EC2'; 'Instance ID' is 'i-0acdf9192e629ffb1' with a refresh button; 'Backup window' has 'Create backup now' selected (with a note 'Starts within 1 hour') and 'Customize backup window' is unselected; 'Retention period' is set to 'Always'; 'Backup vault' is 'Default' with a 'Create new Backup vault' button; and 'IAM role' is 'Default role' (with a note 'If the AWS Backup default role is not present, one will be created for you with the correct permissions.') and 'Choose an IAM role' is unselected.

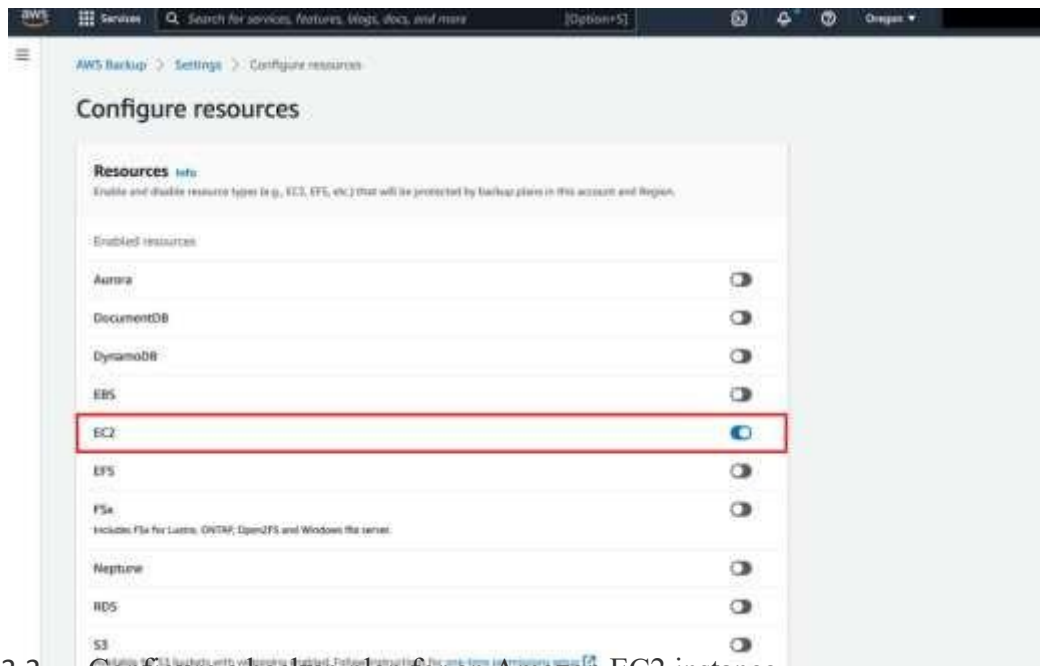
Step 3: Configure an automatic AWS Backup job of an Amazon EC2 instance

3.1 — Configure the services used with AWS Backup

- a. In the left navigation pane in the [AWS Backup console](#), under My account, choose Settings.
- b. On the Service opt-in page, choose Configure resources.

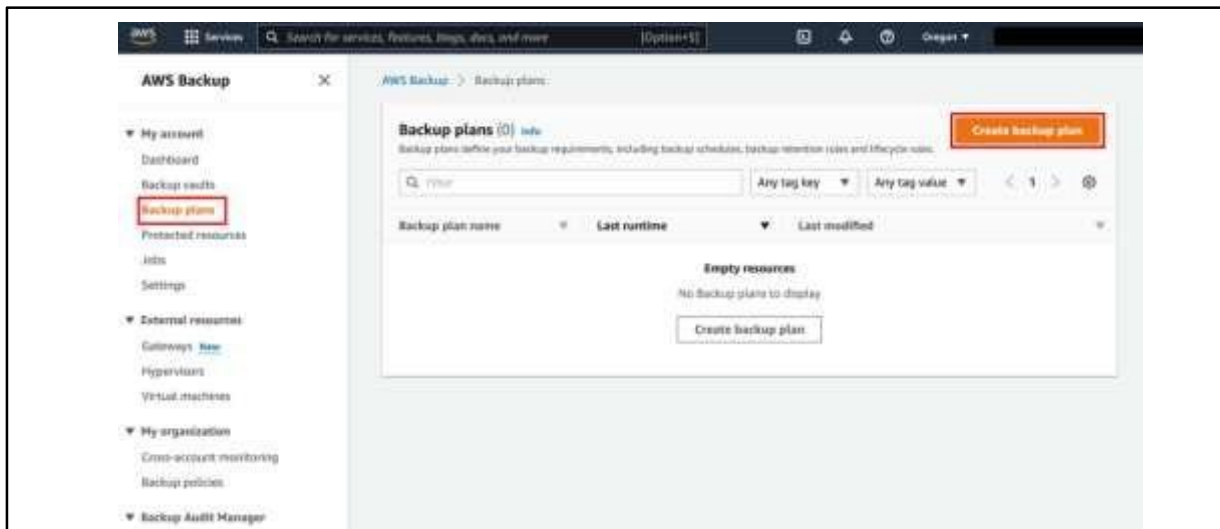


c. On the Configure resources page, use the toggle switches to enable or disable the services used with AWS Backup. Choose Confirm when your services are configured.

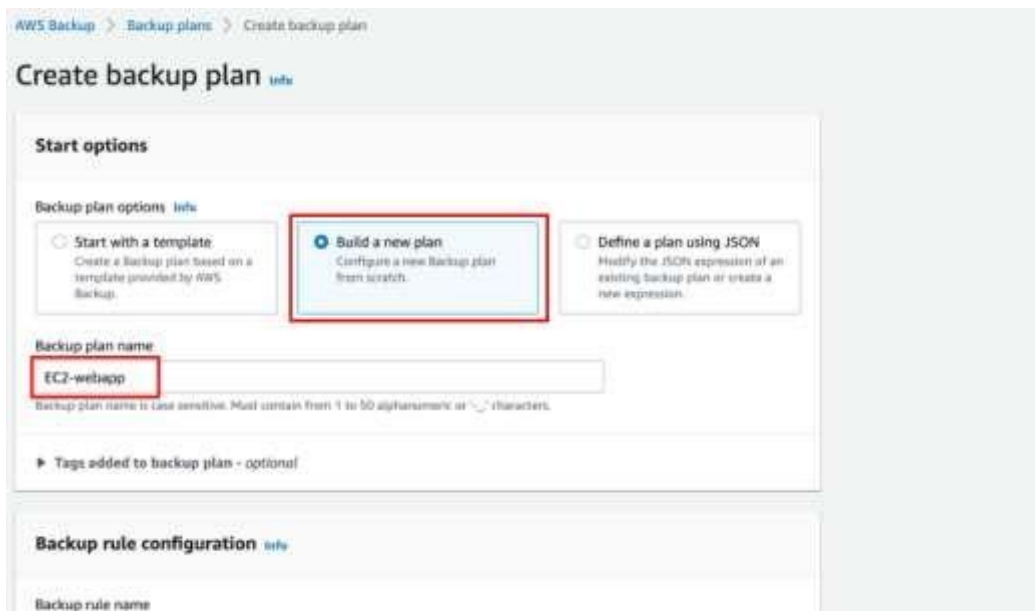


3.2 — Configure a backup plan for an Amazon EC2 instance

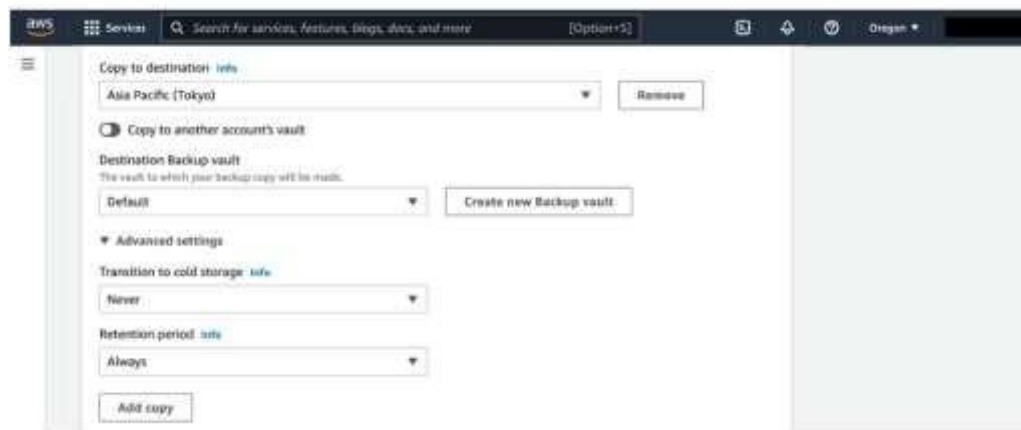
d. In the [AWS Backup console](#), select Backup plans in the left navigation pane under My account, and then Create backup plan.



e. AWS Backup provides three ways to get started using backup plans, select Build a new plan:



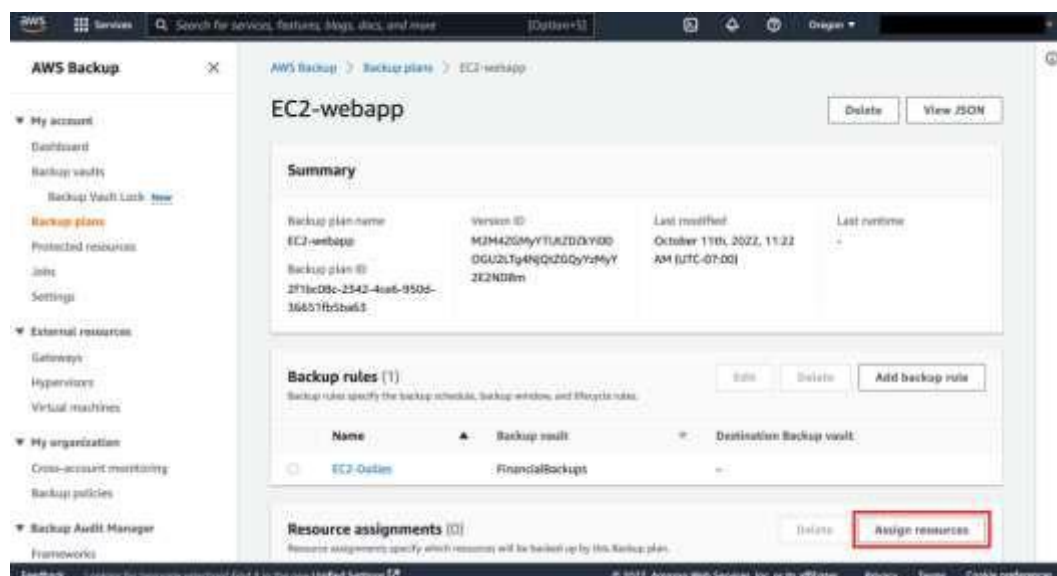
Copy to destination — As part of your backup plan, you can optionally create a backup copy in another AWS Region. Using AWS Backup, you can copy backups to multiple AWS Regions on-demand, or automatically as part of a scheduled backup plan.



3.3 — Assign resources to the backup plan

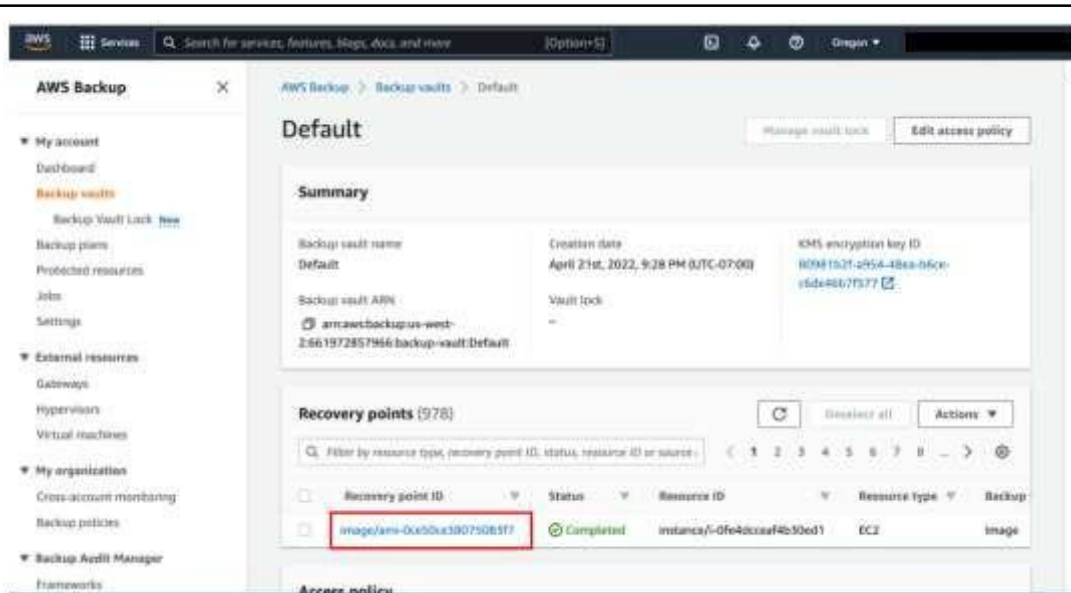
When you assign a resource to a backup plan, that resource is backed up automatically according to the backup plan.

Select the created backup plan and choose the Assign resources button.

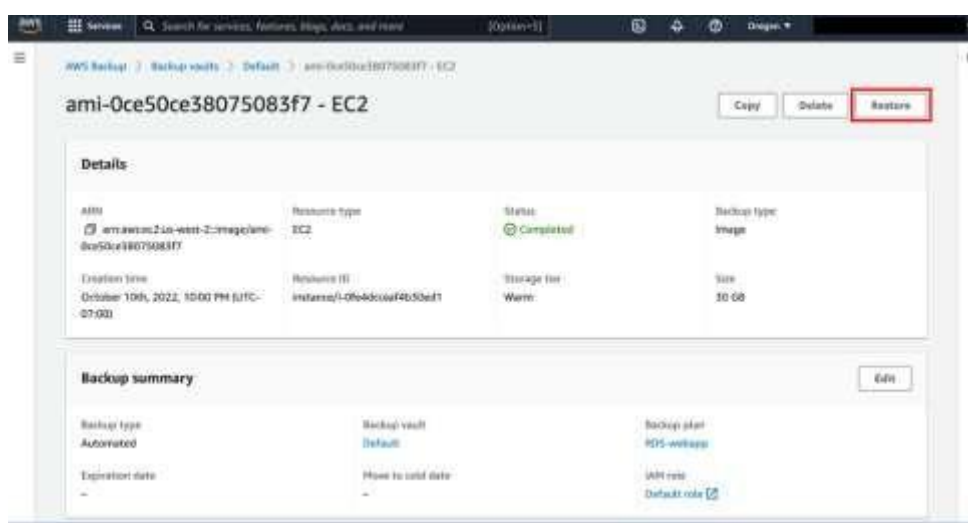


Step 4: Restore an Amazon EC2 instance using AWS Backup.

- Navigate to the backup vault that was selected in the backup plan and select the latest completed backup.



b. To restore the EC2 instance, select the recovery point ARN and choose Restore.



Points to Remember

Cloud computing customer's account concepts

- **Types of customer's account:** Free Tier Account, Pay-As-You-Go (PAYG), Prepaid or Reserved Instances, Enterprise or Business, Developer and Educational, Trial or Promotional
- **Usage and billing** in cloud computing are key components that determine how customers are charged for the services they use. The primary criterion that impact billing include: Compute Power, Storage, Data Transfer (Bandwidth), Networking Services, and Database Services.

- **Security and access control** in cloud computing are essential components to protect data, applications, and services from unauthorized access. It includes Shared Responsibility Model, Identity and Access Management (IAM), Data Protection and Encryption, Network Security, ...

Creating Cloud Computing Account

- Popular cloud computing providers include AWS, GCP, and Azure.

Steps to create account on Amazon Web Services.

- Go to AWS Sign Up.
 - Click on "Create a Free Account."
 - Enter your personal details (name, email, password).
 - Select the Free Tier option.
 - Provide payment information.
 - Verify your identity via phone.
 - Agree to the AWS terms and conditions.
 - Receive confirmation and log in to the AWS Management Console to start using AWS services.
- **Identity and Access Management (IAM)** in cloud computing refers to the framework and technologies used to manage users' identities and control their access to resources within a cloud environment.
 - **Steps to Create IAM User in AWS.**
 1. Login to AWS Management Console and Navigate to IAM
 2. Click on Users in the IAM Dashboard
 3. Click Add user
 4. Fill Up the Details
 5. Provide Required Permission to the User
 6. Add a tag to your user(Optional)
 7. Review User Details and Click Create user
 8. Login as IAM User(Actual IAM user will do this)

Perform Backup and Disaster Recovery

Steps to create an on-demand backup job of an Amazon EC2 instance and protect EC2.

- Login into AWS account
- Go to the AWS Backup console.
- Configure an on-demand AWS Backup job of an Amazon EC2 instance.
- Configure an automatic AWS Backup job of an Amazon EC2 instance.
- Restore an Amazon EC2 instance using AWS Backup.

Learning Outcome 2: Manage Cloud Computing Resources



Indicative contents

2.1 Provisioning Cloud Resources

2.2 Monitoring Cloud Resources

2.3 Optimization of Cloud Resources Utilization

2.4 Applying Cloud Environment Security Measures

Key Competencies for Learning Outcome 2: Manage Cloud Computing Services

Knowledge	Skills	Attitudes
• Description of cloud resources. • Description of scaling strategies • Description of assessment of Requirements for cloud computing resources provisioning. • Identification of cloud monitoring tools • Description of Cloud resources utilization optimization. • Description of cloud environment security measures	• Performing cloud resources provisioning (selection &configure) • Deprosioning cloud resources. • Applying monitoring tools. • Applying Cloud resources utilization optimization	• Being passionate • Having patience • Being creative • Being Innovative • Being attentive



Key readings 2.1.1.: Description of cloud computing resources

1. Definition of Cloud Computing resources

Cloud computing resources refer to the various types of computational services, infrastructure, and tools provided over the internet that allow users to access and utilize computing power, storage, and software without owning or managing the underlying physical hardware.

2. Types cloud resources

2.1. Compute Resources

- Virtual machines (VMs) or containers that run applications and processes.
- Examples: Amazon EC2, Google Compute Engine.

2.2. Storage Resources:

- Services for storing and managing data.
- Examples: Amazon S3, Google Cloud Storage, Azure Blob Storage.

2.3. Network Resources:

- Virtualized networks, load balancers, and IP addresses that enable data transfer and connectivity between different components.
- Examples: Virtual Private Cloud (VPC), Content Delivery Networks (CDNs).

2.4. Database Services:

- Managed database systems for structured, semi-structured, and unstructured data.
- Examples: Amazon RDS, Google Cloud SQL, Azure Cosmos DB.

2.5. Application and Software Services:

- Software or platforms that provide services such as development tools, middleware, and analytics.
- Examples: Google App Engine, Microsoft Power BI, AWS Lambda.

2.6. Security Resources:

- Services for managing access, encryption, firewalls, and other security-related features.
- Examples: AWS IAM, Azure Security Center.



Key readings 2.1.2. Description of assessment of requirements for cloud resources provisioning

1. Role of Assessing the requirements for cloud resources provisioning

Assessing the requirements for cloud resources provisioning is a critical step in ensuring that you allocate the right amount of resources to meet performance, cost, scalability, and security needs. This assessment involves analyzing various aspects of your application or workload to determine the specific types of cloud resources required.

2. Key Factors in Assessing Requirements for Cloud Resources Provisioning:

2.1. Workload Characteristics

•Type of Workload:

- Is it computer-intensive (e.g., machine learning, simulations)?
- Is it storage-heavy (e.g., data warehousing, backups)?
- Is it network-bound (e.g., video streaming, web apps)?

•Expected Traffic:

- Estimate the number of users or transactions over time.
- Predict peak and off-peak periods to ensure resource scalability.

•Application Architecture:

- Are you deploying a monolithic application or microservices?
- Will you use containers or virtual machines?

2.2. Computer Resource Requirements

•CPU and Memory:

- Determine the number of cores and amount of RAM based on processing demands.
- Choose instance types or container sizes that match the workload (e.g., general-purpose, computer-optimized, memory-optimized).

•Processing Power:

- If using AI/ML, will GPUs or TPUs be needed?

•Autoscaling Needs:

- Will your application require automatic scaling based on load or demand?

2.3. Storage Requirements

- Data Volume:**

- Estimate the amount of data that needs to be stored.
- Consider growth rates for future storage needs.

- Type of Storage:**

- Does the workload require block storage (high-performance disk), object storage (for media files, backups), or file storage (shared access)?

- Access Patterns:**

- Is the data frequently accessed (hot storage) or infrequently accessed (cold storage)?
- Do you need low-latency access or high-throughput storage?

2.4. Network and Connectivity

- Network Bandwidth:**

- Evaluate the amount of data that will be transferred between your application, users, and cloud resources.

- Latency:**

- Are there specific latency requirements for real-time applications or low-latency data processing?

- Data Transfer:**

- Will you need to use CDNs for content delivery or load balancers for traffic distribution?

- VPC and Subnets:**

- Will the application require a virtual private cloud with specific subnetting, VPNs, or dedicated interconnect?

2.5. Security and Compliance

- Data Security:**

- Are there encryption requirements for data at rest or in transit?
- Will you need additional security services like a Web Application Firewall (WAF) or Intrusion Detection Systems (IDS)?

- Identity and Access Management:**

- How will users or applications authenticate and authorize access to resources?

- Compliance:**

- Are there compliance regulations like GDPR, HIPAA, or PCI-DSS that influence your resource provisioning (e.g., data residency, encryption)

standards)?

2.6. Cost Constraints

- **Budget:**

- Define budget limits and choose resource types that fit within those limits.

- **Pay-as-you-go vs. Reserved Instances:**

- Determine if you will use on-demand resources or commit to long-term reserved resources for cost savings.

- **Monitoring and Optimization:**

- Set up cost-monitoring tools (e.g., AWS Cost Explorer, Azure Cost Management) to track and manage resource spending.

2.7. Scalability and Availability

- **Horizontal vs. Vertical Scaling:**

- Will you scale by adding more instances (horizontal) or by increasing the size of existing instances (vertical)?

- **High Availability:**

- Does your application need multi-region or multi-zone deployments to ensure uptime?

- **Disaster Recovery:**

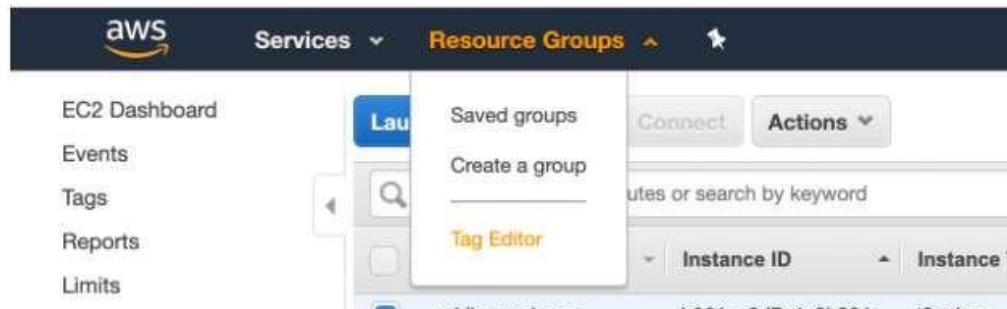
- Do you need backup and disaster recovery resources (e.g., cross-region replication)?



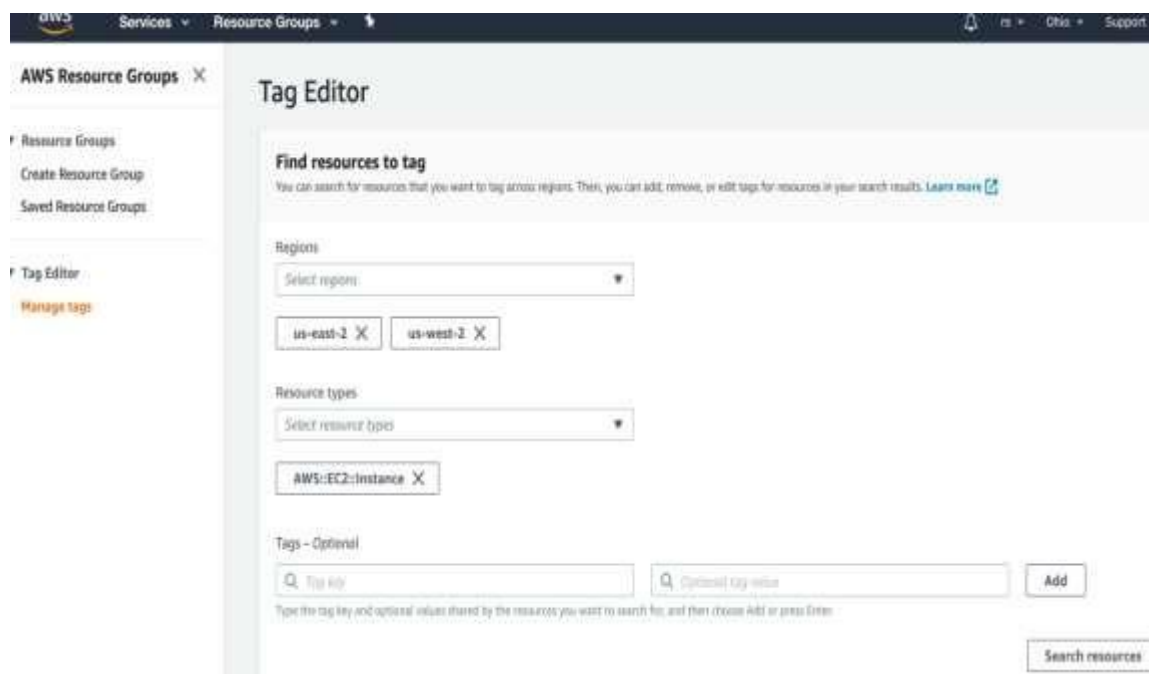
Key readings 2.1.3.: Selection of cloud resources

1. Steps involved in selecting cloud resources

- A. Go to AWS Console at <https://console.aws.amazon.com/>
- B. In the TOP Navigation Pane, click Resource Groups Dropdown
- C. Click Tag Editor



Here we can select either a particular region in which we want to search or select all regions from the dropdown. Then we can select actual resources which we want to search or we can also click on individual resources.



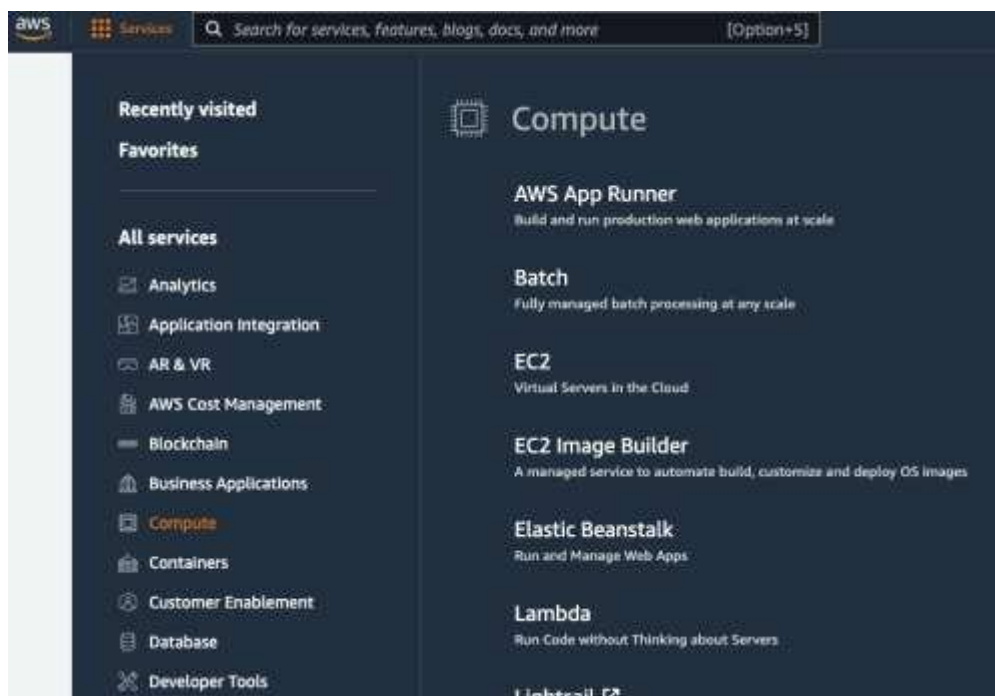


Key readings 2.1.4: Configuring cloud resources

1. Steps to Configure Cloud Resources:

1.1. Configure Virtual Machines (VMs):

1. Log in to the cloud provider's console (e.g., AWS Management Console, Azure Portal).
2. Navigate to the **Compute** section (e.g., AWS EC2, Azure Virtual Machines).



3. Select the **instance type** (general-purpose, compute-optimized, memory-optimized) based on CPU, RAM, and network requirements.

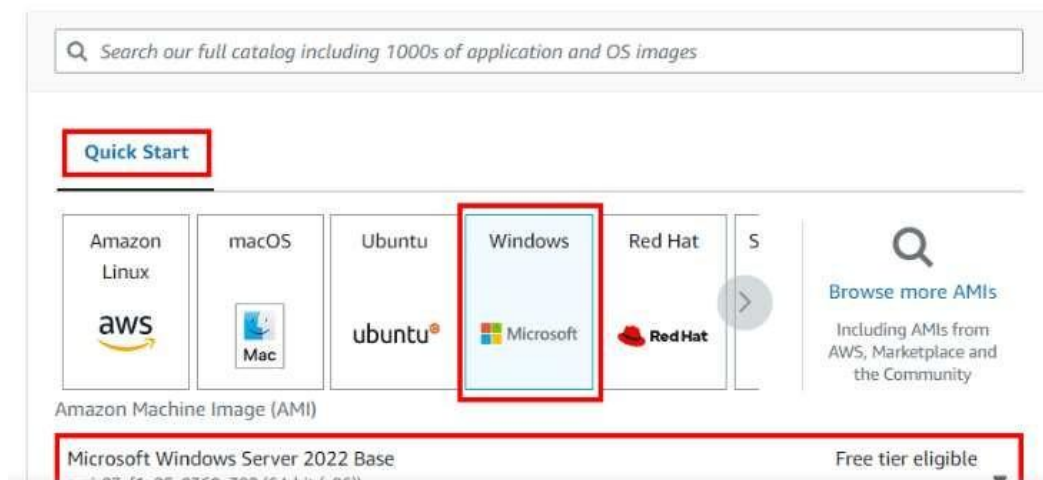
aws EC2 instance types

	General Purpose		Compute Optimized	Memory Optimized		Accelerated Computing	Storage Optimized		
Type	t2	m5	c5	r4	x1e	p3	h1	i3	d2
Description	Burstable, good for changing workloads	Balanced, good for consistent workloads	High ratio of compute to memory	Good for in-memory databases	Good for full in-memory applications	Good for graphics processing and other GPU uses	HDD backed, balance of compute and memory	SDD backed, balance of compute and memory	Highest disk ratio
Mnemonic	t is for tiny or turbo	m is for main or happy medium	c is for compute	r is for RAM	x is for xtreme	p is for pictures	h is for HDD	i is for IOPS	d is for dense

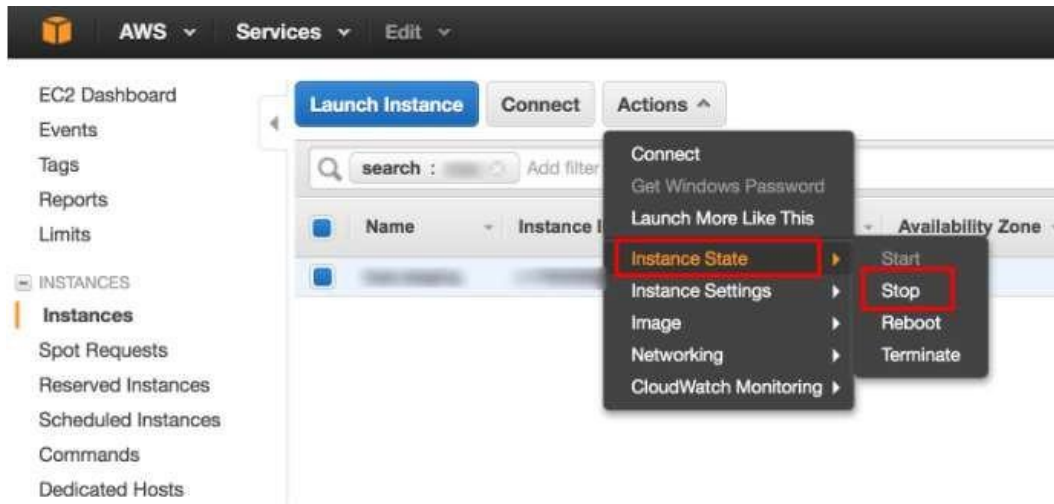
4. Configure the **Operating System** (e.g., Linux, Windows).

▼ Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below.



5. Set the **Instance Size** (e.g., small, medium, large) based on workload needs.



6. Configure **SSH Key Pairs** (for Linux) or **RDP access** (for Windows) to securely access the instance.

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

[Create new key pair](#)

In the Network setting, make sure **RDP traffic** is allowed.

☒ **Allow RDP traffic from**
Helps you connect to your instance

Anywhere
0.0.0.0/0

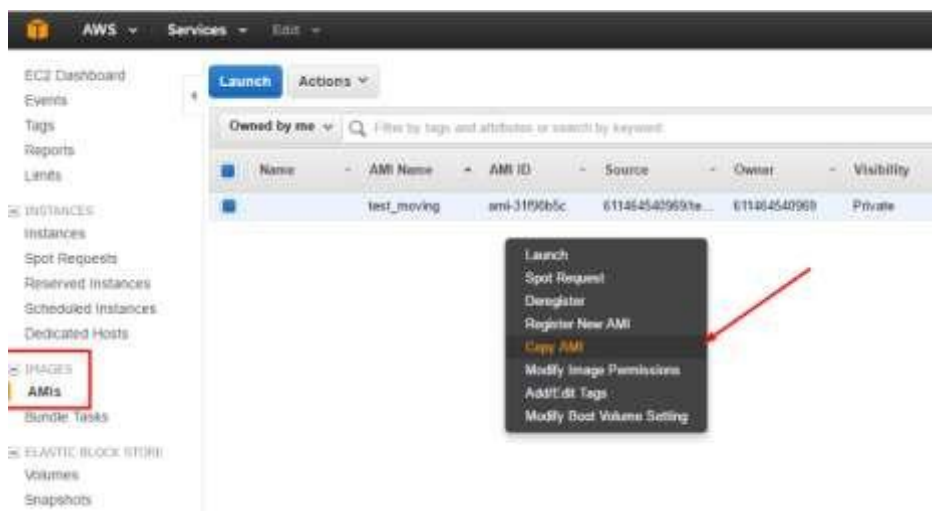
7. Set the **Availability Zone** or **Region** to ensure low latency or meet compliance.

- Create an AMI

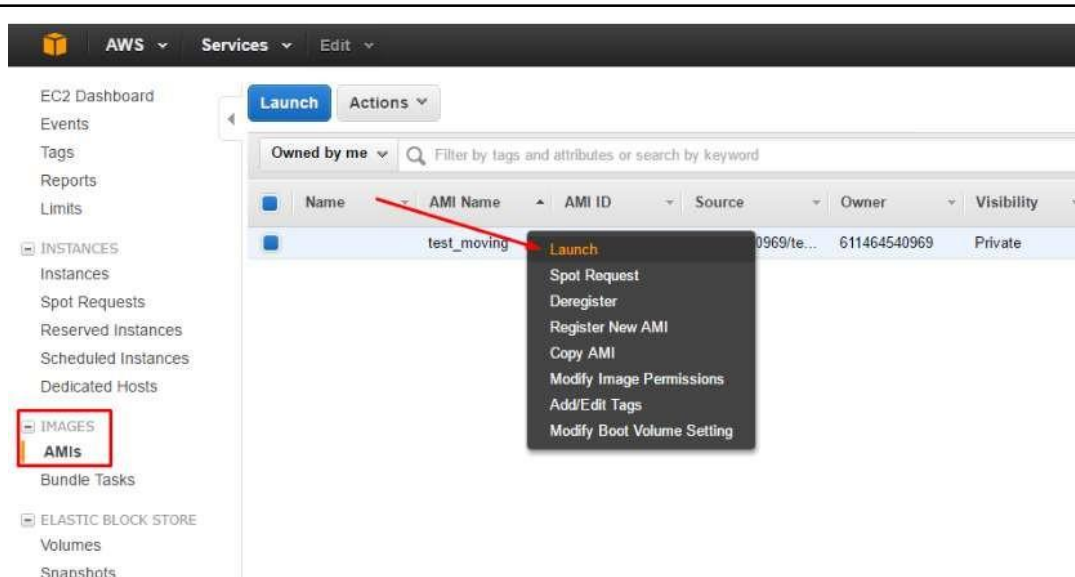
Under the EC2 section, choose Instances, right-click an instance you wish to migrate and choose Image --> Create Image



- Launch created AMI in the desired Availability Zone / VPC / AWS Region



- Select a destination region and a new name for your AMI. Switch to your destination region and go to the next section.



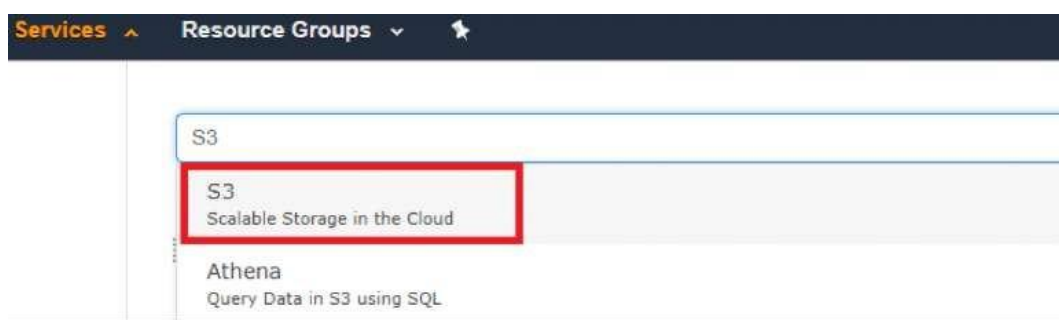
1.2. Configure Storage Resources

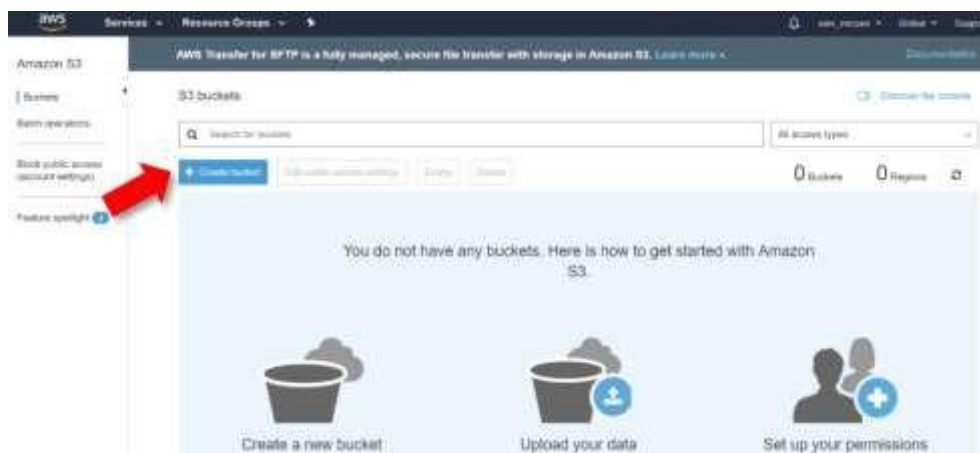
- **Block Storage (e.g., Amazon EBS, Azure Disks):**

1. Choose the **storage type** (SSD, HDD) based on performance needs.
2. Set the storage size (e.g., 100 GB) according to the application's data requirements.
3. Attach storage volumes to virtual machines or instances.
4. Configure **automatic backups** or **snapshots** to ensure data protection.

- **Object Storage (e.g., Amazon S3, Google Cloud Storage):**

1. Create a **storage bucket** for storing unstructured data like logs, backups, or media files.

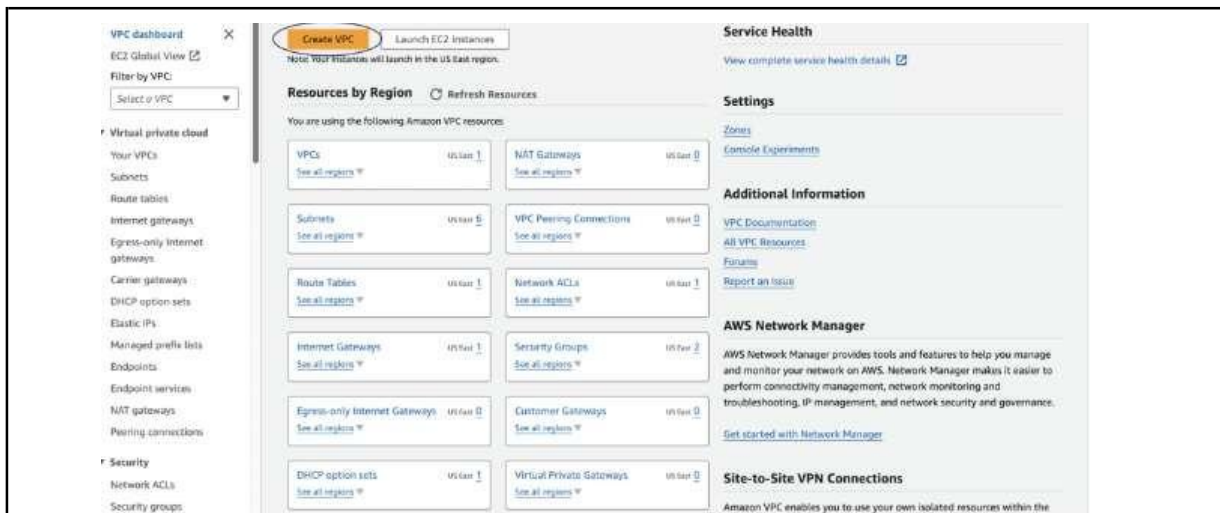




2. Set **storage class** (standard, infrequent access, archival) depending on access frequency.
 3. Configure **bucket permissions** for read/write access using Identity and Access Management (IAM) policies.
 4. Set up **versioning** and **lifecycle policies** to manage storage costs by moving old data to cheaper storage tiers.
- **File Storage (e.g., Amazon EFS, Azure Files):**
 1. Create a **file system** for applications requiring shared file storage.
 2. Mount the file system on multiple VMs/containers to enable shared access.
 3. Configure access control using IAM policies or network security groups.

1.3. Configure Networking Resources

- **Virtual Private Cloud (VPC) or Virtual Network:**
 1. Create a **VPC** or **Virtual Network** to logically isolate your cloud resources.
- Log in to the AWS Management Console.
- Choose CreateVPC



2. Set up **subnets** for different environments (e.g., public, private).
 3. Configure **IP addressing** (CIDR blocks) for each subnet.
- Configure your VPC by specifying a CIDR block (e.g., 10.0.0.0/16). AWS allows the CIDR block to range from a /16 netmask to a /28 netmask.

The screenshot shows the 'VPC settings' form. Under 'Resources to create', the 'VPC and more' option is selected. Under 'Name tag auto-generation', the 'Auto-generate' checkbox is checked, and the value 'project' is entered. Under 'IPv4 CIDR block', the value '10.0.0.0/16' is entered, showing '65,536 IPs'. Under 'IPv6 CIDR block', the 'Amazon-provided IPv6 CIDR block' option is selected. The 'Tenancy' dropdown is set to 'Default'.

4. Set up **Internet Gateways** or **NAT Gateways** for external connectivity and secure access.
- Configure the NAT gateways. It allows the resources in your private subnet to

connect with the internet and other AWS resources.

The screenshot shows the configuration options for a VPC. It includes sections for NAT gateways, Egress only internet gateway, VPC endpoints, and DNS options. The NAT gateways section has three radio buttons: 'None', 'In 1 AZ' (selected), and '1 per AZ'. The Egress only internet gateway section has two radio buttons: 'No' (selected) and 'Yes'. The VPC endpoints section has two radio buttons: 'None' and 'S3 Gateway' (selected). The DNS options section has two checkboxes: 'Enable DNS hostnames' and 'Enable DNS resolution', both of which are checked. There is also a link for 'Additional tags'.

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

☐ None ☒ In 1 AZ ☐ 1 per AZ

Egress only internet gateway [Info](#)
IPv6 only. Allows outbound communication over IPv6 in your private subnets.

☒ No ☐ Yes

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

☐ None ☒ S3 Gateway

DNS options [Info](#)
☒ Enable DNS hostnames
☒ Enable DNS resolution

► Additional tags

- **Load Balancers:**

1. Deploy a **load balancer** (e.g., AWS ELB, Azure Load Balancer) to distribute traffic across multiple instances.
2. Set up **health checks** to monitor instance availability.
3. Configure SSL certificates for secure HTTPS traffic.

- **DNS Configuration:**

1. Use DNS services (e.g., AWS Route 53, Google Cloud DNS) to map domain names to your cloud resources.
2. Set up DNS records (e.g., A records, CNAME records) for routing traffic.

1.4. Configure Databases

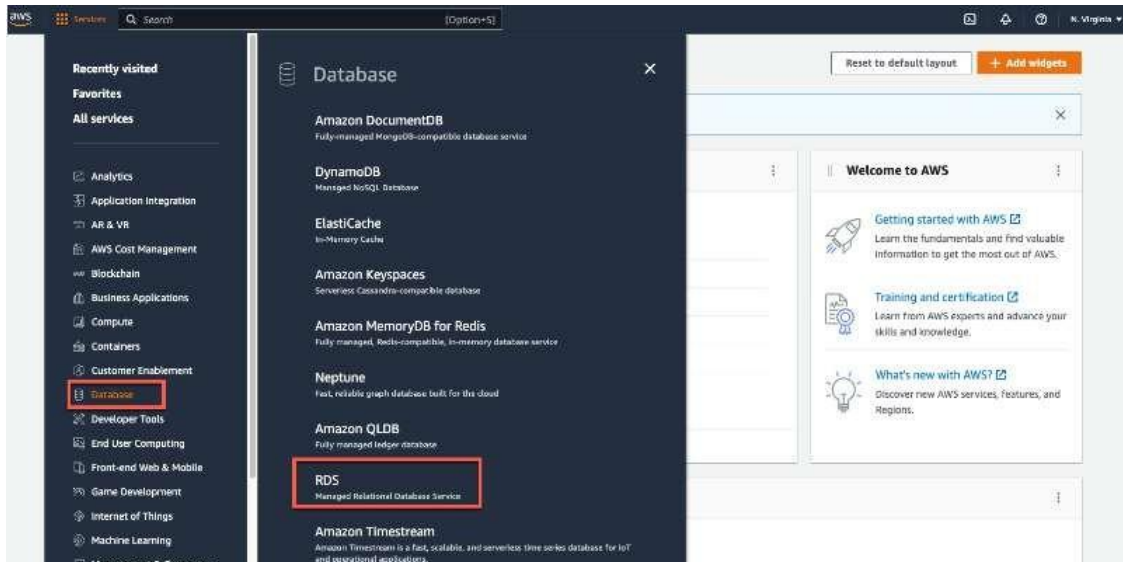
- **Managed Databases (e.g., Amazon RDS, Azure SQL, Google Cloud SQL):**

1. Choose the **database engine** (MySQL, PostgreSQL, SQL Server, etc.).
2. Configure the **instance class** based on database size and transaction volume.
3. Set the **storage size** and IOPS (input/output operations per second).
4. Enable **automatic backups** and **multi-zone replication** for high availability.

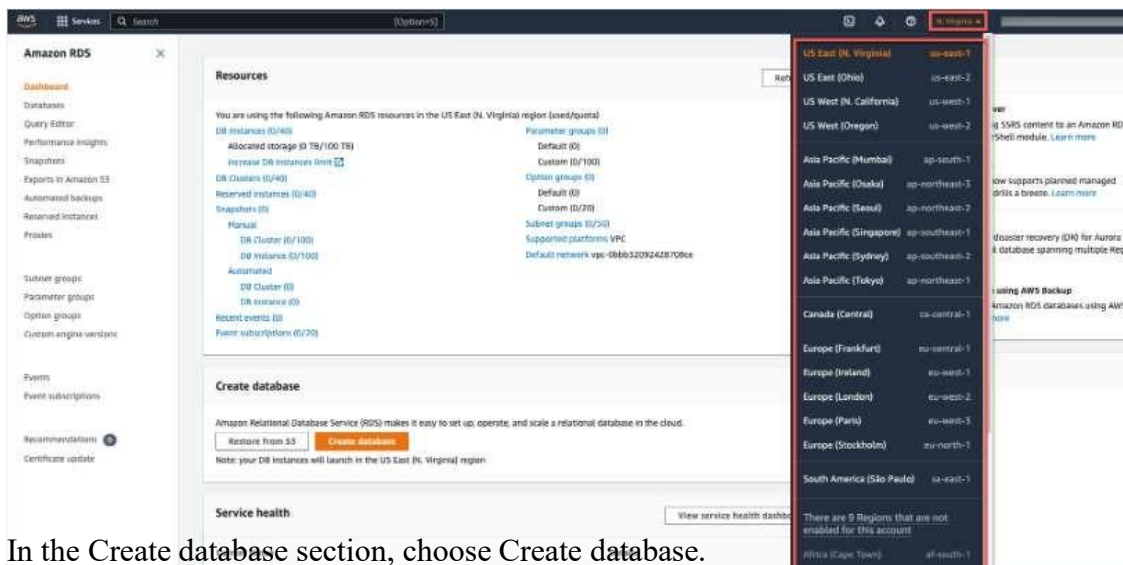
5. Configure **database security** using IAM roles, encryption, and security groups.

Example to create a MYSQL DB Instance

- Open the AWS Management Console



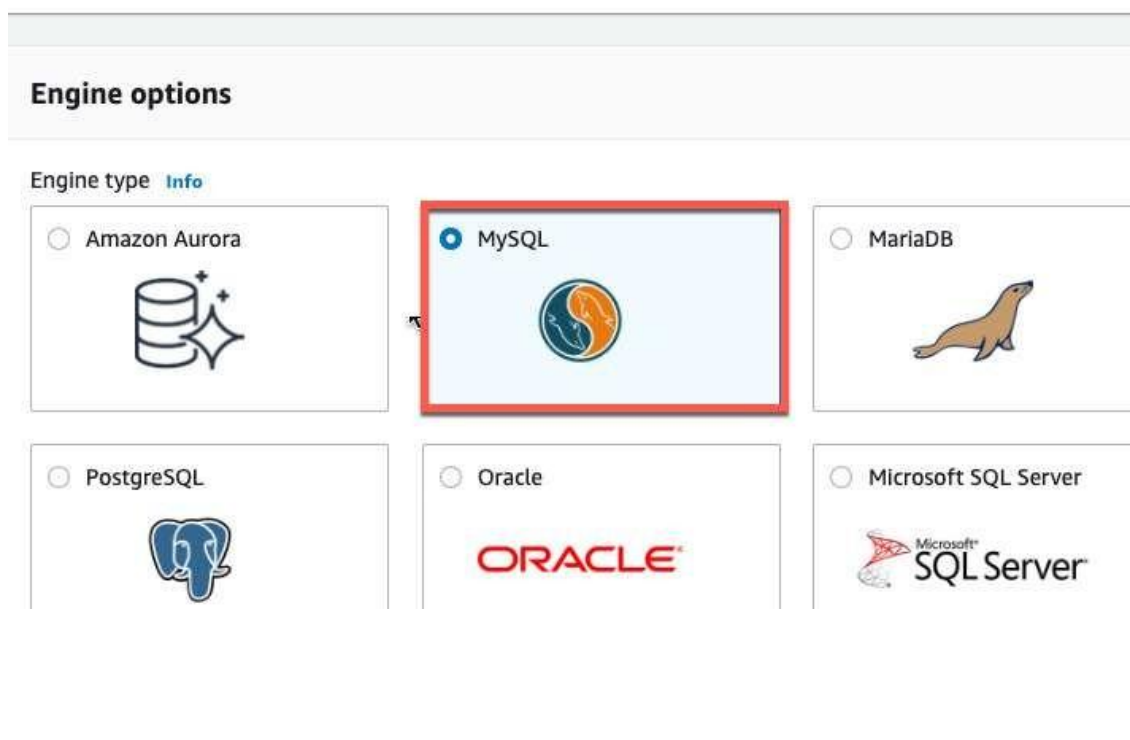
- select the Region in which you want to create the DB instance



- In the Create database section, choose Create database.



- Choose a database creation method



You now have options to select your engine.
Choose a sample template to meet your use case.

☐ **Production**
Use defaults for high availability and fast, consistent performance.

☐ **Dev/Test**
This instance is intended for development use outside of a production environment.

☒ **Free tier**
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.
[Info](#)

Availability and durability

Deployment options [Info](#)

The deployment options below are limited to those supported by the engine you selected above.

- ☐ **Multi-AZ DB Cluster - new**
Creates a DB cluster with a primary DB instance and two readable standby DB instances, with each DB instance in a different Availability Zone (AZ). Provides high availability, data redundancy and increases capacity to serve read workloads.
- ☐ **Multi-AZ DB Instance (not supported for Multi-AZ DB cluster snapshot)**
Creates a primary DB instance and a standby DB instance in a different AZ. Provides high availability and data redundancy, but

- You will now configure your DB instance

Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current Region.

rds-mysql-10minTutorial

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

masterUsername

1 to 16 alphanumeric characters. First character must be a letter.

☐ **Auto generate a password**

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

••••••••••

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), ' (single quote), " (double quote), (at sign).

Confirm master password [Info](#)

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (Includes r and x classes)
- ☒ Burstable classes (includes t classes)

db.t2.micro
1 vCPUs 1 GiB RAM Not EBS Optimized

☐ Include previous generation classes

- Connectivity section where you can provide information that Amazon RDS needs to launch your MySQL DB instance.

Connectivity [Info](#)



Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0bbb32092428708ce)

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB Subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

☒ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐

Choose existing

Choose existing VPC security groups

☒

Create new

Create new VPC security group

New VPC security group name

Enter new VPC security group name

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐

Create an RDS Proxy [Info](#)

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

▼ Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

3306

1.5. Configure Security

- **Identity and Access Management (IAM):**

1. Define user roles and permissions through IAM policies.
2. Implement **least privilege** to restrict access to only necessary resources.
3. Use **Multi-Factor Authentication (MFA)** for secure access to the cloud console.
4. Set up **service roles** for instances or applications needing access to other cloud services (e.g., accessing S3 from an EC2 instance).

- **Network Security:**

1. Configure **Security Groups** and **Network Access Control Lists (NACLs)** to control inbound and outbound traffic to your instances.
2. Set up **firewalls** or **Web Application Firewalls (WAF)** for application security.
3. Implement **encryption** for data at rest (e.g., using AWS KMS or Azure Key Vault) and in transit (e.g., TLS/SSL).



Key readings 2.1.5: Deprovisioning cloud resources

1. Deprovisioning cloud resources refers to the process of releasing or terminating cloud services and resources that are no longer required. It is an essential part of cloud resource management that helps optimize costs and free up unused or unnecessary resources.

2. for Deprovisioning Cloud Resources

1. Identify Resources to Deprovision

- **Audit Active Resources:**
 - Perform an audit of all running resources (compute instances, storage volumes, databases, etc.) to identify which are no longer needed.
 - Example tools: AWS Resource Groups, Azure Resource Manager, Google Cloud Resource Manager.
- **Monitor Resource Utilization:**
 - Review usage data and identify underutilized or idle resources.
 - Example tools: AWS CloudWatch, Azure Monitor, Google Cloud Monitoring.

2. Terminate Compute Resources

- **Shut Down Virtual Machines (VMs):**
 - Safely stop and terminate unused virtual machines (EC2 instances, Azure VMs, Google Compute Engine instances) to avoid incurring additional costs.

3. Delete Unused Storage Resources

- **Block Storage:**

- Detach and delete unused block storage volumes (e.g., Amazon EBS, Azure Managed Disks) that are not attached to any instance.
- **Object Storage:**
 - Remove data from object storage buckets (e.g., AWS S3, Azure Blob Storage) that is no longer required.
 - Ensure that any lifecycle policies or versioning configurations are cleaned up.
- **File Storage:**
 - Delete file storage resources (e.g., Amazon EFS, Azure Files) if they are no longer needed.

4. Tear Down Virtual Networks (VPC):

- Delete any unused virtual private clouds (VPCs), subnets, VPNs, or network peering connections that are no longer required.

5. Delete Unused Databases

- **Managed Databases:**
 - Decommission managed databases (e.g., Amazon RDS, Azure SQL Database) when they are no longer needed. Ensure data backups are created before deletion.
- **NoSQL Databases:**
 - Remove NoSQL databases (e.g., Amazon DynamoDB, Google Firestore) if they are no longer in use or linked to decommissioned applications.



Points to Remember

Description of cloud resource

- **Cloud computing resources** refer to the various types of computational services, infrastructure, and tools provided over the internet that allow users to access and utilize computing power, storage, and software without owning physical hardware.
- **Types of cloud computing resources** are compute resources (like VMs CPU and memory), storage resources (like disk space), and network resources (like bandwidth), database services, application and software services, and security resources.

Description of Assessment of Requirements for cloud resources provisioning

- **Role of Assessing the requirements for cloud resources provisioning**

is a critical step in ensuring that you allocate the right amount of resources to meet performance, cost, scalability, and security needs.

- **Key Factors to consider while Assessing Requirements for Cloud Resources Provisioning:**

Type of Workload, Compute Resource Requirements, Storage Requirements, Network and Connectivity, Security and Compliance, Cost Constraints, Scalability and Availability.

Selection of cloud resource

- **Steps for Selecting cloud resources**

- A. Go to AWS Console at <https://console.aws.amazon.com/>
- B. In the TOP Navigation Pane, click Resource Groups Dropdown
- C. Click Tag Editor
- D. select all regions from the dropdown. Then we can select actual resources which we want to search or we can also click on individual resources.

Configuring cloud resources

- **Steps involved to configure cloud resources:**

Configure Virtual Machines (VMs), Configure Storage Resources, Configure Networking Resources, and configure managed Databases (e.g., Amazon RDS, Azure SQL, Google Cloud SQL)

Deprovisioning cloud resources

- **Deprovisioning cloud resources** is a step in cloud management to ensure cost optimization, free up unnecessary resources.
- **Steps to deprovision cloud resources:** Identify Resources to Deprovision, Terminate Compute Resources, Delete Unused Storage Resources, Tear Down Virtual Networks (VPC), Delete Unused Databases.



Application of learning 2.1.

You are assigned to deploy a high-availability web application on AWS. The application needs to run across multiple Availability Zones (AZs) to ensure uptime in case of hardware or network failure. The application should automatically scale up during traffic points and scale down during off-peak times.

You are required to Select cloud resources, configure those resources, and preform the deprovisioning unnecessary cloud resources.



Key readings 2.2.1. Identification of cloud monitoring tools

1. Digital Ocean monitoring and uptime

A. Digital Ocean monitoring and uptime refers to tracking the performance, resource usage, and health of cloud resources like Droplets (virtual machines), managed databases, and other cloud services. Monitoring helps users detect and resolve issues before they impact service availability or performance.

B. Key features of Ocean monitoring and uptime

- Droplet Monitoring:
 - Resource Metrics: DigitalOcean provides monitoring for individual Droplets, offering metrics like CPU usage, disk I/O, memory usage, and network traffic.
 - Graphs and Dashboards: These metrics are visualized on dashboards, allowing users to easily track trends and spot anomalies.
 - Alerts: Users can set thresholds for key performance indicators (KPIs) like CPU usage or memory, and receive alerts (via email or Slack) when those thresholds are breached.
- Managed Database Monitoring:

- Digital Ocean's managed databases (PostgreSQL, MySQL, Redis, etc.) come with monitoring features that provide insights into database performance and health.
- Metrics like connections, query throughput, replication lag, and disk usage are tracked in real-time.
- Logs:
 - Digital Ocean provides log monitoring through integrations with third-party tools like Loggly and Papertrail for centralized log management, enabling users to track errors, security events, and application performance across all services.

2. AWS Cloudwatch

- It is an AWS tool that enables users to collect, analyze, and visualize real-time data on AWS resources, applications, and services.
CloudWatch is designed to monitor the performance and health of cloud infrastructure, providing insights into system operations and facilitating automated actions based on defined thresholds and alarms.

Key features of AWS

CloudWatch Key Features of

AWS CloudWatch

- **Metrics Monitoring**

CloudWatch collects and stores metrics from AWS services and custom applications, providing real-time data on resource usage and performance.

AWS Service Metrics: CloudWatch automatically collects metrics from AWS services like EC2, RDS, Lambda, and S3. Examples include:

- **EC2 (Elastic Compute Cloud):** CPU utilization, disk reads/writes, network traffic.
- **RDS (Relational Database Service):** Read/write throughput, database connections.
- **Lambda:** Invocation count, duration, error rates.
- **S3 (Simple Storage Service):** Bucket size, object counts, request rates.

Alarms

CloudWatch allows users to set **alarms** based on predefined thresholds for metrics. When a threshold is breached, the alarm can trigger notifications or automated actions.

- **Threshold-based Alerts:** For example, if CPU usage exceeds 80% for 5 minutes, CloudWatch can send alerts via email, SMS, or Amazon SNS (Simple Notification Service).
- **Actions:** Alarms can also initiate automatic actions, such as scaling EC2 instances, restarting services, or invoking Lambda functions.

Logs Monitoring (CloudWatch Logs)

CloudWatch Logs enables users to collect, monitor, and store log data from AWS resources and applications. It aggregates logs from various sources, making them easier to search, analyze, and monitor.

- **Log Sources:** Logs can be collected from EC2 instances, Lambda functions, VPC Flow Logs, CloudTrail, and more.
- **Log Insights: CloudWatch Logs Insights** provides a query engine for running SQL-like queries on log data, enabling users to quickly troubleshoot issues.

CloudWatch Events (Amazon EventBridge)

CloudWatch Events (now integrated with **Amazon EventBridge**) monitors system events and allows users to automate workflows in response to changes in AWS resources.

- **Automated Actions:** Event rules can trigger actions like invoking Lambda functions or sending notifications when an event occurs, such as an EC2 instance terminating or a database backup completing.
- **Custom Event Rules:** Users can define custom rules for monitoring changes in AWS resources or applications and set actions based on event patterns.

3. Microsoft Azure monitor

Microsoft Azure Monitor is a monitoring and observability service provided by **Microsoft Azure**. It allows users to collect, analyze, and act on data from Azure resources, applications, and on-premises environments.

Azure Monitor is designed to maximize the performance and availability of applications and services by providing insights into the operation of cloud and hybrid infrastructures.

4. Google Cloud Operations

Google Cloud Operations (formerly known as Stackdriver) is a comprehensive tools for monitoring, managing, and debugging infrastructure and applications

Key readings 2.2.2. Description of scaling strategies

1. Definition of scaling strategies in cloud computing

In cloud computing, **scaling strategies** refer to the approaches used to increase or decrease cloud resources to handle changing workloads effectively.

There are several **scaling strategies** in cloud computing, and they are primarily divided into two categories: **vertical scaling** and **horizontal scaling**.

2. Vertical Scaling (Scaling Up/Down)

Vertical scaling involves adding more power (CPU, RAM, storage) to an existing server or virtual machine (VM). It is often referred to as "scaling up" when adding resources and "scaling down" when reducing resources.

- **How It Works:**

- When demand increases, the cloud service can upgrade the instance type (or virtual machine type) to a larger one with more resources.
- When demand decreases, the service can be scaled down by reducing the resources allocated to the instance.

- **Use Cases:**

- Applications that require high performance from a single server, such as databases or certain enterprise applications.

- Monolithic applications that cannot be easily distributed across multiple servers.
- **Advantages:**
 - Simplicity: Easier to implement since no changes in architecture are required.
 - Useful for applications that are difficult to distribute or partition.
- **disadvantages:**
 - Limited by the hardware capacity of a single machine. There's a maximum to how much a single instance can scale.
 - Downtime may occur when upgrading/downgrading an instance (though some cloud providers offer features like live migration to reduce this).
 - Costly at larger scales, as upgrading to powerful instances can become expensive.

3. Horizontal Scaling (Scaling Out/In):

Horizontal scaling involves adding more instances (servers, VMs, containers) to handle increased demand. It's known as "scaling out" when adding more instances and "scaling in" when removing instances.

- **How It Works:**
 - When demand increases, additional instances or nodes are provisioned to handle the workload.
 - When demand decreases, the extra instances are terminated, and the workload is distributed across the remaining servers.
- **Use Cases:**
 - **Distributed applications** like web servers, microservices, and containerized applications.
 - **Stateless applications** that can run independently on multiple instances.
 - Systems with high traffic or sudden spikes, like e-commerce platforms or social media.
- **Advantages:**
 - Flexibility and scalability: There's no theoretical limit to the number of instances that can be added.
 - Fault tolerance: With multiple instances, failures of individual instances do not affect the overall application.
 - Cost efficiency: Scale only when needed, and turn off instances when they're no longer required.
- **Disadvantages:**
 - Requires applications to be designed to work in a distributed manner, which can be complex.
 - Load balancing and data consistency across multiple instances need careful management.

- Latency can become an issue if instances are distributed across different geographic locations.

Key readings 2.2.3. Applying cloud monitoring tools

1. introduction

Applying cloud monitoring tools is essential for ensuring the performance, security, availability, and cost-effectiveness of cloud infrastructure and applications.

Cloud monitoring involves the collection and analysis of metrics, logs, and other data points from cloud resources to detect issues, optimize performance, and ensure compliance with operational requirements.

2. Steps in Applying Cloud Monitoring Tools:

- A. **Define Monitoring Objectives:** Before applying cloud monitoring tools, it's important to establish clear objectives for what needs to be monitored. These objectives typically include:
- **Performance Monitoring:** Tracking metrics like CPU usage, memory, disk I/O, and network traffic to ensure that cloud resources are performing optimally.
 - **Security Monitoring:** Detecting unusual access patterns, unauthorized activities, or breaches in security protocols.
 - **Cost Management:** Monitoring cloud usage and costs to prevent over-provisioning and avoid unexpected charges.

- **Availability and Uptime:** Monitoring the availability of cloud services, including downtime and recovery times.
- B. **Choose the Right Cloud Monitoring Tools:** There are many cloud monitoring tools available, each offering different.
- C. **Set Up Metrics:**

Common metrics include:

- **CPU Utilization:** Measures the load on virtual machines or containers.
- **Memory Usage:** Tracks memory consumption to avoid resource exhaustion.
- **Disk I/O & Storage:** Monitors read/write operations and storage capacity.
- **Network Traffic:** Tracks inbound and outbound network traffic to detect bottlenecks.
- **Latency and Response Time:** Measures the time taken for services to respond to requests.
- D. **Implement Alerts and Notifications:** Alerts are a critical component of cloud monitoring, as they notify you of abnormal conditions, failures, or breaches in predefined thresholds.
 - **Threshold-Based Alerts:** Set thresholds for critical metrics (e.g., CPU usage above 90%) and trigger alerts when those thresholds are crossed.
 - **Anomaly Detection:** Some advanced tools use machine learning to detect deviations from normal patterns, allowing proactive response to potential issues.
 - **Custom Alerts:** Configure alerts for specific application events, such as database query failures or high user request latency.
 - **Incident Notifications:** Alerts can be sent via email, SMS, or through platforms like Slack or PagerDuty for instant team notification and incident management.
- E. **Monitor Logs for Detailed Analysis:** Log monitoring is essential for understanding the root cause of issues and ensuring compliance. Cloud monitoring tools often integrate with log management systems to collect, store, and analyze logs.
 - **Amazon CloudWatch Logs (AWS):** Captures log data from AWS services and applications for real-time monitoring and troubleshooting.
 - **Azure Monitor Logs (Microsoft Azure):** Collects and analyzes logs from Azure services and on-premises environments.
 - **Elastic Stack (formerly ELK):** Combines Elasticsearch, Logstash, and Kibana for real-time log collection, storage, and visualization.

Log Analysis helps detect anomalies, security threats, or application errors that may not be evident from metrics alone.

- F. **Security Monitoring:** Security is a critical aspect of cloud monitoring. It ensures

data protection, threat detection, and compliance with industry regulations.

- **Cloud Security Posture Management (CSPM):** Tools that continuously monitor cloud environments for security risks and compliance violations.
- **Intrusion Detection:** Monitoring traffic patterns and access logs for unauthorized access attempts.
- **Encryption Monitoring:** Ensuring that sensitive data at rest and in transit is properly encrypted and adhering to compliance requirements.
- **Audit Trails:** Many tools track actions performed on cloud resources, which are vital for regulatory compliance audits.

3. Guide on how to use AWS Cloud watch free tier as a cloud monitoring tool

A. Understand What the Free Tier Includes:

The **Amazon CloudWatch free tier** provides the following services:

- **10 custom metrics:** You can create and monitor up to 10 custom metrics for free.
- **10 alarms:** Set up to 10 standard resolution alarms without being charged.
- **5 GB of log data ingestion:** Ingest up to 5 GB of log data from your AWS services or custom applications.
- **5 GB of log data archive:** Store up to 5 GB of log data in CloudWatch for archiving purposes.
- **1 million API requests:** This includes any API calls made to CloudWatch to retrieve metrics, logs, or other data.

B. Set Up CloudWatch Monitoring for AWS Services:

CloudWatch provides free basic monitoring for certain AWS services. This includes:

- **Amazon EC2:** Basic monitoring is provided at 5-minute intervals for EC2 instances. You can enable detailed monitoring (1-minute intervals) for a fee, but basic monitoring is free.
- **Amazon RDS, ELB, DynamoDB, and S3:** These services also offer free basic monitoring in CloudWatch, providing visibility into key performance metrics at no additional cost.

To set up basic monitoring:

- Open the **CloudWatch Console** in the AWS Management Console at

<https://console.aws.amazon.com/cloudwatch/>



Sign in as IAM user

Account ID (12 digits) or account alias

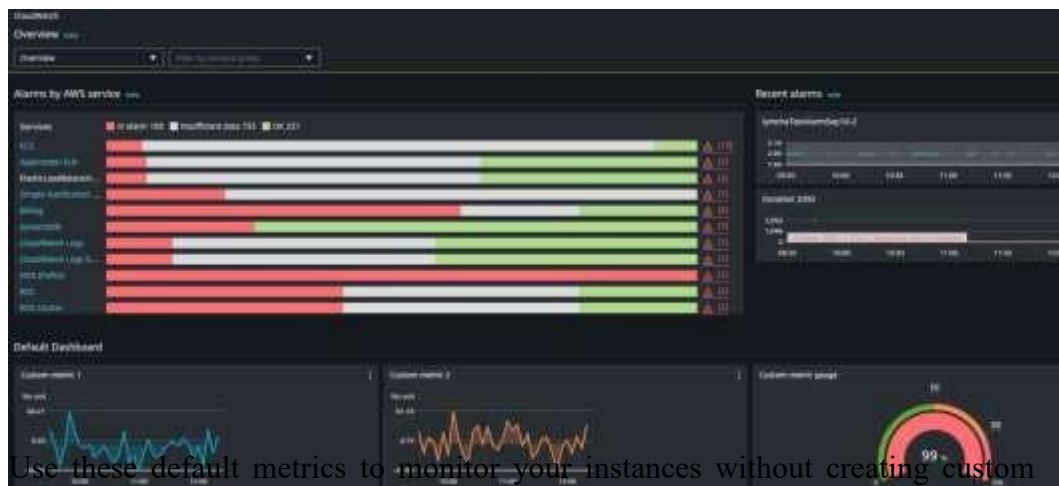
IAM user name

Password

☐ Remember this account

Sign in

- Navigate to **Metrics** and find the default metrics for your AWS services (e.g., EC2, RDS).



- Use these default metrics to monitor your instances without creating custom metrics, as default metrics are free and included in basic monitoring.



Points to Remember

Cloud monitoring tools:

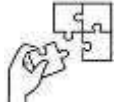
- **Digital Ocean monitoring and uptime** are set of built-in monitoring tools to track the performance of Droplets (VMs), databases, and Kubernetes clusters.
- **CloudWatch:** is an essential service for monitoring, managing, and automating AWS infrastructure and applications.
- **Microsoft Azure Monitor** is a comprehensive monitoring and observability service provided by **Microsoft Azure**.
- **Google Cloud Operations** (formerly known as Stackdriver) is a comprehensive suite of tools for monitoring, managing, and debugging infrastructure and applications on **Google Cloud Platform (GCP)**

Cloud scaling strategies:

- In cloud computing, **scaling strategies** refer to the approaches used to increase or decrease cloud resources to handle changing workloads effectively.
- **Vertical Scaling (Scaling Up/Down)**
Vertical scaling involves adding more power (CPU, RAM, storage) to an existing server or virtual machine (VM).
- **Horizontal Scaling (Scaling Out/In):**
Horizontal scaling involves adding more instances (servers, VMs, containers) to handle increased demand.

Applying cloud monitoring tools:

- **Cloud monitoring** involves the collection and analysis of metrics, logs, and other data points from cloud resources to detect issues, optimize performance, and ensure compliance with operational requirements.
- **Set Up CloudWatch Monitoring for AWS Services:**
 - step 1:** Open the CloudWatch Console in the AWS Management Console at <https://console.aws.amazon.com/cloudwatch/>
 - step 2:** Navigate to Metrics and find the default metrics for your AWS services (e.g., EC2, RDS).
 - step 3:** Use these default metrics to monitor your instances without creating custom metrics, as default metrics are free and included in basic monitoring.



Application of learning 2.2.

MegaTech, is an online e-commerce company, is running its web application on AWS. The company uses Amazon EC2 instances, an RDS database, an Elastic Load Balancer (ELB), and an S3 bucket for static content delivery. The application is critical to the business, so maintaining uptime, performance, security, and cost efficiency is crucial. The company faces the following challenges:

Application performance degrades when there's a traffic spike; the company is facing unexpected AWS bills due to underutilized or idle resources; security is a concern as unauthorized access attempts need to be detected quickly; the operations team needs real-time insights into system health to troubleshoot issues proactively.

You are required to apply AWS CloudWatch to monitor metrics such as CPU Utilization, Memory Usage, Disk I/O & Storage, Network Traffic, Latency and Response Time and more to ensure that the application is running smoothly, stays within budget, and meets SLAs for performance and availability.

Key readings 2.3.1.: Identification of cloud resources utilization optimization

1. Strategies to identify cloud resources utilization optimization

A. Right-Sizing Resources:

- **Analyze Resource Usage:** Use cloud-native tools like **AWS Compute Optimizer**, **Azure Advisor**, or **Google Cloud Recommender** to analyze resource usage and receive recommendations on optimal instance sizes and types.
- **Downscale or Upscale Resources:** Right-size instances and other resources based on their actual workloads. For example, reduce underutilized EC2 or VM instances to smaller sizes, or move to higher-capacity instances if resources are regularly maxed out.
- **Burstable Instances:** For workloads with varying demand, use burstable instance types (like **AWS T3** or **Azure B-series**) that allow for base performance with the ability to burst during peak loads.

B. Leverage Auto Scaling:

- **Horizontal Auto Scaling:** Automatically add or remove instances in response to changing demand using services like **AWS Auto Scaling**, **Google Cloud Autoscaler**, or **Azure VM Scale Sets**. This helps prevent over-provisioning and

ensures optimal use of resources.

- **Vertical Scaling:** Automatically increase or decrease the capacity (e.g., CPU or memory) of an instance based on workload demands. Though vertical scaling has limitations, it can be useful for monolithic applications that cannot scale horizontally.
- **Scheduled Scaling:** Implement scheduled scaling to adjust resources during predictable periods of high or low demand, such as scaling up during business hours and scaling down at night or on weekends.

C. Optimize Storage Usage:

- **Tiered Storage:** Use cloud storage tiers based on access frequency:
 - **S3 Intelligent-Tiering** for automatic transitions between access tiers in AWS.
 - **Azure Blob Storage** or **Google Cloud Storage** classes for optimized data storage costs.
- **Lifecycle Policies:** Define and implement **lifecycle policies** to automatically move infrequently accessed data to cheaper storage classes (e.g., **AWS Glacier**, **Azure Cool Blob Storage**, **Google Nearline**).
- **Right-Size EBS Volumes:** Regularly review and right-size EBS volumes or block storage. For example, ensure volumes aren't over-provisioned and that unattached EBS volumes are deleted.

D. Use Serverless and Managed Services:

- **Serverless Computing:** Utilize serverless services like **AWS Lambda**, **Azure Functions**, or **Google Cloud Functions** to automatically scale based on demand without managing infrastructure. You pay only for the compute time consumed, making it highly cost-effective for bursty or variable workloads.
- **Managed Databases and Services:** Migrate workloads to **managed services** like **Amazon RDS**, **Azure SQL Database**, or **Google Cloud Spanner**. These services automatically handle scaling, patching, and optimization, reducing the need for manual intervention.

E. Use Spot Instances and Preemptible VMs:

- **Spot Instances (AWS) or Preemptible VMs (Google Cloud):** For workloads that can tolerate interruptions, like batch processing or rendering jobs, use Spot Instances or Preemptible VMs. These instances are significantly cheaper than On-Demand instances, often saving up to 90%.
- **Spot Fleets or VM Scale Sets:** Set up fleets or scale sets that automatically allocate Spot Instances alongside On-Demand Instances, optimizing costs while

ensuring service continuity.

F. Containerization and Orchestration:

- **Containerization:** Use containers to encapsulate applications and their dependencies. Containers allow you to run multiple, isolated workloads on the same infrastructure, optimizing resource usage. Tools like **Docker** and **Podman** help package applications into containers.
- **Container Orchestration:** Use orchestration tools like **Kubernetes** (managed by **Amazon EKS**, **Azure AKS**, or **Google GKE**) to dynamically scale and manage containerized workloads. This enables optimal resource utilization by ensuring containers use only the resources they need.

G. Implement Cost Optimization Tools:

- **Reserved Instances and Savings Plans:** For predictable workloads, purchase **Reserved Instances (RIs)** (AWS, Azure, Google Cloud) or **Savings Plans** (AWS) for significant discounts (up to 72%) compared to On-Demand pricing.
- **Spot Instances and Savings Plans:** Set up a **hybrid approach** that combines Reserved Instances, On-Demand Instances, and Spot Instances to optimize both cost and availability.

H. Continuous Monitoring and Alerts:

- **Cloud Monitoring Tools:** Use **AWS CloudWatch**, **Azure Monitor**, or **Google Cloud Operations Suite** to monitor resource utilization and receive alerts when resources are underutilized or over-provisioned. Create alarms for high CPU usage, memory usage, or disk I/O to act immediately.
- **Optimize Workloads Based on Metrics:** Regularly analyze performance metrics to identify underused or overused resources and adjust them accordingly.

I. Implement Data Transfer Optimization:

- **Reduce Data Transfers Between Regions:** Minimize inter-region data transfer costs by keeping related resources (like databases and instances) in the same region or availability zone.
- **Use Content Delivery Networks (CDNs):** Utilize CDNs like **Amazon CloudFront**, **Azure CDN**, or **Google Cloud CDN** to cache frequently accessed content closer to end-users, reducing data transfer costs and improving performance.
- **Use Private Networking Options:** For AWS, use **VPC endpoints** to reduce data transfer charges between VPCs and services such as S3.

J. Optimize Load Balancing and Traffic Management:

- **Auto-tune Load Balancers:** Use managed load balancers like **AWS Elastic Load Balancer (ELB)**, **Azure Load Balancer**, or **Google Cloud Load Balancer** to

distribute traffic efficiently and reduce the need for manual intervention.

- **DNS-Based Routing:** Use **Route 53** or **Azure Traffic Manager** for DNS-based routing to distribute traffic globally, optimize resource utilization, and route based on latency or location.

K. Set Policies for Idle Resources:

- **Automatically Turn Off Idle Resources:** Use automation to identify and shut down idle resources, such as development or test environments, outside working hours. AWS Lambda or Azure Automation can be used to stop idle instances automatically.
- **Unattached and Unused Resources:** Regularly audit and remove unattached or unused resources, such as **EBS volumes**, **load balancers**, **IP addresses**, or **databases** that are no longer needed.

L. Use Cloud-Native Optimization Tools:

- **AWS Trusted Advisor:** Regularly review **AWS Trusted Advisor** reports for insights on cost optimization, performance, and fault tolerance.
- **Azure Advisor:** Use **Azure Advisor** to get personalized recommendations for optimizing your cloud resources.
- **Google Cloud Recommender:** Leverage **Google Cloud Recommender** to receive suggestions for optimizing cloud costs and performance based on historical usage.

2. Best practices for cloud resources utilization optimization

A. Right-Sizing Resources

- **Use the Correct Instance Types:** Regularly evaluate your workload requirements and choose the instance types that match the CPU, memory, and storage needs. Avoid over-provisioning.
- **Use Monitoring Tools:** Leverage tools like **AWS Compute Optimizer**, **Azure Advisor**, or **Google Cloud Recommender** to get instance size recommendations based on actual usage.
- **Resize Based on Usage:** Continuously monitor resource utilization (CPU, memory, disk, network) and resize underused or overused instances accordingly.

B. Auto Scaling

- **Set Up Auto Scaling:** Implement auto-scaling to automatically adjust compute resources (instances, containers) based on actual demand. Scale up during peak traffic and scale down during idle periods.
- **Horizontal Scaling:** Prefer horizontal scaling (adding more instances) instead of vertical scaling (upgrading instance sizes) for better fault tolerance and

scalability.

- **Leverage Scheduled Scaling:** Schedule scaling based on predictable workload patterns (e.g., scale up during business hours and scale down overnight).

C. Use Managed and Serverless Services

- **Adopt Serverless Architectures:** Use serverless services like **AWS Lambda**, **Azure Functions**, or **Google Cloud Functions** to automatically scale based on workload and eliminate the need for managing infrastructure.
- **Managed Services:** Use managed services such as **Amazon RDS**, **Azure SQL Database**, and **Google Cloud Spanner** for databases, which handle scaling, patching, and backups for you.

D. Optimize Storage Utilization

- **Use Tiered Storage:** Choose the appropriate storage class based on access patterns. For example:
 - **AWS S3 Intelligent-Tiering** for automatic storage class transitions.
 - **Azure Blob Storage** with Hot, Cool, and Archive tiers.
 - **Google Cloud Storage classes** (Standard, Nearline, Coldline, Archive).
- **Lifecycle Policies:** Implement data lifecycle management to automatically move infrequently accessed data to cheaper storage tiers or delete obsolete data.
- **Optimize Volumes:** Right-size your block storage volumes (EBS, Azure Disks) to prevent over-provisioning and ensure that unused or unattached volumes are removed.

E. Implement Spot and Preemptible Instances

- **Use Spot Instances:** Use **AWS Spot Instances**, **Google Preemptible VMs**, or **Azure Spot VMs** for cost savings when running non-critical, fault-tolerant workloads. They can save up to 90% compared to On-Demand instances.
- **Spot Fleet and Auto Scaling:** Create spot fleets or auto-scaling groups that dynamically allocate Spot Instances along side On-Demand Instances for a cost-effective balance.

F. Cost Optimization Plans

- **Use Reserved Instances or Savings Plans:** For predictable workloads, commit to **Reserved Instances (RIs)** (AWS, Azure) or **Savings Plans** (AWS) to benefit from significant discounts (up to 72%).
- **Mix Pricing Models:** Adopt a mix of On-Demand, Reserved, and Spot pricing models to optimize costs based on different workload requirements.

G. Continuous Monitoring

- **Use Cloud Monitoring Tools:** Use tools like **AWS CloudWatch**, **Azure Monitor**, or **Google Cloud Monitoring** to track resource usage and performance. Set up alarms to detect high CPU, memory, or network usage.
- **Monitor Costs Regularly:** Use cloud cost management tools like **AWS Cost Explorer**, **Azure Cost Management**, and **Google Cloud Billing Reports** to monitor and analyze costs, ensuring resources are optimized.
- **Proactive Alerts:** Set up proactive alerts for cost overruns, underutilized resources, or resource bottlenecks.

H. Network Optimization

- **Optimize Data Transfer:** Minimize data transfer between regions and availability zones to reduce costs. Keep resources within the same region whenever possible.
- **Use Content Delivery Networks (CDNs):** Use **AWS CloudFront**, **Azure CDN**, or **Google Cloud CDN** to cache content closer to users and reduce network load.
- **Private Network Connectivity:** Use **VPC Peering** or **Direct Connect** (AWS), **ExpressRoute** (Azure), or **Cloud Interconnect** (Google Cloud) to reduce costs on data transfer within the cloud network.

I. Optimize Load Balancing

- **Auto-Tune Load Balancers:** Use managed load balancers like **AWS Elastic Load Balancer (ELB)**, **Azure Load Balancer**, or **Google Cloud Load Balancer** to dynamically distribute traffic across instances.
- **Leverage DNS-Based Load Balancing:** Use services like **AWS Route 53**, **Azure Traffic Manager**, or **Google Cloud DNS** to route traffic based on latency, geographic location, or performance.

K. Use Containerization

- **Run Containers Efficiently:** Use containerization platforms like **Docker** to package applications and optimize resource utilization by running multiple containers on the same instance.
- **Kubernetes Orchestration:** Implement container orchestration tools like **Kubernetes** with managed services like **Amazon EKS**, **Azure AKS**, or **Google GKE** to manage container workloads and automatically scale based on demand.

L. Decommission Idle Resources

- **Identify and Remove Idle Resources:** Regularly audit and remove idle or unattached resources such as EC2 instances, EBS volumes, IP addresses, or databases that are not in use.
- **Shutdown Unused Resources:** Set up automation (e.g., **AWS Lambda**, **Azure Automation**, **Google Cloud Functions**) to automatically shut down development



Key readings 2.3.2: Application of cloud resources utilization optimization

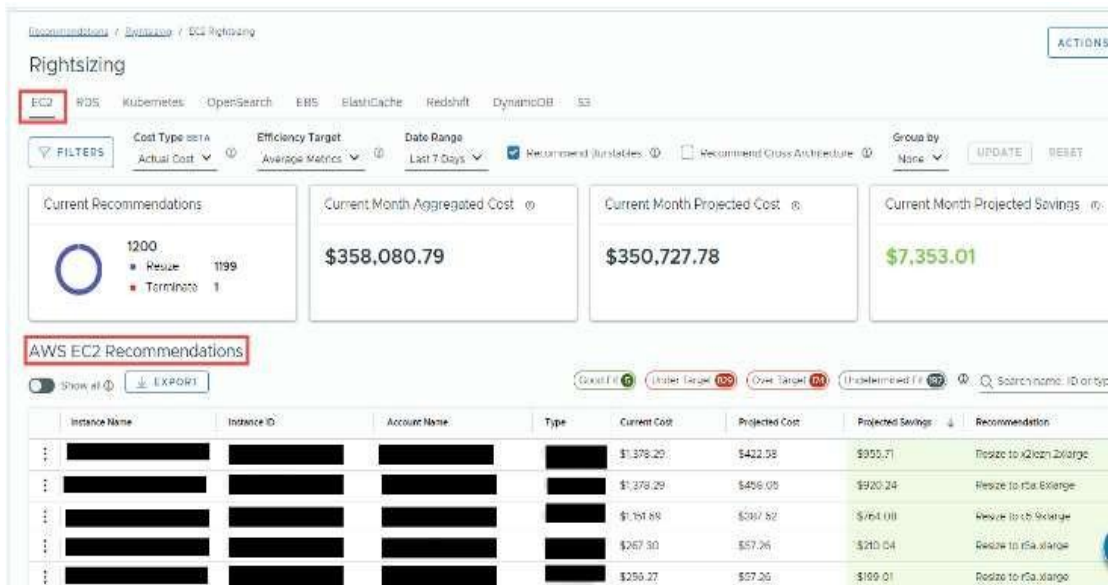
While applying cloud resources utilization optimization pass through these steps:

1. Right sizing Resources

- **What it is:** Rightsizing involves adjusting cloud resources (e.g., CPU, memory, storage) to match the actual workload needs.
- **How to apply:**
 - **Analyze usage patterns:** Use monitoring tools (e.g., AWS CloudWatch, right sizing tool, Azure Monitor) to analyze CPU, memory, and network usage.
 - **Select appropriate instance types:** For instance, AWS offers various EC2 instance types (e.g., compute-optimized, memory-optimized) tailored for different workloads.

Example using right sizing tool

- Launch right sizing tool
- Recommendations > Rightsizing.



2. Auto-scaling

- **What it is:** Automatically adjust the number of instances or resources based on demand.
- **How to apply:**
 - **Horizontal scaling:** Add/remove virtual machines (VMs) or containers based on incoming traffic.
 - **Vertical scaling:** Increase/decrease the power (CPU, memory) of an individual VM or container.
 - **Examples:** AWS Auto Scaling, Azure Virtual Machine Scale Sets, Kubernetes Horizontal Pod Autoscaler (HPA).

Example of applying Auto Scaling on AWS

Step 1: Sign into the AWS Management Console

Create an account and sign into the console. With Amazon EC2, you pay only for what you use. If you are a new AWS customer, you can get started with Amazon EC2 for free. For more information, see AWS Free Tier.

Step 2: Create a launch template

In the Amazon EC2 Dashboard, choose "Launch Templates" to create a launch template, specifying a name, AMI, instance type, and other details. Below are some guidelines on setting up your first launch template.

Choose an Amazon Machine Image (AMI): We recommend the Amazon Linux 2

AMI (free-tier eligible).

Choose an instance type: We recommend the t2.micro (free-tier eligible). Security group: You have the option to configure your virtual firewall.

Step 3: Create an Auto Scaling group

Using the Auto Scaling wizard, create an Auto Scaling group specifying a name, size, and network for your Auto Scaling group.

Step 4: Add Elastic Load Balancers (Optional)

When you set up Auto Scaling with Elastic Load Balancing, you can automatically distribute incoming application traffic across Amazon EC2 instances within your Auto Scaling groups to build highly available, fault-tolerant applications.

Associate your load balancer with your Auto Scaling group to distribute traffic for your application across a fleet of EC2 instances that can scale with demand.

You must first create a load balancer.

Step 5: Configure Scaling Policies (Optional)

Configure scaling policies for your Auto Scaling group.

3. Workload Scheduling

- **What it is:** Schedule workloads to run during non-peak times or distribute workloads efficiently across different instances.
- **How to apply:**
 - **Spot Instances:** Use AWS Spot Instances, Google Preemptible VMs, or Azure Spot VMs for cost-efficient computing.
 - **Batch Processing:** Schedule large jobs (e.g., data processing) to run during low-demand periods.
 - **Serverless computing:** Opt for serverless models (AWS Lambda, Azure Functions) for event-driven workloads to scale automatically and reduce idle resource costs.

Example of using AWS Spot Instances

- Log in to the AWS Management Console, then click the “Amazon EC2” tab.
- Click on “Spot Requests” in the navigation pane on the left.

- Click on “Pricing History” to open a view of historical pricing selectable by instance type. This will help you choose a maximum price for your request. Pricing shown is specific to the Availability Zone selected. If no Availability Zone is selected, you will see the prices for each Availability Zone in the Region.
- Click on “Request Spot Instances” and proceed through the Launch Instance Wizard process, choosing an AMI and instance type. Enter the number of Spot Instances you would like to request, your maximum price and whether the request is persistent or not. After choosing your key pair and security group(s), you are ready to submit your Spot Instance request.

4. Containerization and Orchestration

- **What it is:** Containers isolate applications and optimize resource usage by running multiple containers on the same VM with minimal overhead.
- **How to apply:**
 - **Docker and Kubernetes:** Deploy applications in lightweight containers using Docker. Use Kubernetes for orchestrating containers and automatically managing scaling and resource allocation.
 - **CI/CD pipelines:** Integrate containers into continuous integration/continuous delivery (CI/CD) pipelines to automate deployments and optimize usage.

5. Reserved Instances and Savings Plans

- **What it is:** Purchase cloud resources at a discounted rate for long-term use.
- **How to apply:**
 - **AWS Reserved Instances:** Commit to using AWS EC2 or RDS instances for 1 or 3 years in exchange for reduced rates.
 - **Azure Reserved VM Instances:** Commit to using Azure VMs for a certain period for cost savings.
 - **Savings Plans:** AWS also offers savings plans that allow more flexibility than reserved instances but with similar cost savings.

6. Monitoring and Cloud Cost Management

- **What it is:** Continuous monitoring and optimizing cloud spend by identifying underutilized resources or wasteful spending.
- **How to apply:**
 - **Monitoring tools:** Use built-in cloud monitoring tools such as AWS Cost Explorer, Azure Cost Management, or third-party tools like CloudHealth.
 - **Alerts:** Set up budget alerts when costs approach a certain threshold.
 - **Cost optimization recommendations:** Cloud providers often provide cost

optimization recommendations (e.g., AWS Trusted Advisor).

7. Storage Optimization

- **What it is:** Manage storage efficiently by selecting appropriate storage types and applying policies like data archiving.
- **How to apply:**
 - **Tiered storage:** Use different storage classes (e.g., AWS S3 Standard, S3 Infrequent Access, Glacier) based on access patterns.
 - **Data lifecycle policies:** Automatically move data between different storage tiers as it ages.
 - **Deduplication and compression:** Apply data deduplication and compression techniques to reduce storage footprint.

8. Network Optimization

- **What it is:** Ensure optimal data transfer rates while minimizing network costs and latency.
- **How to apply:**
 - **Content Delivery Networks (CDNs):** Use CDNs (e.g., AWS CloudFront, Azure CDN) to cache content closer to users and reduce latency.
 - **Virtual Private Networks (VPNs):** Use VPNs or Direct Connect options to optimize secure communication between on-premise data centers and cloud resources.
 - **Load balancing:** Implement load balancing to distribute traffic across multiple instances, improving performance and uptime.

9. Implement Serverless Architectures

- **What it is:** Serverless architectures eliminate the need to manage server resources by running code in response to events.
- **How to apply:**
 - **AWS Lambda:** Automatically scale based on the number of requests, paying only for compute time consumed.
 - **Azure Functions:** Similar to Lambda, pay only for the time your function runs, which reduces idle compute costs.
 - **Microservices:** Design applications using microservices architecture to leverage serverless technologies efficiently.

10. Use AI/ML for Predictive Scaling

- **What it is:** Use machine learning algorithms to predict future workloads and

resource demands.

- **How to apply:**
 - **AWS Predictive Scaling:** Predict future scaling needs based on historical data and adjust resources accordingly.
 - **Custom solutions:** Develop machine learning models to forecast resource usage based on patterns, then automate scaling and allocation processes.



Points to Remember

Cloud Resources Utilization

- **Cloud resource utilization optimization strategies** are right-sizing resources, using autoscaling, leveraging serverless architectures, and eliminating unused
- **Best practices of Cloud resource utilization optimization** involve applying the above methods right-sizing resources, using auto-scaling, leveraging serverless architectures, and eliminating unused resources.

Application of cloud resources utilization

Cloud resource utilization optimization involves various strategies to maximize performance and minimize costs in cloud environments. The application cloud resources utilization optimization includes the following steps:

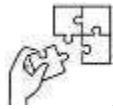
1. **Rightsizing Resources:** Adjust resource allocations (CPU, memory)
2. **Auto-scaling:** Automatically adjust resource levels (horizontally or vertically).
3. **Workload Scheduling:** Use cost-efficient resources like spot instances or schedule workloads during non-peak times.
4. **Containerization and Orchestration:** Utilize containers (e.g., Docker, Kubernetes) to optimize resource use and streamline deployments.

Step 1: Monitoring and Cost Management: Continuously monitor cloud usage and costs to identify optimization opportunities.

Step 2: Storage Optimization: Implement tiered storage and lifecycle policies to efficiently manage and archive data.

Step 3: Network Optimization: Reduce latency and costs through CDNs, load balancing, and optimized network configurations.

Step 4: Serverless Architectures: Use serverless models (e.g., AWS Lambda) to pay only for compute time, reducing idle resource costs.



Application of learning 2.3.

Your company provides a SaaS platform that serves multiple clients with varying usage patterns. Some clients use the platform heavily during business hours, while others access it sporadically throughout the day. Despite this variation, your cloud infrastructure is over-provisioned, leading to high costs during low-usage periods. You need to optimize cloud resources to reduce costs while ensuring scalability for peak usage. Apply rightsizing, auto-scaling and storage optimization strategies to reduce costs while maintaining performance and scalability.



Key readings 2.4.1: Description of cloud environment security measures

1. Definition of security measures as used in cloud computing services.

In cloud computing, security measures refer to the protocols, policies, technologies, and practices implemented to protect cloud-based systems, data, and infrastructure from unauthorized access, cyberattacks, data breaches, and other security threats.

2. Types of security measures as used in cloud computing services.

A. Physical Security

- Refers to the protection of the actual cloud data centers where the hardware is stored.
- Includes security guards, surveillance systems, biometric access controls, and climate control to prevent unauthorized physical access or damage to the servers.

B. Network Security

- Protects the cloud infrastructure and services from unauthorized network access and attacks.
- **Examples:**
 - **Firewalls:** Monitor and filter traffic between the cloud network and the internet.
 - **Intrusion Detection and Prevention Systems (IDPS):** Detect and prevent suspicious or malicious activities.
 - **Virtual Private Networks (VPNs):** Provide secure communication channels for data transmission.
 - **DDoS Protection:** Defend against Distributed Denial of Service attacks that overwhelm cloud services with traffic.

C. Data Security

- Ensures the confidentiality, integrity, and availability of data stored or processed in the cloud.
- **Examples:**
 - **Encryption:** Data is encrypted both in transit and at rest, ensuring only authorized parties can access it.
 - **Data Masking:** Obscures data to prevent exposure of sensitive information.
 - **Tokenization:** Replaces sensitive data with unique identification symbols or tokens.

D. Identity and Access Management (IAM)

- Controls who has access to cloud resources and enforces authentication and authorization policies.
- **Examples:**
 - **Multi-Factor Authentication (MFA):** Adds an extra layer of security by requiring multiple forms of verification.
 - **Role-Based Access Control (RBAC):** Restricts access to cloud resources based on user roles and responsibilities.
 - **Single Sign-On (SSO):** Allows users to authenticate once and gain access to multiple cloud services.

E. Application Security

- Focuses on securing cloud-based applications from vulnerabilities and threats.

Examples:

- **Code Reviews and Penetration Testing:** Identifying vulnerabilities in application code before deployment.
- **Web Application Firewalls (WAFs):** Protect applications from common web-based attacks such as SQL injection and cross-site scripting (XSS).
- **API Security:** Ensures secure use and communication of cloud APIs.

F. Compliance and Legal Measures

- Ensures that cloud services comply with relevant laws, regulations, and industry standards.
- **Examples:**
 - **Compliance Audits:** Regularly auditing the cloud environment to ensure compliance with standards like ISO 27001, SOC 2, GDPR, HIPAA, etc.
 - **Data Residency Policies:** Ensuring data is stored in specific locations to comply with local regulations.

G. Operational Security

- Focuses on protecting the day-to-day operations of cloud infrastructure and services.
- **Examples:**
 - **Security Information and Event Management (SIEM):** Collecting and analyzing security logs for detecting threats in real-time.
 - **Patch Management:** Regularly applying patches and updates to cloud systems to fix vulnerabilities.
 - **Incident Response:** Preparing a plan to respond to security breaches or incidents quickly.

H. Disaster Recovery and Business Continuity

- Ensures that cloud services can recover from security incidents, hardware failures, or natural disasters without significant downtime.
- **Examples:**
 - **Data Backups:** Regularly backing up data to ensure it can be restored in case of a failure.
 - **Disaster Recovery Plans:** Strategies for recovering services and data in the event of a disaster.
 - **Redundancy:** Using multiple data centers to ensure availability and resilience.

I. User Security Awareness

- Involves educating cloud users about security best practices to prevent security incidents caused by human error.
- **Examples:**
 - **Security Training:** Educating users about phishing attacks, password management, and safe data sharing.
 - **Regular Updates on Policies:** Keeping users informed about cloud security policies and practices.

3. Benefits of cloud measures as used in cloud computing services.

A. Data Protection & Privacy

- **Encryption:** Cloud security often includes robust encryption mechanisms that protect data both at rest (stored data) and in transit (data being transmitted). This prevents unauthorized access to sensitive data.
- **Data Loss Prevention (DLP):** Cloud providers implement DLP strategies to monitor and protect data, ensuring that confidential information is not inadvertently exposed or lost.

B. Compliance with Regulatory Standards

- **Compliance Certifications:** Cloud service providers comply with a variety of industry standards and certifications such as ISO 27001, GDPR, HIPAA, and SOC 2, helping businesses meet legal and regulatory requirements.
- **Audit Trails:** Cloud platforms typically maintain detailed logs and audit trails, ensuring transparency and traceability for compliance and security auditing purposes.

C. Identity and Access Management (IAM)

- **Access Control:** IAM services allow businesses to manage user access, ensuring that only authorized personnel can access specific data or applications.
- **Multi-factor Authentication (MFA):** MFA enhances security by requiring multiple forms of verification before users can access cloud resources, reducing the risk of unauthorized access.
- **Role-Based Access Control (RBAC):** Organizations can assign permissions based on user roles, ensuring that users only have access to the resources needed for their tasks, which minimizes the attack surface.

D. Threat Detection & Prevention

- **Intrusion Detection and Prevention Systems (IDPS):** Cloud security measures

often include IDPS, which monitor and detect unusual activities or malicious attempts to breach the system.

- **Security Information and Event Management (SIEM):** SIEM tools collect and analyze security data in real time, identifying potential threats or vulnerabilities and enabling rapid response to incidents.
- **Firewalls & DDoS Protection:** Cloud platforms implement advanced firewalls and Distributed Denial of Service (DDoS) protection to block malicious traffic and prevent service disruptions.

E. Automated Security Patching & Updates

- **Continuous Patch Management:** Cloud providers ensure that their systems are regularly updated with the latest security patches, protecting against known vulnerabilities and emerging threats without requiring manual intervention.
- **Automatic Software Updates:** Cloud services handle software updates, reducing the risk of attacks that exploit outdated software or unpatched systems.

F. Disaster Recovery and Backup

- **Automated Backups:** Cloud security measures include regular backups, ensuring that data can be recovered in the event of accidental deletion, data corruption, or cyberattacks like ransomware.
- **Geo-redundancy:** Many cloud platforms store data across multiple geographically distributed locations, ensuring data availability even in the event of regional disasters or data center failures.

G. Centralized Security Management

- **Unified Security Controls:** Cloud providers offer centralized dashboards and tools that allow businesses to manage security policies and monitor activity across all of their cloud services.
- **Unified Threat Intelligence:** Cloud platforms benefit from large-scale global operations, allowing them to gather and share intelligence on the latest threats, providing timely updates and protective measures to all customers.

H. Cost Efficiency

- **Reduced Cost of Security Infrastructure:** Cloud security measures allow businesses to avoid the high costs associated with building and maintaining on-premise security infrastructure (e.g., firewalls, security appliances).
- **Pay-as-you-go Security:** Companies can leverage the security tools offered by cloud providers without upfront investments, paying for the specific security services they need.

I. Improved Resilience and High Availability

- **Redundancy & Failover:** Cloud security measures ensure that data is replicated

across multiple locations, providing resilience in case of failure or attack. This guarantees high availability and minimizes downtime.

- **Zero Trust Architecture:** A growing number of cloud providers adopt Zero Trust principles, assuming no trust within the network perimeter and continuously verifying users, devices, and applications.

J. Enhanced Collaboration with Secure Environments

- **Secure File Sharing:** Cloud platforms ensure that files and documents shared between teams and across geographic locations are encrypted and accessible only by authorized users, ensuring safe collaboration.
- **Granular Permissions for Collaboration:** Cloud services allow businesses to set detailed permissions for different users or teams, ensuring that collaboration can occur without compromising security.

K. Flexibility and Agility

- **Security-as-a-Service (SECaaS):** Many cloud providers offer security as a service, allowing businesses to scale their security measures as needed and customize them to their unique environment without heavy lifting.
- **Rapid Response to Emerging Threats:** Cloud providers can deploy security fixes and updates almost instantaneously across their platform, ensuring that all users are protected from new threats faster than traditional on-premise security.

L. Real-Time Monitoring & Analytics

- **Proactive Threat Monitoring:** Continuous monitoring of networks, applications, and data helps in detecting unusual activity and responding in real-time to prevent potential security breaches.
- **Actionable Insights:** Advanced analytics provided by cloud platforms can help businesses identify vulnerabilities, optimize security measures, and create more effective incident response plans.

4. Cloud environment security measures

A. Identity and Access Management (IAM)

- **Access Controls:** IAM frameworks help manage who can access specific cloud resources. This involves setting permissions, roles, and policies that determine what actions users can perform on particular resources.
- **Multi-Factor Authentication (MFA):** MFA adds an extra layer of security by requiring users to provide two or more verification factors to gain access to cloud resources, ensuring stronger authentication.
- **Least Privilege Principle:** This principle limits user permissions to the minimum necessary to perform tasks, reducing the risk of insider threats or accidental data

exposure.

B. Data Encryption

- **Encryption at Rest:** Data stored in the cloud is encrypted using algorithms like AES-256. This ensures that even if unauthorized parties gain access to the stored data, it remains unreadable without the decryption key.
- **Encryption in Transit:** Data moving between cloud services or between a client and a cloud provider is encrypted, often using SSL/TLS protocols. This protects sensitive data from being intercepted during transmission.
- **Key Management:** Cloud providers offer services to securely manage and store encryption keys, allowing users to control how and when data can be decrypted.

C. Network Security

- **Virtual Private Cloud (VPC):** A VPC isolates a portion of a public cloud and allows users to create private networks within the cloud. This segmentation improves security by reducing exposure to public networks.
- **Firewalls:** Cloud providers implement firewall solutions that can be configured to allow or deny traffic based on predefined security rules. These firewalls can be used to create network perimeters that restrict access to certain applications or services.
- **Intrusion Detection & Prevention Systems (IDPS):** IDPS tools monitor cloud networks for unusual activity, such as unauthorized access or malicious behavior, and automatically block or alert administrators to potential threats.

D. Security Information and Event Management (SIEM)

- **Log Monitoring:** Cloud environments generate logs that track system activity, access patterns, and security events. SIEM tools analyze these logs in real time to identify suspicious activity or potential security threats.
- **Automated Alerts:** SIEM solutions can be configured to trigger automated alerts or actions when security anomalies or policy violations are detected, allowing for quick response to threats.
- **Centralized Management:** Cloud platforms often provide centralized dashboards for monitoring security events across the entire environment, simplifying management and response processes.

E. Data Loss Prevention (DLP)

- **Policy-based Protection:** DLP tools prevent sensitive information (e.g., financial data, personally identifiable information) from being accidentally or maliciously shared, leaked, or accessed by unauthorized users.
- **Content Monitoring:** DLP solutions monitor and analyze data moving within the cloud environment to ensure that confidential data is not transferred outside of

secure zones.

- **Data Masking:** In certain scenarios, DLP solutions apply masking techniques to hide sensitive data in cloud applications while still allowing legitimate users to perform necessary tasks.

F. Application Security

- **Secure Development Lifecycle (SDLC):** Cloud providers enforce secure development practices, including regular security testing, code review, and vulnerability scanning of applications deployed in the cloud.
- **Web Application Firewalls (WAFs):** WAFs protect cloud-based applications by filtering and monitoring HTTP traffic, blocking common web exploits such as SQL injection and cross-site scripting (XSS).
- **Container Security:** For cloud-native applications, security measures around containers (e.g., Docker, Kubernetes) ensure that workloads run in isolated environments, protecting the host system and other applications from malicious behavior.

G. Automated Patching and Updates

- **Patch Management:** Cloud providers regularly apply security patches to their underlying infrastructure and services, ensuring that known vulnerabilities are addressed quickly. This minimizes the risk of exploits based on outdated software.
- **Automated Updates:** Security updates are often applied automatically, reducing the chances of leaving systems exposed to newly discovered vulnerabilities. This is especially crucial in environments where resources are shared across multiple users (multi-tenancy).

H. Disaster Recovery and Business Continuity

- **Data Backup and Redundancy:** Cloud providers ensure data is backed up regularly and stored across multiple geographic locations. This geo-redundancy ensures that data remains available even if a failure occurs in one region.
- **Disaster Recovery Plans:** Cloud platforms offer disaster recovery (DR) solutions that enable businesses to quickly restore services in the event of a failure. This can include automated failover mechanisms and backup restoration processes.
- **High Availability:** Many cloud providers guarantee high availability with Service Level Agreements (SLAs), ensuring that critical applications and data are always accessible.

I. Zero Trust Security Model

- **Continuous Verification:** In a zero-trust architecture, no user, device, or service is trusted by default. Every access request is continuously verified based on identity, device health, location, and behavior.
- **Microsegmentation:** This involves dividing the cloud environment into smaller segments, each with its own access controls. This minimizes the impact of a potential breach, limiting the attack surface.
- **Granular Permissions:** Zero Trust implements granular permissions based on users' roles and the specific resources they need, ensuring that no broad access is given.

J. Distributed Denial of Service (DDoS) Protection

- **Traffic Scrubbing:** Cloud providers offer DDoS protection services that analyze incoming traffic, filtering out malicious traffic before it reaches critical applications. This ensures uninterrupted service even during an attack.
- **Auto-scaling to Handle Traffic Spikes:** Cloud infrastructure can scale automatically to handle large volumes of traffic, preventing DDoS attacks from overwhelming services.

K. Endpoint Security

- **Device Authentication:** Endpoint security measures ensure that only trusted devices can connect to the cloud environment. This can include certificate-based authentication or the use of secure tunnels (e.g., VPNs).
- **Mobile Device Management (MDM):** Cloud providers often integrate with MDM solutions that allow businesses to enforce security policies on employee devices, ensuring that cloud access is limited to secure and compliant devices.

L. Cloud Security Posture Management (CSPM)

- **Security Misconfiguration Detection:** CSPM tools continuously monitor cloud environments for misconfigurations, such as improperly secured storage buckets or overly permissive access policies. This reduces the risk of data exposure.
- **Compliance Monitoring:** CSPM ensures that cloud resources remain compliant with internal security policies and industry regulations (e.g., GDPR, PCI-DSS).
- **Remediation Guidance:** These tools provide automated or guided remediation steps to fix security issues, ensuring that organizations can rapidly address vulnerabilities.

M. Governance, Risk, and Compliance (GRC)

- **Cloud Governance Policies:** Effective GRC measures ensure that cloud environments are managed according to defined policies, including security protocols, access rules, and audit practices.
- **Risk Assessment:** GRC measures include regular risk assessments to evaluate

potential vulnerabilities and determine the best way to mitigate them.

- **Compliance Reporting:** Cloud environments often have built-in tools that help organizations generate compliance reports, ensuring that they meet the necessary security standards.



Points to Remember

In cloud computing, **security measures** refer to the protocols, policies, technologies, and practices implemented to protect cloud-based systems, data, and infrastructure from unauthorized access, cyberattacks, data breaches, and other security threats.

- **Types of security measures include:**

Physical Security, Network Security, Data security, Identity and Access Management (IAM), Application Security, Operational Security, Disaster Recovery and Business Continuity.

- **Benefits of security measures include:**

Data Protection & Privacy, Compliance with Regulatory Standards, A. Identity and Access Management (IAM), Threat Detection & Prevention, Automated Security Patching & Updates, Disaster Recovery and Backup, Centralized Security Management, Cost Efficiency

- **Cloud environment security measures include:**

Identity and Access Management (IAM), Data Encryption, Network Security

Security Information and Event Management (SIEM), Data Loss Prevention (DLP), Application Security, Automated Patching and Updates, Disaster Recovery and Business Continuity

Zero Trust Security Model, Distributed Denial of Service (DDoS) Protection, Endpoint Security, Cloud Security Posture Management (CSPM), Governance, Risk, and Compliance (GRC).



Application of learning 2.4

A local business needs to set up a small e-commerce website for its businesses. The website will have basic functionality, including product listings, a shopping cart, and an order processing system. The business owner has heard about cloud computing and wants to know about the security of cloud environment. The business owner is looking for an IT who can assist him to answer the following questions:

1. Identify the cloud environment security measures.
2. Identify the benefits cloud environment security measures.

Books:

Cloud Maintenance

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Indicative contents

3.1 Applying Cloud Resources Backup Strategies

3.2 Applying Patches

3.3 Fixing Errors in Cloud Computing Services

Key Competencies for Learning Outcome 3: Perform Maintenance

Knowledge	Skills	Attitudes
• Description of backup • Description of patches • Description of cloud errors	• Implementing cloud backup strategies • Implementing different types of patches • Applying error and fixing solutions	• Being patient • Being innovative • Being creative • Being passionate • Being attentive • Having critical thinking.



Key readings 3.1.1: Description of cloud backup

1. Definition of cloud backup

Cloud backup refers to the process of storing copies of your data on remote servers over the internet, rather than locally on a physical device like a hard drive or external storage. These remote servers, often maintained by third-party providers, are typically housed in data centres. Cloud backup provides an extra layer of protection for your data in case of hardware failure, data corruption, theft, etc.

Why use cloud backup?

- **Accessibility:** You can access your backed-up data from anywhere with an internet connection.
- **Security:** Cloud providers often have robust security measures to protect your data from loss, theft, or damage.
- **Reliability:** Cloud servers are typically highly redundant, meaning if one fails, another can take over.

- **Scalability:** Cloud backup can easily scale to accommodate your growing data needs.
- **Cost-effective:** Cloud backup can be more cost-effective than traditional on-site backup solutions.

Example of data backed up to the cloud:

- **Files and folders:** Documents, images, videos, and other personal or business files.
- **Applications:** Software programs and their associated data.
- **Systems:** Operating systems and configurations.

Types of backups

1. Full backup

A **full backup** is a type of data backup that involves making a complete copy of all the data in a system, including files, folders, databases, or entire systems. Every time a full backup is performed, all of the selected data is backed up in its entirety, regardless of whether it has changed since the last backup.

2. incremental backup

An **incremental backup** is a type of data backup where only the files that have changed since the last backup are copied to the storage medium. This method is more efficient than full backups, especially for frequent updates, as it reduces the amount of data that needs to be transferred.

3. Differential Backup

Differential Backup is another type of data backup strategy that copies all files that have changed since the last *full* backup. This means that each differential backup includes all the files that have been modified, created, or deleted since the most recent full backup.

4. Mirror Backup

Mirror Backup is a type of backup where an exact replica (or mirror) of the source data is created in real time or on a scheduled basis. This backup method mirrors the source data exactly as it is, reflecting all changes made, including modifications, deletions, or additions.

5. Hybrid Backup

Hybrid Backup combines the benefits of both on-premises and cloud backup strategies. It involves storing a portion of your data locally and another portion in the cloud. It involves backing up data to a local device (such as a server or external storage) and simultaneously or subsequently replicating that backup to the cloud. This dual-layer approach provides the benefits of both quick local access and the security of off-site cloud storage.

6. Cloud-to-Cloud Backup

Cloud-to-Cloud Backup is a backup strategy where data from one cloud service is backed up to another cloud service, rather than being stored locally or on physical hardware.

Benefits of backup

Cloud backup offers several advantages over traditional on-premises backup solutions:

Accessibility

- **Remote Access:** Access your data from anywhere with an internet connection.
- **Device Independence:** Access your data from various devices, including computers, smartphones, and tablets.

Scalability

- **On-Demand Storage:** Easily increase or decrease storage capacity as needed.
- **Cost-Effective:** Pay only for the storage you use.

Reliability

- **Redundancy:** Cloud providers typically have multiple data centers to ensure data availability.
- **Disaster Recovery:** Quickly recover data in case of a disaster or data loss.

Security

- **Robust Security Measures:** Cloud providers often implement advanced security measures to protect your data.
- **Regular Updates:** Cloud providers keep their security software up-to-date.

Cost-Effectiveness

- **Reduced Hardware Costs:** No need for on-premises backup hardware.
- **Simplified Management:** Less time and effort spent on managing backup infrastructure.

Efficiency

- **Automation:** Many cloud backup solutions offer automated backup schedules and features.
- **Integration:** Easily integrate with other cloud services and applications.

Compliance

- **Data Retention:** Meet industry-specific data retention requirements.
- **Data Privacy:** Adhere to data privacy regulations like GDPR and CCPA.

Identification of backup strategies

Backup strategy refers to a comprehensive plan outlining how an organization will protect its data from loss or corruption. It involves identifying critical data, selecting appropriate backup methods, determining backup frequency, and defining recovery procedures.

1. Assess Your Cloud Backup Needs

- **Data Types:** Identify the types of data that need to be backed up to the cloud (e.g., files, databases, applications).
- **Data Volume:** Estimate the amount of data to be backed up and the growth rate over time.
- **Criticality:** Determine the importance of different types of data and the impact of their potential loss.
- **Compliance:** Check for any regulatory requirements related to data protection and retention that apply to your organization.

2. Define Recovery Objectives

- **Recovery Time Objective (RTO):** Determine how quickly you need to restore data from the cloud to minimize downtime.
- **Recovery Point Objective (RPO):** Define the acceptable amount of data loss, indicating how frequently backups should be taken.

3. Evaluate Cloud Backup Methods

- **Full Cloud Backup**
- **Description:** A complete backup of all data to the cloud.
- **Pros:** Simple to restore as all data is contained in one backup set.
- **Cons:** Can be time-consuming and expensive due to the large amount of data transferred.
- **Best For:** Initial backups or periodic full backups as part of a broader strategy.
- **Incremental Cloud Backup**
- **Description:** Backs up only the data that has changed since the last backup (full or incremental).
- **Pros:** Efficient in terms of storage and bandwidth; quicker backups after the initial full backup.
- **Cons:** Recovery can be more complex as multiple incremental backups need to be restored.
- **Best For:** Frequent changes in data with a focus on storage efficiency.
- **Differential Cloud Backup**
- **Description:** Backs up data that has changed since the last full backup.
- **Pros:** Simpler recovery compared to incremental backups; requires fewer backup sets.
- **Cons:** Backup size grows over time; more storage required compared to incremental backups.
- **Best For:** When simpler recovery is preferred over incremental backups.
- **Continuous Data Protection (CDP)**
- **Description:** Continuously captures changes to data and backs them up in near real-time.
- **Pros:** Minimal data loss; allows recovery to any point in time.
- **Cons:** Requires high storage and bandwidth; more complex to manage.

- **Best For:** Critical applications where near-instant recovery is needed.
- **Snapshot Backup**
- **Description:** Captures the state of data at specific points in time, typically stored in the cloud.
- **Pros:** Fast recovery for specific data points; efficient for virtual environments.
- **Cons:** Limited to point-in-time recovery; can require substantial storage for multiple snapshots.
- **Best For:** Virtual machines and systems needing quick recovery.
- **Cloud-to-Cloud Backup**
- **Description:** Backs up data from one cloud service to another cloud service.
- **Pros:** Provides additional redundancy and protection for cloud-based data.
- **Cons:** May involve additional costs and complexity; requires managing multiple cloud services.
- **Best For:** Data stored in cloud services needing extra protection.
- **Hybrid Cloud Backup**
- **Description:** Combines on-premises backup solutions with cloud storage.
- **Pros:** Offers both local and cloud-based backups, balancing quick recovery with off-site protection.
- **Cons:** Can be complex and costly to manage; requires integration of multiple backup solutions.
- **Best For:** Organizations needing quick local recovery and off-site redundancy.
- **4. Consider Key Cloud Backup Features**
- **Scalability:** Ensure the cloud backup solution can easily scale with your growing data needs.
- **Security:** Verify that the solution provides strong encryption for data in transit and at rest.
- **Compliance:** Ensure the provider meets relevant regulatory requirements for data protection and retention.
- **Cost:** Evaluate pricing models, including storage costs, data transfer fees, and potential additional charges.
- **Integration:** Check how well the backup solution integrates with your existing systems and workflows.
- **Automation:** Look for features that automate the backup process to reduce manual intervention and errors.
- **5. Implement and Test**
- **Implementation:** Set up the chosen cloud backup solutions, ensuring all critical data is covered.
- **Testing:** Regularly test backups to verify that data can be restored successfully and meet recovery objectives.
- **6. Monitor and Maintain**

- **Monitoring:** Continuously monitor backup processes to ensure they are running smoothly and identify any issues.
- **Maintenance:** Keep backup systems and policies updated based on changes in data volume, technology, or business needs.

7. Backup Policies and Procedures

- **Retention Policies:** Define how long backup data is retained before deletion or archival.
- **Access Controls:** Implement measures to control who can access and manage backup data.
- **Documentation:** Maintain clear documentation of backup procedures, policies, and disaster recovery plans.



Key readings 3.1.2: Implementing cloud backup strategies.

Implementing cloud backup strategies involves deploying and configuring solutions that ensure your data is securely backed up to the cloud, and can be quickly restored when needed. Here's a step-by-step guide to implementing cloud backup strategies:

To implement cloud backup strategies effectively,
Follow these steps:

1. **Assess Your Data Needs:**
 - Identify critical data and applications.
 - Determine your recovery time objectives (RTO) and recovery point objectives (RPO).

2. **Select a Cloud Backup Solution:**

- Research providers (e.g., AWS Backup, Google Cloud Backup, Microsoft Azure Backup).
- Choose based on features, security, scalability, and cost.

3. **Design Your Backup Plan:**

○ **Backup Types:**

- **Full Back up:** Complete copy of all data.
- **Incremental Backup:** Backups only changes since the last backup.
- **Differential Backup:** Backups changes since the last full backup.

- **Frequency:** Decide on backup intervals (e.g., daily, weekly).

- **Retention:** Set policies for how long to keep backups.

4. **Automate Backup Processes:**

- Configure automated backups to run at scheduled intervals.

5. **Implement Security Measures:**

- Use encryption for data at rest and in transit.
- Set up access controls to ensure only authorized users can access backups.

6. **Regularly Test Backups:**

- Conduct periodic restore tests to ensure data can be recovered successfully.

7. **Document Backup Procedures:**

- Create detailed documentation of backup processes, including roles and responsibilities.

8. **Monitor and Maintain:**

- Continuously monitor backup performance and address issues promptly.
- Update the backup strategy as needed to adapt to changes in data volume or technology.

9. **Ensure Compliance:**

- Confirm that your backup strategy adheres to relevant legal and regulatory requirements.

Example: TechCo's Cloud Backup Strategy

1. Assess Data Needs

- **Critical Data:** Customer databases, transaction records, product inventory, and website files.
- **RTO:** 2 hours (TechCo needs to restore operations within 2 hours of data loss).
- **RPO:** 1 hour (TechCo can tolerate losing up to 1 hour of data).

2. Select a Cloud Backup Solution

- **Provider:** TechCo chooses **Microsoft Azure Backup** due to its seamless integration with their existing Azure infrastructure, strong security features, and scalability.
- **Plan:** Selects a service plan that includes automated daily backups and support for encryption.

3. Design Backup Plan

- **Backup Types:**
 - **Full Backup:** Weekly full backups every Sunday at midnight.
 - **Incremental Backup:** Daily incremental backups every night at 1 AM.
- **Frequency:**
 - **Full Backups:** Weekly.
 - **Incremental Backups:** Daily.
- **Retention:**
 - **Daily Backups:** Retained for 30 days.
 - **Weekly Backups:** Retained for 1 year.

4. Automate Backup Processes

- **Configuration:** Set up Azure Backup to automate the backup schedule:
 - Full backups on Sundays at midnight.
 - Incremental backups every day at 1 AM.
- **Notifications:** Configure alerts to notify the IT team of backup completion or failures.

5. Implement Security Measures

- **Encryption:** Data is encrypted using Azure's built-in encryption for data at rest and in transit.
- **Access Controls:** Use Azure Active Directory to manage access with role-based permissions, ensuring only authorized personnel can access backup settings and data.

6. Regularly Test Backups

- **Testing Schedule:** Perform monthly restore tests by restoring a sample of data to a staging environment to ensure the backup and recovery processes are working correctly.
- **Adjustments:** Make any necessary adjustments based on test results.

7. Document Backup Procedures

- **Documentation:** Create a detailed backup and recovery plan including:
 - Backup schedule and types.
 - Roles and responsibilities for managing backups.
 - Step-by-step recovery procedures.

8. Monitor and Maintain

- **Monitoring Tools:** Use Azure Monitor to track the status of backups and set up alerts for failures or issues.
- **Review:** Conduct quarterly reviews of backup performance and make adjustments as needed based on data growth or changes in business requirements.

9. Ensure Compliance

- **Regulations:** Verify that the backup strategy complies with relevant regulations

like GDPR for data protection and retention.

- **Audits:** Schedule regular audits to ensure ongoing compliance and adjust policies as necessary.



Points to Remember

Cloud backup defined as the process of storing copies of data on remote servers hosted on the internet (the "cloud").

- **Types of cloud backup** include Full back up, Incremental backup, Differential backup, Mirror Backup, Hybrid Backup, Cloud-to-Cloud Backup.
- **Benefits of cloud backup** include Accessibility, Scalability, Reliability, Security, Cost-Effectiveness, Efficiency, and Compliance.
- **Identification of backup strategies** include Assess Your Cloud Backup Needs, Define Recovery Objectives, Evaluate Cloud Backup Methods, Consider Key Cloud Backup Features, Implement and Test, Monitor and Maintain, Backup Policies and Procedures.
- **Steps to implement cloud backup strategies**
 1. Assess Your Data Needs.
 2. Select a Cloud Backup Solution.
 3. Design Your Backup Plan.
 4. Automate Backup Processes.
 5. Implement Security Measures.
 6. Regularly Test Backups.
 7. Document Backup Procedures.
 8. Monitor and maintain.
 9. Ensure Compliance



Application of learning 3.1.

You are the IT manager of a mid-sized e-commerce company, **Shop Now Online**, which processes hundreds of customer transactions daily. The company stores customer data, order history, payment information, and product inventories on its local servers. Recently, the company experienced a system failure due to a hardware malfunction, which led to a significant data loss and downtime.

As the IT manager of the company, you are tasked to implement cloud backup strategies to prevent future data loss and ensure business continuity.

Key readings 3.2.1: Description of Patches in cloud computing services

In the context of cloud computing services, "**patches**" refer to updates or modifications applied to cloud-based software, services, or infrastructure to fix bugs, resolve security vulnerabilities, enhance performance, or introduce new features. Cloud computing services typically involve multiple layers of software, including operating systems, applications, and middleware. Patches ensure that these components remain secure, reliable, and functional.

Types of Patches

Cloud computing services often require regular updates to address security vulnerabilities, improve performance, or introduce new features. These updates are known as **patches**.

Here are the primary types of patches commonly encountered in cloud environments:

1. Operating System Patches

- **Updates to the core OS:** These patches address security vulnerabilities, bug fixes, and performance improvements within the operating system itself (e.g., Linux, Windows).

- **Examples:** Kernel updates, driver updates, security patches.
- 2. Application Patches**
- **Updates to specific software:** These patches target individual applications running on the cloud infrastructure (e.g., databases, web servers, custom applications).
 - **Examples:** MySQL updates, Apache updates, application-specific bug fixes.
- 3. Firmware Patches**
- **Updates to hardware components:** These patches address issues or vulnerabilities in the firmware of hardware components (e.g., network devices, storage devices).
 - **Examples:** Router firmware updates, storage controller firmware updates.
- 4. Security Patches**
- **Prioritized updates:** These patches address critical security vulnerabilities that could be exploited by malicious actors.
 - **Examples:** Patches for known vulnerabilities like Heartbleed, Shellshock.
- 5. Performance Patches**
- **Optimizations:** These patches aim to improve the performance or efficiency of the cloud infrastructure or specific applications.
 - **Examples:** Patches to enhance database query performance, network throughput, or application response times.
- 6. Feature Updates**
- **New functionalities:** These patches introduce new features or enhancements to the cloud services.
 - **Examples:** New storage features, additional networking capabilities, or improved integration with other services.

Working operations

The working operations of patches in cloud computing involve a series of processes that ensure software, infrastructure, and applications within the cloud environment remain up-to-date, secure, and functional. These processes, often automated or semi-automated, involve detecting, scheduling, and applying patches across distributed cloud systems. Here's a breakdown of how patches operate within cloud computing environments:

1. Patch Detection and Identification

- **Vulnerability Scanning:** Cloud providers and organizations use automated tools to continuously scan for vulnerabilities in their infrastructure and applications.
- **Patch Release:** Once a vulnerability or issue is identified, software vendors or cloud service providers develop and release patches. This could include security updates, bug fixes, or performance improvements.
- **Patch Notification:** Cloud providers typically notify users of available patches,

either through management consoles, email alerts, or integrated patch management services.

2. Patch Management Tools

Cloud providers offer patch management tools to automate the patching process for cloud-based systems:

- **AWS Systems Manager Patch Manager**
- **Google Cloud OS Patch Management**
- **Azure Update Management**

These tools help automate the detection, scheduling, and application of patches, ensuring that systems remain up-to-date with minimal manual intervention.

3. Patch Testing and Validation

Before a patch is applied to a live environment, it is often tested in a **staging environment** to ensure it doesn't cause any disruptions:

- **Compatibility Testing:** Ensures that the patch works with existing software, operating systems, and infrastructure.
- **Rollback Mechanism:** In case of issues, the system should have a rollback process to revert to the previous state if a patch causes problems.

Some cloud environments allow users to apply patches to specific virtual machines (VMs) or instances in isolated environments to validate them before applying to production environments.

4. Patch Deployment (Scheduling and Rollout)

Patches are deployed in a controlled manner, often using automated systems:

- **Automated Patch Deployment:** Cloud providers or customers use patch management services to automate patch deployment across multiple virtual machines, applications, or databases.
- **Scheduling Patches:** Patches can be scheduled to occur during maintenance windows or times of low system usage to minimize disruptions to business operations.
- **Phased Rollouts:** In large-scale environments, patches are often rolled out in phases, starting with a small subset of servers or instances, to ensure that there are no major issues before full-scale deployment.

5. Patch Application

The actual application of patches depends on the service model (IaaS, PaaS, SaaS) and the cloud architecture:

- **IaaS (Infrastructure as a Service):** Customers are responsible for patching their own virtual machines (VMs), operating systems, and applications. The cloud provider ensures that the underlying hardware and hypervisor are patched.
 - Example: Patching an Ubuntu server hosted on AWS.
- **PaaS (Platform as a Service):** The cloud provider manages patching of the

platform's underlying infrastructure and middleware. Customers are typically responsible for patching their application code.

- Example: Google App Engine handles operating system patching, while developers patch their applications.
- **SaaS (Software as a Service):** The cloud provider is responsible for all patching, including both the application and infrastructure. Customers automatically benefit from the latest version without needing to manage patches.
- Example: A CRM platform like Salesforce applies patches without user intervention.

6. Patch Monitoring and Compliance

Once patches are applied, it's crucial to monitor the system for potential issues:

- **Post-Patch Monitoring:** After the patch is applied, system administrators monitor logs, performance metrics, and error reports to ensure the patch was successfully applied and didn't introduce new problems.
- **Audit and Compliance:** Many organizations require compliance reports to verify that patches have been applied across all systems. Cloud providers often include patch auditing tools to generate reports for security and compliance purposes.

7. Rollback and Recovery

If a patch causes issues or disrupts services, cloud environments offer rollback mechanisms:

- **Snapshot Backups:** In IaaS environments, snapshots of VMs are taken before patching. If a patch fails, the system can be restored to the pre-patch state.
- **Rollbacks in PaaS/SaaS:** Some cloud platforms offer rollback features that allow reverting to previous stable versions of the software.

Key readings 3.2.2: Implementing different types of patches

The implementation of different patch types in cloud computing involves specific processes and strategies, depending on the type of patch (security, bug fix, performance, etc.), the cloud service model (IaaS, PaaS, SaaS), and the environment being patched.

Below is a guide on how various patch types are implemented across cloud computing environments.

1. Security Patches

Purpose: Address vulnerabilities that could be exploited by attackers.

Implementation Steps:

1. **Vulnerability Detection:** Security patches are often driven by security assessments or threat detection tools that identify vulnerabilities in the system (e.g., a vulnerability in a cloud-based web server or database).
2. **Patch Development:** The software vendor or cloud provider releases a patch that fixes the security flaw.
3. **Notification:** Cloud providers notify users about the available patch via

dashboards or email alerts.

4. **Testing:** In a staging environment, the patch is applied to a subset of servers or instances to ensure it doesn't introduce new issues.
5. **Scheduled Deployment:** Security patches are typically scheduled for immediate application, but can be scheduled during a maintenance window if the risk is lower.
6. **Automation:** Patch management services (e.g., AWS Systems Manager Patch Manager) automate the deployment of security patches to virtual machines, containers, and services.
7. **Monitoring:** After applying the patch, monitoring tools are used to verify that the patch is successful and that the system functions normally.

Example:

- **AWS:** Automatically applies critical security patches to managed services like RDS (Relational Database Service).
- **Azure:** Azure Update Management applies security patches to Virtual Machine(s) based on predefined baselines.

2. Bug Fix Patches

Purpose: Correct software errors or glitches that cause instability or incorrect behaviour.

Implementation Steps:

1. **Bug Identification:** Bugs are identified through error logs, system monitoring, or user reports.

2. **Patch Development:** Software engineers or cloud vendors develop a fix for the bug and package it as a patch.
3. **Testing and Validation:** The bug fix is tested in a development or staging environment to ensure that it resolves the issue without affecting other functionalities.
4. **Controlled Rollout:** The patch is deployed to a small subset of production instances first to ensure it behaves as expected.
5. **Full Deployment:** Once the patch is validated, it is rolled out to all affected systems. Automated patch management tools can be used to apply bug fix patches across multiple environments.
6. **Post-Patch Testing:** Users monitor the system to ensure the bug is resolved and no new issues have been introduced.

Example:

- **PaaS Platform (e.g., Google App Engine):** A bug in the cloud-based application runtime is fixed by Google engineers, tested internally, and then deployed across all users' environments.

3. Performance Patches

Purpose: Optimize system performance, resource utilization, or speed.

Implementation Steps:

1. **Performance Bottleneck Identification:** Through performance monitoring tools (e.g., AWS CloudWatch or Azure Monitor), performance issues like slow response times or inefficient resource usage are identified.
2. **Patch Creation:** Engineers develop a patch that addresses the bottleneck, such as memory leaks, slow database queries, or inefficient load balancing.
3. **Testing in Non-Production:** Performance patches are tested in a non-production environment to measure the improvement and ensure there are no negative side effects.
4. **Deployment:** The patch is deployed in phases across production environments, starting with low-priority or non-critical systems.
5. **Monitoring and Fine-Tuning:** After applying the patch, systems are monitored for changes in performance. If additional fine-tuning is needed, follow-up patches or configurations may be applied.

Example:

- **Google Cloud:** Updates to the Google Cloud SQL service may include performance patches that improve query execution times, automatically applied to user databases.

4. Feature or Functional Patches

Purpose: Introduce new capabilities or enhance existing ones.

Implementation Steps:

1. **Feature Request and Development:** New features or enhancements are developed based on user requests or internal product roadmaps.
2. **Patch Packaging:** The feature is packaged as an update or patch and released as part of a software or service update.
3. **User Notification:** Cloud providers notify users about the availability of the new feature or enhancement, often through release notes, dashboards, or emails.
4. **Optional vs. Mandatory Deployment:** Depending on the feature, the patch might be optional (users can choose whether to enable it) or automatically applied.
5. **Training and Documentation:** Documentation and guides are often provided to help users adopt the new features.

Example:

- **AWS Lambda:** AWS rolls out new features, such as enhanced monitoring capabilities, automatically to all users, without requiring manual intervention.

5. Compliance Patches

Purpose: Ensure cloud environments meet updated regulatory standards or industry requirements.

Implementation Steps:

1. **Identification of New Compliance Requirements:** Regulatory bodies or industry groups issue new compliance standards (e.g., GDPR, HIPAA).
2. **Patch Creation:** Cloud service providers or software vendors develop patches that bring systems into compliance, such as data encryption standards or logging requirements.
3. **Notification:** Customers are notified of the patch and its implications for compliance.
4. **Deployment:** Compliance patches are typically mandatory and deployed across all affected systems, ensuring that organizations adhere to updated regulations.
5. **Audit and Verification:** After applying the compliance patch, systems are audited to verify that they now meet the necessary legal and regulatory requirements.

Example:

- **Azure:** Updates to Azure Key Vault to meet evolving compliance standards for encryption and data protection, automatically applied to all regions.

Example of security patches implementation

Implementing security patches in cloud computing services is essential to protect systems from vulnerabilities and potential exploits.

Here's a detailed step-by-step guide for effectively applying security patches in a cloud environment:

1. Identify and Assess Patch Requirements

- **Monitor for Vulnerabilities:** Regularly check for security advisories and vulnerability reports from cloud providers (e.g., AWS, Azure, Google Cloud), application vendors, and security teams.
- **Vulnerability Scanning:** Use automated tools to scan systems, virtual machines (VMs), containers, and applications for vulnerabilities.
- **Risk Assessment:** Assess the impact of the identified vulnerabilities. Prioritize patches based on the severity (e.g., critical, high, medium, low) and the risk they pose to business operations.

Tools:

- AWS Inspector, Azure Security Centre, Google Cloud Security Command Centre.

2. Plan and Schedule Patching

- **Patch Policy:** Define your patching policy, including frequency (e.g., weekly, monthly), criticality levels (e.g., critical patches to be applied immediately), and acceptable downtime windows.
- **Test Patches in Staging Environment:** Before deploying patches to production, test them in a non-production or staging environment to ensure they don't disrupt services or cause conflicts.
- **Create Backups:** Take snapshots or backups of systems to allow for quick recovery in case something goes wrong during patching.

Best Practices:

- Schedule patching during maintenance windows or low-traffic periods.
- Establish rollback procedures for patch failures.

3. Automate Patch Deployment

- **Configure Automated Patching:** Leverage cloud-native tools to automate patching for critical updates and reduce manual intervention.
- **Use Patch Management Tools:** Set up cloud provider tools to automatically install security patches on virtual machines, applications, and other resources.
- **Define Maintenance Windows:** Set specific maintenance windows to avoid service interruptions during critical hours.

Tools:

- **AWS Systems Manager Patch Manager:** Automates patching of Windows and Linux systems on EC2 instances.
- **Azure Update Management:** Manages patching across Azure virtual machines.
- **Google Cloud OS Patch Management:** Handles patch management for VMs in Google Cloud.

4. Apply Security Patches

- **Deploy Security Patches:** Roll out patches across the targeted cloud environment (e.g., VMs, containers, applications, or databases). This may include automatic or manual deployment, depending on the cloud setup.
- **Patch Testing in Production:** If needed, test patches in small production environments before rolling them out widely, especially for critical applications.

Example:

- In AWS, you can deploy patches across EC2 instances using Patch Manager and specify the types of updates to apply (security updates only, all updates, etc.).

5. Monitor Patch Deployment

- **Track Patch Progress:** Monitor the patching process to ensure patches are being applied successfully across all targeted instances and resources.
- **Handle Patch Failures:** If any patches fail to apply, identify and troubleshoot the issue, either by retrying the deployment or manually resolving conflicts.
- **Check Logs and Alerts:** Use monitoring and logging services to track the success of patch deployment and capture any errors or exceptions.

Tools:

- CloudWatch (AWS), Azure Monitor, Google Cloud Operations Suite.

6. Validate Patch Installation

- **Verify Patch Status:** Once the patches are applied, verify that the affected systems and resources are properly patched. This can be done through patch compliance reports provided by cloud-native tools.
- **Conduct Post-Patch Security Scans:** Run vulnerability scans after the patches are applied to ensure that the vulnerabilities are closed and the system is secure.
- **Monitor System Performance:** Check the performance and functionality of the

patched systems to ensure that there are no adverse effects after patching.

Tools:

- AWS Security Hub, Azure Security Center, Google Security Command Center.

7. Document and Report Patch Deployment

- **Document Changes:** Keep records of all applied patches, including systems updated, vulnerabilities fixed, and any associated downtime.
- **Compliance Reporting:** Generate compliance reports for audits and regulatory requirements, ensuring the organization's patch management process adheres to security standards (e.g., PCI-DSS, HIPAA).
- **Analyse Failures and Successes:** Review any issues encountered during the patching process and adjust future patch management strategies based on the findings.

Tools:

- AWS Config, Azure Compliance Manager, Google Cloud Asset Inventory.

8. Review and Refine Patch Management Policies

- **Evaluate Effectiveness:** Regularly review the effectiveness of your patch management process to identify areas for improvement.
- **Update Patch Schedules:** Based on feedback and system performance, adjust the frequency or process for patching to optimize security and minimize disruptions.
- **Continuous Monitoring:** Implement continuous monitoring and vulnerability scanning to identify new security risks and ensure patches are applied promptly.



Points to Remember

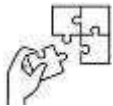
Description of patches

- A **patch** defined as software update that fixes bugs, security vulnerabilities, or adds new features to a system.
- **Types of patches include** Operating System Patches, Application Patches, Firmware Patches, Security Patches, Performance Patches, Feature Updates.
- **Working operations**
 1. Patch Detection and Identification
 2. Patch Management Tools
 3. Patch Testing and Validation

4. Patch Deployment (Scheduling and Rollout)
5. Patch Application
6. Patch Monitoring and Compliance
7. Rollback and Recovery

Steps to implement different types of patches (security patch)

1. Identify and Assess Patch Requirements.
2. Plan and Schedule Patching.
3. Automate Patch Deployment.
4. Apply Security Patches.
5. Monitor Patch Deployment.
6. Validate Patch Installation.
7. Document and Report.
8. Review and Refine Policies.



Application of learning 3.2

You are the Security Operations Manager at a cloud service provider that hosts applications for several financial institutions. Recently, your security team received a notification about a **critical vulnerability** in the web application framework used by many of your clients' applications. This vulnerability could lead to **remote code execution (RCE)**, allowing attackers to execute malicious code on the affected servers, posing a significant risk to sensitive financial data.

As Security Operations Manager, you are requested to implement different types of patches in the cloud computing services to prevent the above stated issues.



Key readings 3.3.1.: Description of errors in cloud computing services

Cloud backup error:

A cloud computing service error defined as any issue or malfunction that occurs when a cloud service is unable to perform as expected. Cloud services provide on-demand access to computing resources (such as servers, storage, databases, networking, and software) over the internet. When an error occurs, it can affect the availability, performance, or functionality of these services.

Types of cloud computing service errors

Some common types of cloud computing service errors include:

1. Service Availability Errors:

These errors occur when the cloud service is down or unavailable. This can happen due to hardware failures, network issues, or outages in the data center where the cloud services are hosted. For example, a cloud-based application might become inaccessible if the underlying infrastructure fails.

Example: Users cannot access an application because the cloud provider's data centre is experiencing downtime.

2. Authentication and Authorization Errors:

These occur when users or applications are unable to access cloud services due to issues with credentials, permissions, or security policies.

Example: A user cannot log into a cloud platform because their credentials are invalid or there is an issue with identity management systems (such as AWS IAM or Azure Active Directory).

3. Resource Limits Exceeded:

Many cloud providers enforce resource limits or quotas (such as CPU, memory, or network bandwidth). Errors can occur if the service exceeds these limits, often leading to degraded performance or failure to create new instances.

Example: An application might fail to deploy because the maximum number of virtual machines (VMs) allowed by the account has been reached.

4. Configuration Errors:

Cloud services often require specific configurations to work correctly. If there is a misconfiguration in networking, storage, security, or any other cloud service setting, errors can arise.

Example: A database service might fail because it is misconfigured to reject incoming connections from the application.

5. Service-Specific Errors:

Different cloud services have their own specific errors. For example, a cloud storage service (e.g., AWS S3, Azure Blob Storage) might return errors related to permissions, object limits, or bucket availability.

Example: Trying to upload a file to cloud storage but receiving a "403 Forbidden" error due to improper access permissions.

6. Latency and Performance Issues:

Sometimes, errors are related to high latency, timeouts, or slow performance in cloud services, which can affect the user experience. These are often caused by overloaded servers, network congestion, or poor configurations.

Example: A database query takes too long to return results, leading to timeouts in the application that relies on it.

7. **Billing and Subscription Errors:**

Cloud services operate on a pay-as-you-go model, and errors can occur if the account's billing or subscription is not up to date. Services might stop functioning if a payment fails or if the subscription period expires.

Example: Cloud services are suspended due to unpaid bills, causing the entire infrastructure to become unavailable.

8. **Data Consistency or Storage Errors:**

In distributed cloud storage systems, errors can occur with data replication, storage failures, or data corruption. This might lead to missing data or inconsistencies across different regions or services.

Example: An application that stores user data in a distributed database experiences data loss because one of the storage nodes failed.

Identification of different error fixing solutions

1. Automated Remediation

In cloud computing, **Automated Remediation** refers to systems that automatically detect and resolve errors or issues without human intervention. It is particularly useful in cloud environments, where the scale and complexity of infrastructure make manual error resolution impractical. Automated remediation solutions in cloud services typically involve the use of monitoring, alerts, and predefined actions to address common issues.

Here is some automated remediation techniques commonly used in cloud computing services:

- **Auto-Scaling for Resource Issues**

Problem: Resources (e.g., CPU, memory, storage) are over-utilized or under-utilized, causing performance degradation or unnecessary costs.

Automated Solution:

Auto-Scaling Policies: Cloud platforms like AWS, Azure, and Google Cloud offer auto-scaling features that automatically add or remove resources (e.g., VMs, containers) based on workload demands.

Example: If CPU usage goes beyond 80%, the system automatically adds more instances to handle the load.

2. Automated Security Remediation

Problem: Security vulnerabilities such as open ports, insecure configurations, or detected attacks.

Automated Solution:

Security Monitoring and Remediation Tools: Cloud services integrate security solutions that automatically detect threats (e.g., AWS GuardDuty, Azure Security

Center) and can trigger actions like blocking IP addresses, closing ports, or rotating credentials.

Example: A security tool detects unauthorized access attempts and automatically updates firewall rules to block the offending IP address.

Automated Backup and Disaster Recovery

Problem: Data corruption or loss due to accidental deletion, software bugs, or hardware failure.

Automated Solution:

Automated Backup and Restore: Cloud platforms offer automatic backups (e.g., AWS Backup, Google Cloud Backup) that regularly create snapshots or backups of critical data. If data loss occurs, automated recovery scripts can restore the latest backup.

Example: Daily backups are automatically created, and in the event of data corruption, the system triggers a restore from the most recent backup.

2. Manual Troubleshooting

Manual troubleshooting in cloud computing services involves identifying, diagnosing, and resolving issues that may occur within cloud environments, such as resource misconfigurations, service outages, network failures, or security issues. Here are steps and key strategies for manually troubleshooting errors in cloud computing services:

Identify the Problem

Gather Information: Start by collecting detailed information about the issue. This may include error messages, application logs, user reports, system metrics, or alerts from monitoring tools.

Categorize the Issue: Try to determine the nature of the problem:

Performance Issues: Slow response times, latency, or resource bottlenecks.

Connectivity Issues: Network failures, dropped packets, or unresponsive services.

Service Failures: Instances, databases, or services failing to start or respond.

Configuration Errors: Misconfigured services, firewall rules, or permissions.

Check Service Health Dashboards

Cloud Service Status Pages: Check if the issue is caused by an outage or

degradation of service at the cloud provider level.

Analyse Logs and Monitoring Data

Application Logs: Inspect logs generated by your application to detect errors or failures. Logs can provide insights into misconfigurations, exceptions, or crashes.

Inspect Resource Configuration

Misconfigured Instances or Services: Check the configuration of your cloud resources, such as virtual machines (VMs), databases, load balancers, or storage. Misconfigured resources can lead to service failures or degraded performance.

Network Troubleshooting

Ping and Traceroute: Use basic network troubleshooting tools like ping and traceroute to check if there is network connectivity between services or to an external resource.

Ping: Can confirm whether a particular service or instance is reachable.

Traceroute: Helps identify where network issues (e.g., packet loss or high latency) are occurring along the path between your cloud resources.

3. Software Updates and Patching

Software updates and patching are critical maintenance tasks in cloud computing environments to ensure systems are secure, perform well, and remain free from known vulnerabilities.

Examples of Software Update and Patching Errors in Cloud Computing

- **Update Failures**

The update or patch installation fails midway, resulting in incomplete installations or rollback of changes.

Causes:

Network connectivity issues: Poor connectivity during the update process can result in download failures or timeouts.

Insufficient resources: Lack of sufficient memory, CPU, or disk space on virtual machines (VMs) can prevent updates from completing.

Version incompatibility: The new software patch may not be compatible with the current system configuration or dependencies.

Solution:

- **Check connectivity:** Ensure that the network connection is stable throughout the update process. For cloud-based instances, verify the availability of the required bandwidth.
- **Resource scaling:** Monitor and, if necessary, temporarily increase the resource allocation (e.g., CPU, memory, disk space) of the instances during the update process.
- **Pre-check compatibility:** Review patch release notes to check for compatibility issues and dependencies. In case of incompatibility, use version control to revert to previous stable configurations.

- **Failed Rollbacks**

After a failed update, an attempted rollback to the previous version of the software or system state also fails, leaving the system in an unstable state.

- **Causes:**
 - **Incomplete backups:** No proper snapshots or backups were taken before the update, making rollback impossible.
 - **System dependency changes:** Changes to system libraries or dependencies during the update can prevent rollbacks from working correctly.

- **Solution:**

- **Backup before updates:** Always take snapshots or backups before applying updates or patches. Cloud platforms like AWS (EC2 snapshots), Azure (Backup Service), and Google Cloud (Snapshots) can automate this process.
 - **Use atomic updates:** In complex environments, use atomic or transaction-based updates that ensure changes are only applied if they can be completed successfully.
 - **Test rollbacks in staging:** Test rollback procedures in a staging environment to ensure that the rollback mechanism works correctly in case of failure.
- **Scaling Resources**

Scaling resources in cloud computing services refers to adjusting computing power, storage, and other resources to meet fluctuating demands. Cloud environments allow for both **vertical scaling** (increasing resources on existing instances) and **horizontal scaling** (adding more instances to distribute load).

- **Data Recovery and Backup Restoration**

Data recovery and backup restoration are critical functions in cloud computing, ensuring data integrity and availability after accidental deletions, system failures, or cyberattacks.

Error solution Documentation

This documentation outlines common errors encountered in cloud computing environments, their root causes, the procedures to address these issues, and the final solutions.

1. Auto-Scaling

Failure Error:

Auto-scaling fails to initiate during traffic spikes, causing service disruption.

Root Cause:

- Incorrect or overly conservative scaling policies (CPU, memory thresholds).
- Slow instance provisioning due to high startup times.
- Misconfigured auto-scaling group or triggers.

Procedures:

1. Analyse Traffic Patterns:

- Use historical data to identify peak times and corresponding system load.

2. Examine Scaling Configuration:

- Check auto-scaling triggers and policies to ensure they are properly configured for your workloads.

3. Test Auto-Scaling:

- Run load tests to simulate traffic spikes and verify that auto-scaling responds appropriately.

Solution:

- **Adjust Scaling Triggers:** Set lower CPU/memory thresholds to trigger faster scaling.
- **Optimize Instance Startup Time:** Pre-warm instances or use containerized services to reduce initialization delays.
- **Enable Monitoring:** Use tools like AWS CloudWatch or Azure Monitor to track scaling actions and trigger alerts for failures.

2. High Network

Latency Error:

Users experience slow application response times due to high latency.

Root Cause:

- Geographic distance between users and cloud servers.
- Inefficient routing or lack of content delivery network (CDN) usage.
- Bandwidth bottlenecks or improper network configurations.

Procedures:**1. Measure Latency:**

- Use tools like ping or traceroute to measure network latency.

2. Identify Bottlenecks:

- Use cloud-native monitoring tools (AWS VPC Flow Logs, Azure Network Watcher) to pinpoint network issues.

3. Test Routing Optimization:

- Use CDNs to simulate content delivery and measure improvements in latency.

Solution:

- **Implement CDN:** Use a CDN to cache and deliver content from edge locations closer to users.
- **Geographically Distribute Resources:** Deploy instances in multiple regions and route users to the nearest one using global load balancers.
- **Optimize Network Settings:** Tune network settings for better bandwidth and reduced latency between virtual networks.

3. Database Performance**Degradation Error:**

Database queries are slow, causing application performance issues.

Root Cause:

- Inefficient database queries or lack of proper indexing.
- Insufficient database capacity to handle high traffic.
- Replication lag between master and replicas.

Procedures:**1. Monitor Database Performance:**

- Use performance insights tools (e.g., AWS RDS Performance Insights) to monitor slow queries and database load.

2. Review and Optimize Queries:

- Analyse slow-running queries and add necessary indexing or optimizations.

3. Check Replication Health:

- Monitor replication lag to ensure that replicas are in sync with the master database.

Solution:

- **Query Optimization:** Rewrite inefficient queries and add indexes to frequently queried columns.
- **Scale the Database:** Scale vertically (larger instance) or horizontally (add read replicas) to handle increased traffic.

- **Monitor and Fix Replication Lag:** Use synchronous replication to minimize lag and ensure data consistency.

4. Data Inconsistencies or

Loss Error:

Data inconsistencies across regions or databases, or data loss due to failed backups.

Root Cause:

- Replication issues or failed synchronization across multiple regions or replicas.
- Incomplete backups or insufficient backup schedules.
- Storage device failures.

Procedures:

1. **Audit Backup and Replication Configurations:**
 - Review backup schedules, replication settings, and disaster recovery setups.
2. **Run Data Consistency Checks:**
 - Use database integrity tools to check for data inconsistencies or corruption.
3. **Test Backup Restoration:**
 - Simulate backup restorations to ensure they are functioning correctly.

Solution:

- **Enhance Backup Strategies:** Automate and schedule backups with versioning enabled to ensure data safety.
- **Implement Cross-Region Replication:** Use global replication services to keep data synchronized across regions.
- **Monitor Backup and Replication Health:** Set up alerts for backup failures or replication issues and respond in real-time.

5. Service Outage or

Downtime Error:

Critical services experience outages, leading to unplanned downtime.

Root Cause:

- Single points of failure (e.g., no redundancy).
- Misconfigured health checks or delayed service recovery.
- Software bugs or resource exhaustion (CPU, memory).

Procedures:

1. **Identify Affected Services:**
 - Use cloud monitoring dashboards (AWS CloudWatch, Azure Monitor) to identify which services are impacted.
2. **Review Logs and Health Checks:**
 - Analyze logs and health check configurations to determine if failures occurred due to misconfiguration.
3. **Test Failover Systems:**
 - Simulate service failures to ensure redundancy mechanisms (e.g., multi-AZ

deployments) work as expected.

Solution:

- **Multi-AZ and Multi-Region Deployments:** Deploy resources across multiple availability zones and regions to ensure redundancy.
- **Automate Health Checks and Auto-Recovery:** Use automated health checks and configure auto-recovery for failing instances.
- **Implement Disaster Recovery (DR):** Develop a DR plan using cloud-native solutions like AWS Disaster Recovery or Azure Site Recovery.

6. Unauthorized Access or Security

Breach Error:

Unauthorized access or exposure of sensitive data due to security misconfigurations.

Root Cause:

- Misconfigured Identity and Access Management (IAM) roles or permissions.
- Publicly accessible cloud resources (e.g., S3 buckets, databases).
- Lack of encryption or security measures.

Procedures:

1. **Audit IAM Policies:**
 - Review IAM roles, permissions, and access logs to identify over-permissioned users or exposed resources.
2. **Review Public Accessibility:**
 - Check for any public access to sensitive resources (e.g., S3 buckets, VMs).
3. **Perform Security Tests:**
 - Run security assessments using tools like AWS GuardDuty or Azure Security Center to detect vulnerabilities.

Solution:

- **Implement the Principle of Least Privilege:** Reduce permissions by refining IAM policies and using role-based access control (RBAC).
- **Restrict Public Access:** Use VPCs, firewalls, and security groups to restrict access to sensitive resources.
- **Enable Encryption:** Encrypt data at rest and in transit and manage encryption keys with services like AWS KMS or Azure Key Vault.

7. High Cloud Costs Due to Over-

Provisioning Error:

Unexpectedly high cloud service bills due to over-provisioned resources.

Root Cause:

- Oversized instances or unused resources.
- Unoptimized storage or excessive scaling.
- Lack of cost controls or budgeting mechanisms.

Procedures:

1. Analyse Resource Utilization:

- Use cloud cost management tools (e.g., AWS Cost Explorer, Azure Cost Management) to identify underutilized resources.

2. Set Cost Alerts and Budgets:

- Configure budgets and set up cost alerts to monitor spending and prevent overruns.

3. Test Resource Scaling:

- Simulate load tests and observe the auto-scaling behaviour to ensure resources are not over-provisioned.

Solution:

- **Right-Size Resources:** Scale down instances based on usage or switch to spot instances where possible.
- **Optimize Storage:** Use tiered storage for infrequently accessed data to reduce storage costs.
- **Implement Cloud Budgets:** Set strict budget limits and configure automatic alerts to detect when spending approaches predefined thresholds.

8. Failed Backups or Snapshot

Issues Error:

Backup jobs fail to complete, or snapshots are not created as expected, risking data loss.

Root Cause:

- Network issues, resource limitations, or misconfigured backup settings.
- Insufficient storage space for backups or snapshots.
- Lack of alerts or notifications for failed backups.

Procedures:

1. Monitor Backup Jobs:

- Use backup logs and monitoring tools to identify failures in backup processes.

2. Check Storage Limits:

- Verify that sufficient storage is available for backups and snapshots.

3. Test Backup Restorations:

- Perform periodic restore tests to verify that backups are functional and data integrity is maintained.

Solution:

- **Automate Backups:** Use cloud-native tools like AWS Backup or Azure Backup to automate backups and set alerts for failures.
- **Enable Versioning and Redundant Backups:** Implement versioning for cloud storage and configure redundant backups across multiple regions.
- **Regular Backup Validation:** Conduct regular tests to ensure backups are functional and can be restored successfully.



Key readings 3.3.2: Applying errors and fixing solutions

1. Fixing Scaling Resource Errors

- **Problem:** Ineffective scaling policies, slow instance provisioning, resource exhaustion, or network limitations.
- **Solution:**
 - **Adjust Scaling Policies:** Ensure auto-scaling thresholds are realistic and fine-tune cooldown periods. Use predictive or proactive scaling, when possible, to preemptively handle expected traffic spikes.
 - **Optimize Instance Provisioning:** Use containers or lightweight instances to speed up scaling. Pre-warm instances to be available during known high-traffic periods.
 - **Monitor Network Bandwidth:** Ensure the underlying network can support the scaled infrastructure. Upgrade network bandwidth as needed, and use content delivery networks (CDNs) to offload static content.

2. Fixing Data Recovery and Backup Restoration Errors

- **Problem:** Slow recovery times, incomplete backups, or data corruption.
- **Solution:**
 - **Ensure Backup Integrity:** Regularly test restore operations to ensure backups are complete and uncorrupted. Automate backup verification with cloud-native tools

like **AWS Backup**, **Azure Backup**, or **Google Cloud Backup**.

- **Optimize for Fast Recovery:** Store critical backups in high-performance storage tiers (e.g., hot or warm storage) for quicker recovery. Use incremental or differential backups to speed up the restoration process.
- **Establish Proper Permissions:** Use cloud identity and access management (IAM) to ensure only authorized users and services can restore backups. Implement encryption key management for secure data access.

3. Fixing Software Updates and Patching Errors

- **Problem:** Failed updates, compatibility issues, or downtime during patching.
- **Solution:**
 - **Use Automated Patch Management:** Utilize cloud-native services for automated updates (e.g., **AWS Systems Manager Patch Manager**, **Azure Update Management**, **Google OS Patch Management**) to schedule and apply updates across instances.
 - **Test Patches in a Staging Environment:** Implement a staging environment to test patches before rolling them out to production. This avoids compatibility issues and reduces the likelihood of failure.
 - **Enable Rolling Updates:** Use rolling updates and **canary deployments** to minimize downtime by updating instances gradually instead of all at once.

4. Fixing Security and Permissions Errors

- **Problem:** Unauthorized access, data breaches, or misconfigured access control.
- **Solution:**
 - **Implement Least Privilege Access:** Use role-based access control (RBAC) and IAM to grant the minimum required permissions. Regularly audit roles and permissions to prevent unauthorized access.
 - **Encrypt Data:** Ensure data is encrypted both at rest and in transit using cloud-native encryption services (e.g., **AWS KMS**, **Azure Key Vault**, **Google Cloud KMS**).
 - **Enable Multi-Factor Authentication (MFA):** Require MFA for access to critical cloud resources to add an extra layer of security.

5. Fixing Load Balancer and Traffic Management Errors

- **Problem:** Uneven traffic distribution, poor performance, or downtime due to load balancer issues.
- **Solution:**
 - **Optimize Load Balancer Configuration:** Use cloud-native load balancers (e.g., **AWS Elastic Load Balancer**, **Azure Load Balancer**, **Google Cloud Load Balancer**) and configure them to distribute traffic efficiently. Ensure health checks are in place to route traffic away from unhealthy instances.
 - **Use Global Load Balancing:** For multi-region deployments, implement global load balancing to distribute traffic based on region, improving performance and reliability.
 - **Configure Auto-Scaling with Load Balancer:** Ensure that your auto-scaling policies are aligned with the load balancer, so instances are added or removed based on real-time traffic demands.

6. Fixing Resource Quota and Limit Errors

- **Problem:** Cloud provider-imposed limits on resources (e.g., compute instances, vCPUs) prevent scaling.
- **Solution:**
 - **Monitor and Manage Quotas:** Use built-in monitoring tools to track resource usage against quotas (e.g., **AWS Service Quotas**, **Azure Subscription Limits**, **Google Cloud Quotas**).
 - **Request Quota Increases:** If needed, submit requests to increase quotas for critical resources to avoid hitting limits during scaling or recovery operations.
 - **Use Resource Tags:** Implement tagging to track resource usage by department, project, or workload, ensuring efficient use of cloud resources.

7. Fixing Network and Latency Issues

- **Problem:** High latency or network bottlenecks reduce application performance.
- **Solution:**
 - **Upgrade Network Bandwidth:** Use enhanced networking options provided by the cloud vendor (e.g., **AWS Elastic Network Adapter**, **Azure Accelerated**

Networking, Google Cloud Premium Tier Networking) to improve network performance.

- **Optimize Traffic with CDNs:** Offload static content to CDNs to reduce latency and network strain. Use **AWS CloudFront**, **Azure CDN**, or **Google Cloud CDN** to cache content closer to end-users.
- **Implement Virtual Private Cloud (VPC) Peering:** Ensure that instances across regions or zones are efficiently connected by configuring VPC peering or using **Direct Connect** (AWS), **ExpressRoute** (Azure), or **Interconnect** (Google Cloud).

8. Fixing Application Performance and Availability Errors

- **Problem:** Application slowdowns, crashes, or availability issues.
- **Solution:**
 - **Use Auto-Scaling:** Enable auto-scaling (horizontal or vertical) for applications to ensure they can handle fluctuating workloads.
 - **Implement Health Monitoring:** Use cloud-native monitoring services (e.g., **AWS CloudWatch**, **Azure Monitor**, **Google Cloud Monitoring**) to track application performance metrics in real-time. Set up alerts for critical performance thresholds.
 - **Implement High Availability (HA):** Ensure application redundancy by deploying in multiple availability zones (AZs) or regions, and use failover mechanisms to maintain availability in case of a zone failure.

9. Fixing Billing and Cost Overruns

- **Problem:** Unexpected or high cloud costs due to inefficient resource utilization.
- **Solution:**
 - **Implement Cost Monitoring:** Use cost management tools like **AWS Cost Explorer**, **Azure Cost Management**, or **Google Cloud Billing** to monitor and optimize cloud spending.
 - **Enable Auto-Shutdown for Idle Resources:** Use cloud automation tools to shut down unused or underutilized instances automatically.
 - **Use Spot Instances:** For non-critical workloads, use spot instances (e.g., **AWS Spot Instances**, **Azure Spot VMs**, **Google Preemptible VMs**) to reduce costs.



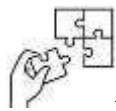
Points to Remember

Description of cloud computing service errors.

- A **cloud computing service error** is defined as any issue or malfunction that occurs when a cloud service is unable to perform as expected.
- **Types of cloud computing service errors** include Service availability Error, Authentication and Authorization Errors, Resource Limits Exceeded, Configuration Errors, Service-Specific Errors.
- **Error fixing solutions** include Automated Remediation, Manual troubleshooting, Software updates and patching, Scaling resources, and Data recovery and backup restoration.
- **Error solution Documentation** include Root Cause, Procedures, Procedures, and solution

Steps to apply errors on cloud computing services.

1. Identify the Error
2. Diagnose the Cause
3. Fix the solution



Application of learning 3.3.

As a cloud systems engineer at a growing e-commerce company, you are responsible for maintaining the company's cloud-based infrastructure hosted on a major cloud provider (AWS, Azure, or GCP). Recently, customers have reported intermittent downtime and slow performance on the website during peak hours, impacting the user experience and revenue.

As a cloud system engineer, you are required to apply error-fixing solutions to fix the above stated issues.

END

